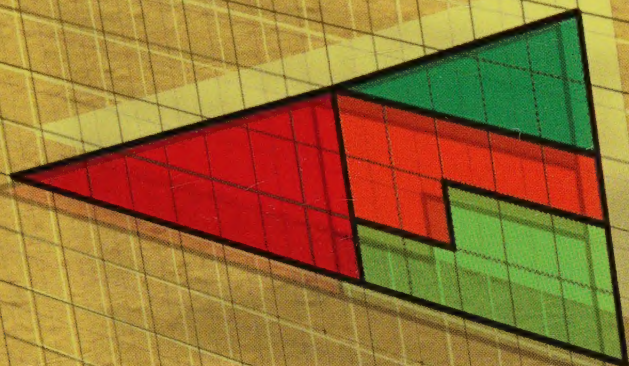


PARADOXES IN MATHEMATICS



**STANLEY J.
FARLOW**

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PARADOXES IN MATHEMATICS

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There's more than one way to define a paradox, and this intriguing book offers examples of every kind. Stanley J. Farlow, a prominent educator and author, presents a captivating mix of mathematical paradoxes: the kind with surprising, nonintuitive outcomes; the variety that rely on mathematical sleight-of-hand to impress the unwary observer; and the baffling type with a solution that passes all understanding.

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STANLEY J. FARLOW

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Dedication
Ali Ozluk (1952–2012)

Twenty years ago my colleague, good friend, and very good number theorist Ali Ozluk came into my office and suggested we write a book together titled something like, *Believe It or Not: Paradoxes in Mathematics*, which detailed many of the unexpected and surprising results in mathematics. I thought the idea was splendid so I started writing some short pieces describing what I thought were some unusual and startling results in mathematics. But Ali was deeply involved in his research in number theory, working on the distribution of the roots of the Riemann zeta function, in which he made several important contributions, so he put his suggestion on hold. To make a long story short, while Ali was deep in number theory, I wrote an entire book on his suggestion and told Ali that we should publish it together, but being the gracious person he was, he insisted that I publish it on my own.

Unfortunately, Ali died unexpectedly on March 1, 2012 and although he never saw his suggestion come to fruition, it is my great privilege to dedicate this book to my good friend and very good number theorist, Ali Ozluk. We miss you, Ali.



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Introduction

As a goodwill gesture for buying this overpriced book, the author would like to present you with the last piece of his famous blueberry pie, as shown in Figure 1. Feel free to choose either Pie A or Pie B. You say you prefer Pie A? Certainly, it is the larger piece. Or is it? Before you make your final decision, you may want to turn the page upside down and take another peek. You see, things are not always as they seem; your brain may be deceiving you.

According to Martin Gardner in his book, *Riddles of the Sphinx*,¹ this optical illusion, which shows Pie A as the larger piece when the book is right-side up, and Pie B as the larger piece when the book is upside down, is due to the science fiction writer Fred Flanagan. According to Mr. Flanagan, it isn't really an illusion at all, *it just looks like one*.

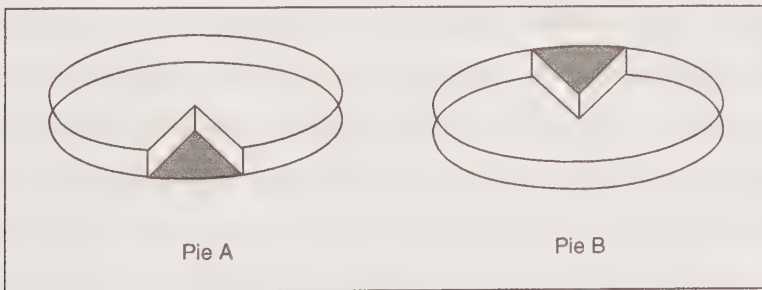


Figure 1

¹ Martin Gardner, *Riddles of the Sphinx*. New Mathematical Library, Vol. 32. 1987.

To psychologists, an optical illusion is something we perceive visually in a deceptive and misleading manner, which is another way of saying information taken in by the eye and processed by the brain doesn't quite jibe with the object under observation. The fact of the matter is our brains sometimes add to what our eyes report and sometimes what it adds is not, well, let's say, 100% accurate. But we shouldn't always blame our brains. Sometimes our eyes are the culprit; they sometimes filter out information and leave it to our brains to fill in the gaps, which it sometimes does incorrectly.

The history of civilization is full of "obvious truths" which later were proven to be obvious falsehoods. No one gave Aristotle a second thought when he said, "Heavy falls fast, light falls slow." It took 2,000 years before a man from Pisa, Italy proved him wrong.

This book is about paradoxes in mathematics, which in a way is a paradox in itself. Also, people are not in complete agreement on what is a paradox. For some, a paradox refers to a phenomenon that's baffling and the resolution of the phenomenon is beyond current understanding. The word "paradox" comes from Greek: para = beyond and doxa = opinion. Other people relax the meaning of the word and argue that a paradox resolved is *still* a paradox. But others argue that once a paradox is solved it is no longer a paradox; it is simply part of an accepted belief system or theory. Many paradoxes in logic, such as Russell's Paradox, which played a major role in the development of logic and axiomatic set theory, are of this type. They may not be considered paradoxes to professional logicians, but they are still conundrums to the general public. (Just ask someone to resolve Russell's Barber

Paradox.) Other people adopt a more liberal meaning of a paradox to include phenomena whose outcomes are surprising and nonintuitive, but whose solution can be resolved easily with mathematics. These phenomena are more of the *believe it or not* type of paradoxes. The Rope-Around-the-Earth Paradox in this book is this type. The most liberal interpretation of a paradox is the mathematical illusion, where a mathematical sleight of hand is performed on an unwary observer. The tale of Uncle George's Will in this book is such an example. Although most people would not consider this to be a paradox, the puzzled look of the observer is still the same. This book includes all these interpretations of a paradox, and it will be the reader's duty to determine into which category a given story falls.

The study of paradoxes broadens the mind. Some are of the "aha" category, where after some thought you say, "Why didn't *I* think of that?" There are others, however, such as the Two-Envelope Paradox, which (arguably) has never been completely resolved, although several people have claimed, quite vociferously, to have done so. The really good paradoxes are those which we can't resolve without broadening our understanding of the underlying theory or foundations of the paradox. Many paradoxes are responsible for the clarification of basic concepts and are responsible for the introduction of new ideas. E.T. Bell once said,² "One of the endlessly alluring aspects of mathematics is that its thorniest paradoxes have a way of blooming into beautiful theories."

² E. T. Bell. *The Development of Mathematics*. 2nd Edition. McGraw-Hill, 1945.

Paradoxes in logic, such as the Liar's Paradox and Russell's Barber Paradox, forced logicians to modify existing theories to rid themselves of troubling inconsistencies. The Liar Paradox, credited to the Greek philosopher Eubulides of Miletus, who lived around 400 B.C., states:

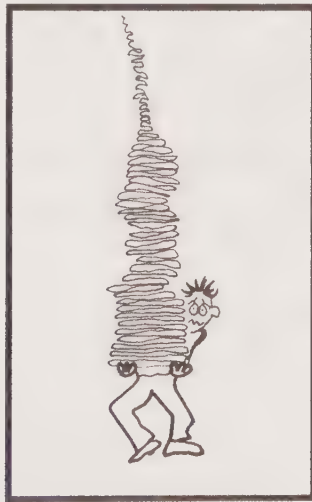
- A man says, I am lying.
- If this statement is a lie, then in fact he is *not* lying.
- If this statement is not a lie, then in fact he *is* lying.
- In other words, he is lying *and* he is not lying.

This is an example of the many types of paradoxes you will discover in this book. Enjoy.

—Jerry Farlow

Paper Folding: *Do Not Try This at Home*

For our first paradox, we present the famous Paper Folding Paradox, one that has been presented in various venues for a thousand years, but nevertheless never fails to impress the unwary observer. The interesting thing about this paradox is that even after the solution is presented and proven positively, unequivocally to be 100% true, there are still nonbelievers who just don't find it in their hearts to accept the conclusion. So, here's the problem. Try to guess the answer, or at least a guesstimate in the general ballpark.



Suppose you take a piece of paper, like the one you are looking at right now, and suppose for simplicity's sake it is $\frac{1}{100}$ th of an inch thick (or 0.01 inch). You then fold it on top of itself 50 times. Of course, the page in front of you can only be folded about five times since it becomes too small, but assume you had a humongous sheet of paper large enough

that could be folded 50 times. The question then arises, how thick will the folded paper be when finished?

Go ahead and give it your best shot. How many inches, feet, or yards will it be? Of course you *could* get out a pencil and paper and actually compute it, but use the God-given intuition you were born with and take a crack at it.

OK, assuming you have now arrived at your estimate, let's do some grade-school arithmetic and find the real answer. It is very easy: every time you fold the paper, the resulting sheet doubles in thickness from the previous thickness, so if you fold it once it will be twice as thick as the original paper, and if you fold it a second time it will be twice the previous thickness or four times the original thickness, and so on. Hence, if the original paper is 0.01 inches thick, and then after one folding the paper will be 0.02 inches thick, after the second folding it will be 0.04 inches, and after the third folding it will be 0.08 inches, and so on. You get the idea: you do this fifty times and you're done. It is really very simple.

Now, here is where it gets interesting, since most people guess the final thickness will be about a foot, maybe two, and a few might even guess up to a hundred feet. If you get out your calculator and multiply the original thickness of 0.01 inches by 2 fifty times, you will get

$$\begin{aligned}
 \text{FINAL THICKNESS} &= (0.01) \times \underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{50 \text{ 2's}} \\
 &= (0.01) \times 1,125,899,906,842,624 \\
 &= 11,258,999,068,426 \text{ inches}
 \end{aligned}$$

Wow! To convert this number into something a little more palatable, let's convert it to feet or maybe even miles. So if we divide it by 12 to convert to feet, and then by 5,280 to convert to miles, we find the thickness in miles to be

$$\text{FINAL THICKNESS} = \frac{11,258,999,068,426}{12 \times 5280} = 177,698,848 \text{ miles}$$

Double wow! In other words, it's roughly twice the distance from the earth to the sun, or the distance to the moon and *back* 700 times. In the words of Mr. Ripley, believe it or not!

Table 1 Paper-Folding Results		
Number of Foldings	Thickness (inches)	Thickness (miles)
0	0.01	0.00000016
1	$0.01 \times 2 = 0.02$	0.00000032
2	$0.01 \times 4 = 0.04$	0.00000064
3	$0.01 \times 8 = 0.08$	0.00000128
4	$0.01 \times 16 = 0.16$	0.00000256
5	$0.01 \times 32 = 0.32$	0.00000512
6	$0.01 \times 64 = 0.64$	0.00001024
7	$0.01 \times 128 = 1.28$	0.00002048
8	$0.01 \times 256 = 2.56$	0.00004096
9	$0.01 \times 512 = 5.12$	0.00008192
10	$0.01 \times 1024 = 10.24$	0.00016384
11	$0.01 \times 2048 = 20.48$	0.00032768
12	$0.01 \times 4096 = 40.96$	0.00065536
13	$0.01 \times 8192 = 81.92$	0.00131072
14	$0.01 \times 16,384 = 163.84$	0.00262144
15	327.68	0.00524288
16	655.36	0.01048576
17	1310.72	0.02097152
18	2,621.44	0.04194304
19	5,242.88	0.08388608

20	10,485.76	0.16777216
21	20,971.52	0.33554432
22	41,943.04	0.67108864
23	83,886.08	1.34217730
24	167,772.16	2.68435460
25	335,554.32	5.36870910
...	...	...
30	10,737,418	169
35	343,597,384	5,423
40	10,995,116,277	173,534
45	351,843,720,888	5,553,089
50	11,258,999,068,426	177,698,848

So, how was your guesstimate? If you are like most people you were waaaaaaaaay off. Many people in fact do not even believe the final result even when presented with grade-school arithmetic. It might be interesting to look at Table 1 to determine the number of foldings required to arrive at your guesstimate. For example, the thickness passes the two-foot mark after twelve foldings. Just try it with a piece of paper.

Now, the question arises, if the final thickness is this humongous number, what are the length and width of the final piece of paper? Well, we will tell you right now that in order to fold a sheet of paper 50 times, it requires more than just a 9-by-11-inch sheet. In fact we will start with a square sheet of paper 100 miles by 100 miles. After we fold it on top of itself the dimension will be 100 miles by 50 miles, and after the second folding the dimensions will be 50 miles by 50 miles, exactly half the size of the original paper. Continuing in this manner, we see that after every two foldings the dimensions get cut in half. Thus, if we start with a sheet 100 miles on each side, then after 50 foldings we arrive at a square sheet of paper, the dimensions of which are 100 miles divided by 2 twenty-five times.

So, get out your calculator again and start hitting the divide key. If you do this, you will discover the final dimensions to be

$$\text{FINAL DIMENSION} = \frac{100}{\underbrace{2 \times 2 \times 2 \cdots \times 2}_{25 \text{ twos}}} = 0.000002980232239 \text{ miles}$$

or if you prefer inches, we multiply the above result by $12 \times 5,280 = 63,360$ which yields the final dimensions of the paper as

$$\text{FINAL DIMENSION} = 0.1888275146 \text{ inches.}$$

Hmmmmmmmmmm. One suspects that to actually be able to do this you would need an even larger sheet of paper, say the size of Texas or 1,000 miles by 1,000 miles, which would make the final dimensions 10 times larger, or a 1.888275146-by-1.888275146-inch sheet of paper 177,698,848 miles thick. Hmmmmmmmmmm, we might need an even larger sheet.

The question then arises, *why* does the conclusion seem so non-intuitive? If you are like most people, your intuition underestimates the answer by a couple of zillion miles. My theory is that firstly, none of us have *really* doubled a number *that* many times, maybe 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ... and then we stop. After the numbers get really, really large, one side of our brain says to double, but our intuitive brain is probably adding. Well, that's my theory and I'm sticking to it!

A sequence of numbers like the doubling sequence

$$1, 2, 4, 8, 16, 32, 64, \dots$$

where each number in the sequence is a fixed multiple of the previous one, is called a *geometric sequence* and although it is not necessary to multiply by 2, all geometric sequences grow faster than you think, even if the multiplier is just a tiny bit larger than 1.

Dutch and the Canarsies :

Here is an interesting geometric sequence¹

$$1, (1.08), (1.08)^2, (1.08)^3, (1.08)^4, \dots$$

which has a relatively small multiplier of 1.08, but nevertheless the sequence grows faster than you realize. To place this sequence in a historical setting, recall the story of the Dutch explorers who offered the Canarsie Indians \$24 for the island of Manhattan in 1624. Let us assume the Canarsies could have deposited the \$24 in Canarsie Trust that paid 8% annual interest, compounded annually. The question then arises, what would be the value of their deposit today, say in 2013, 389 years later.

The answer can be found the same way the bank computes it: simply multiply the initial deposit of \$24 by 1.08 each of the 389 years, which is denoted mathematically as

$$\text{VALUE OF ACCOUNT AFTER 389 YEARS} = \$24(1.08)^{389}$$

To save you the time of getting out your calculator and computing this, we found this value to be

¹ The exponents on 1.08, like $(1.08)^2, (1.08)^3$ represent squares and cubes of 1.08.

$$\text{VALUE OF ACCOUNT AFTER 389 YEARS} = \$241,019,469,461,110.27$$

In other words, \$241 trillion dollars! That's more than the real-estate value of Manhattan today! Well, you say who can get 8% interest from a bank today. If you only got 4% then the present value of the \$24 would be

$$\begin{aligned}\text{VALUE OF ACCOUNT AFTER 389 YEARS} &= \$24 \times (1.04)^{389} \\ &= \$101,433,223.86\end{aligned}$$

Here's another example: *E. coli*

Now, if you want a more relevant example of geometric growth, suppose you didn't cook that hamburger long enough and suppose you ingested a single *Escherichia coli* bacterium, and suppose for the sake of argument every bacteria divides into two bacteria every 20 minutes. What we have here is a geometric sequence where every hour the multiplier is $2 \times 2 \times 2 = 8$, which means (assuming your body does not start killing off the little buggers) that after 24 hours the number of bacteria will have grown to

$$1 \times 8^{24} = 4,722,366,482,869,645,213,696 \text{ bacteria}$$

or roughly 4 *thousand billion billion*. Hopefully your immune system would have kicked in before that time, resulting in a much smaller number.

The Inverse Problem

Suppose now we ask the *inverse* problem: that is, to find the number of times you must *fold* a piece of a paper 0.01 inch thick so that the final thickness is 177,698,848 miles. In this problem, most people would over-guess the correct answer, which we saw was 50 times, to upwards of a million times. This inverse problem is the *logarithm problem* of the previous exponential problem. That is, if the paper is one inch thick, and if we fold it 50 times, its final thickness will be

$$2^{50} = \underbrace{2 \times 2 \times \cdots \times 2}_{50\ 2\text{'s}} = 1,125,899,906,824,624 \text{ inches}$$

which in mathematical language, says that the *logarithm* (to the base 2) of the final thickness is 50, which is written as

$$50 = \log_2(1,125,899,906,842,624).$$

In other words, although 2^n grows *faster* than we think as n grows, the logarithm $\log_2 n$, which is the exponent of 2, grows *slower* than expected.

Reference: Google geometric sequence, paper-folding problem

Rope Around the Earth

OK, so you underguessed the answer to the paper-folding problem by a couple of zillion miles, but don't worry, we're now going to give you a chance to redeem yourself with the rope-around-the-earth problem or paradox if you like.

This problem is as simple as the paper-folding paradox, but the answer is even *more* nonintuitive, if that's possible. In this problem, we string a rope around the earth at the equator, three feet above the ground, and then drop the rope to the ground and pull up the slack, where we assume the earth is a perfect sphere with a circumference of 25,000 miles. The \$64,000 question then is, how much slack will there be in the rope? Keep in mind the rope goes *all* the way around the world, going west from Brazil to Ecuador, across the Pacific Ocean to the Indonesian islands, on through Singapore and Sumatra, across the Indian Ocean all the way to Africa, then finally across the Atlantic back to Brazil. So if the rope starts three feet off the ground, it is obvious there should be a great deal of slack in the rope after we drop it to the ground, right?

OK, make your guesstimate

and keep in mind your paper-folding guess was waaaaaaaaaaaaaay low. To help you get a feel for the problem, we have supplied photos

showing the rope as it goes around the earth passing through Africa and Indonesia.

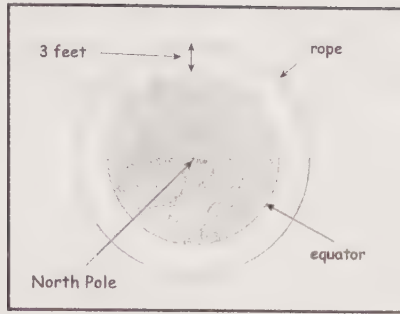


Rope around the earth passing thru Saharan Africa



Rope around the earth off an Indonesian island

OK, let's solve this problem using simple middle-school mathematics to see how close you came this time. In Figure 1 we have drawn a diagram of the earth viewed from above the North Pole in which the equator appears as a circle, and the rope three feet above the ground is drawn as a slightly larger circle.



Rope around the earth three feet off the ground
Figure 1

You will now recall from 10th-grade geometry, the circumference of a circle (which we denote by C) is given by the formula

$$C = 2\pi R$$

where

C = circumference of a circle

R = radius of a circle

$\pi = 3.1415927\dots$

So, to find the slack in the rope, we simply take the circumference of the rope when it is 3 feet above the ground and subtract from it the circumference of rope on the earth. That is, the difference between the quantities

$$\begin{aligned} \text{LENGTH OF THE ROPE 3 FEET ABOVE THE GROUND} &= 2\pi(R+3) \\ \text{CIRCUMFERENCE OF THE EARTH} &= 2\pi R \end{aligned}$$

where in this equation the radius R of the earth is given in feet. Computing this difference, we get

$$\begin{aligned}
 \text{Slack in the rope} &= 2\pi(R+3) - 2\pi R && \text{(basic 9th-grade algebra)} \\
 &= 2\pi R + 2\pi(3) - 2\pi R && \text{(first and last terms cancel)} \\
 &= 6\pi \\
 &\doteq 18.85 \text{ feet}
 \end{aligned}$$

Conclusion: The slack in the rope is slightly less than 19 feet. In other words, GOTCHA! Just how many hundred miles *did* you overguess?

So, what have we said? Well, in general if a rope is h feet off the ground and dropped to the ground, the resulting slack in the rope is

$$\text{SLACK IN ROPE } h \text{ FEET OFF THE GROUND} = 2\pi h \text{ feet}$$

or roughly $2\pi \doteq 6.28$ times the distance the rope is above the ground.

It is interesting to note that the answer does *not* depend on the size of the earth, only the distance that the rope is above the ground. In other words, knowing the circumference of the earth is irrelevant.

Epilogue:

So how many hundred miles did you overguess? Most people guess a couple of hundred miles. And why did you guess too much? Well, it has to do with the nature of a circle. There just isn't very much slack in a circle. (For example, try the same problem with a square, triangle, or a "lesser" symmetric object.)

Now, if you actually went to South America and strung a rope around the equator three feet in the air and dropped it to the ground, the real

rope would have a lot more slack than 19 feet since *real* earth has a lot of ups and downs. Maybe a guess of 50 miles, who knows, would be close to the real answer.

Since the slack depends only on the distance the rope is above the ground and not the size of the earth, this means if you string a rope three feet around an orange and let it collapse, the slack would still be the same $6\pi \doteq 19$ feet. Wow! In fact, if you string a rope three feet in the air around the sun or around any spherical object, the slack would be the same 19 feet! The earth is larger than the orange, but there is less “curvature” per mile as you go around the earth, causing the slack around the earth to be the same slack if the rope is going around an orange.

Of course, if you strung the rope four feet in the air around the earth, an orange, or the sun, the slack would be

$$\text{SLACK FROM 4 FEET} = 2\pi h = 2\pi(4) = 8\pi \doteq 25 \text{ feet.}$$

There are other ways this problem can be restated. Maybe you can think of some yourself. A popular variation of this problem is the Jogging Problem. Suppose you and a friend are jogging around a 400-meter track or any oval track, and suppose you are jogging 3 feet on the outside. The question is, how much farther do you run per lap? Of course, you already know the answer, realizing that the only time you run farther is at the two ends of the track, and putting the ends together results in a circle. Hence, you are like the rope 3 feet above

the earth, and so per lap, the extra distance you run will be the same **2π feet**. That is

$$\text{EXTRA DISTANCE YOU RUN PER LAP} = 2\pi h = 2\pi(3) = 6\pi \doteq 19 \text{ ft}$$

In other words, the same distance as the slack in the rope around the earth. Believe it!

Also, this problem could be stated in *reverse*.¹ That is, if you had a rope resting on the ground all the way around the equator, and if you somehow added 19 feet to the rope, how much higher could you raise the rope above the ground? Of course, most people would say some small number like 0.001 inches, but in fact the rope could be raised to 3 feet above the ground.

ΔΧΣΩΞΥΥΘΠ

¹ As in many mathematics problems, one problem leads to others, often to new ideas and theories. Can you work out this problem using a square?

Oh No! The Monty Hall Problem!

Intrigue ... *check*

Mystery ... *check*

Jealousy ... *check*

Sex ... *check*

...well maybe not sex, but what the heck, we're trying to sell books here. Yet, if there ever was a problem that caused more than just a little angst, especially to professional mathematicians, it's the Monty Hall problem, or for lack of a better name, the Monty Hall embarrassment. In 1990, Marilyn vos Savant published the problem¹ in her "Ask Marilyn" column in *Parade* magazine. Ms. vos Savant is noteworthy for being listed in the *Guinness Book of World Records Hall of Fame* for having the "highest I.Q. ever recorded," which no doubt puts a bull's-eye on her back for many people.

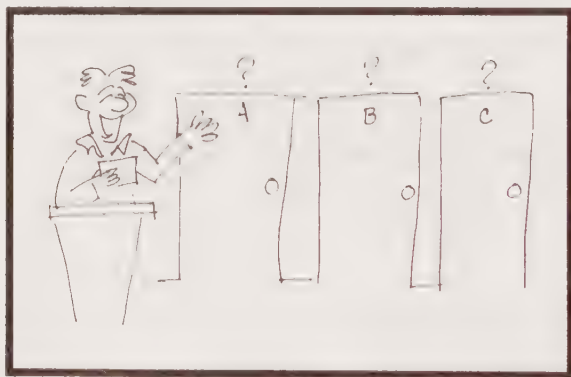
But the solution to the problem dished up by Ms. vos Savant in her *Parade* column was like poking a hornet's nest. She received upward of 10,000 letters, many written on university stationery with Ph.Ds suffixed to authors' names, telling her she might be smart but, my dear, she was *no* mathematician. But after some reflection and more than a little professional chagrin, her detractors were forced to eat a hearty dish of crow. You see, when it comes to probability, things can

¹ Actually, variations of this problem have been around for a long time.

get very nonintuitive, and the Monty Hall problem is possibly the *creme de la creme* of counterintuitive examples.

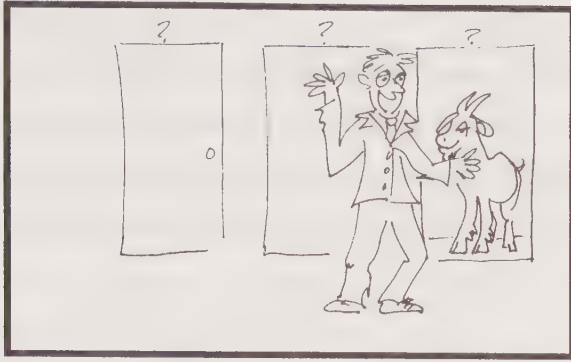
If your intuition gives you the correct answer to this problem, you might have a future as a professional probabilist. If not, you are like the rest of us bozos.

In the Monty Hall problem, suppose you are a contestant on the Monty Hall show, *Let's Make a Deal*, and you have the opportunity to win a new car if you are able to guess behind which of three doors it is parked. Behind the other two doors are booby prizes consisting of a couple of goats. Well, booby prizes unless maybe you've had a lifetime ambition to be a goatherd.



Car behind one door, goats behind the others

Now, here is the dilemma, or paradox if you like. Suppose you pick Door A as the location of the car, but after making this selection. Monty opens one of the other doors showing a goat and offers you the chance to switch from Door A to the other unopened door.



Monty shows a goat.

And therein lies the dilemma. Will you increase your odds of winning if you switch from Door A to the other door, or should you stick with Door A? Or in fact does it make any difference at all? What are the chances of winning if you don't switch, and what are the chances of winning if you do switch? Decisions, decisions, decisions.

So make up your mind and we will analyze this dilemma.

OK, so make your decision, and remember a new Chevy Impala is on the line. Now, if you are like most people you will say the answer is obvious. After Monty revealed a goat, there are two unopened doors and hence there is a 50-50 chance the car is behind one of the two doors. Hence, it doesn't make a hoot of difference whether you stick with Door A or switch to the other door, you have a 50-50 chance with either. It's hard to argue with reasoning like that.

But, like we said, in probability, things are not always what they seem.

To solve this dilemma, we have drawn Table 1, which gives a complete picture of what is going on. It shows your fate (Winning or losing) in case the car is behind Door A, B, or C in case you decide to switch or stick. Here GOAT represents the GOAT that Monty shows you, and GOAT and CAR (in bold type) represent your winnings if you switch or stay. And finally, the last two columns give the results of winning or losing, depending on whether you stick or switch.

Outcomes for switching and sticking					
Location	Door A	Door B	Door C	switch	stick
Behind A	CAR	GOAT	GOAT	LOSE	WIN
Behind B	GOAT	CAR	GOAT	WIN	LOSE
Behind C	GOAT	GOAT	CAR	WIN	LOSE

Table 1

So what do you conclude from the table? It's very simple. In two of the three locations of the car, you win if you switch from Door A to the other unopened door. In other words, you have a $\frac{2}{3}$ chance of winning the car if you switch and $\frac{1}{3}$ chance of winning if you stick. Believe it or not!

Let's look more carefully at what happens in case the car is behind each of the three doors.

Car Behind Door A

If the car is behind Door A, Monty will reveal the goat behind Door B or C. Let's say he opens Door C. (It doesn't really make any difference which door he opens.) In either case if you stick with Door A you win, and if you switch (to Door B) you lose.

Car Behind Door B

Now, if the car is behind Door B. Monty of course will reveal the goat behind Door C (he's NOT going to reveal the car. he's not THAT nice of a guy), so if you stick with Door A you lose, but if you switch (to Door B) you win.

Car Behind Door C

If the car is behind Door C. Monty shows the goat behind Door B (again he will not show the car behind Door C), so if You stick with Door A you lose, and if you switch to Door C you win.

Several readers of the "Ask Marilyn" column noted that the problem was ill-posed in the sense that the question doesn't say if Monty *always* offers the contestant the chance to switch or just sometimes. For example, if he *only* allows the contestant to switch when the contestant has correctly chosen the prize door, this changes the probabilities dramatically since if the contestant now switches, he or she will not increase the probability of winning. In fact the contestant will lose every time!

Now if you *still* don't have a good feel why the probability of winning increases from $\frac{1}{3}$ to $\frac{2}{3}$ when you switch, look at it this way. Two-thirds of the time the car is behind Door B or C, and those times Monty basically *gives* you the car since by opening one door, you know the car is behind the other door! What a guy, huh?

ΔΧΣΩΞΨΥΘΠ

The Birthday Paradox: You're NOT Going to Believe This

Sometimes a problem is referred to as a paradox when certain aspects of the problem are overlooked. One paradox that falls into this category is the Birthday Paradox. The name of the paradox refers to the storyline of the paradox, although there are other scenarios for the same phenomenon, and after reading this story you might consider making up some yourself. But everyone loves a party so let's suppose you are at a birthday party, and there are 23 people at the party. Now maybe you shouldn't bring this up at your next party or else people will start avoiding you, but that being said, the problem that arises is to determine the probability that (at least) two persons at the party have the same birthday. We are not talking about the year but the day and month of the year. We don't care *what* day of the year, just any day.

We won't tell you the answer but people guess either too high or too low. You will find out yourself if you are a "high guesser" or a "low guesser."

So, go ahead and make your guesstimate or prediction if you like.

What is the probability that in a group of 23 persons 2 persons have the same birthday?

In fact, make a prediction for the number of persons required for the probability of two persons having matching birthdays to be $\frac{1}{2}$. How many would it take, 100, 150, or maybe 183? What is the tipping point where the odds are $\frac{1}{2}$ of having matching birthdays at the party?

The solution of the Birthday Paradox is quite easy if you know a couple of basic rules of probability. For example, when you flip a coin the probability of getting a head is $\frac{1}{2}$ since one out of two possible outcomes is a head, and if you flip a coin two times the probability of getting both heads is the product¹

$$\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4} = 0.25$$

On the other hand, if a million people buy a ticket in the lottery and you buy one, then of course the probability of you winning is 1 divided by a million, or

$$\text{PROBABILITY OF YOU WINNING} = \frac{1}{1,000,000} = 0.000001$$

Now, the second rule of probability is the *complement rule*, which says the probability of something happening is one minus the probability it doesn't happen. For example, if the probability that you win the lottery is

¹ We are assuming here the result of the second toss is independent of the result of the first toss.

$$\text{PROBABILITY OF WINNING THE LOTTERY} = 0.00000001$$

then the complement rule says the probability of *not* winning is

$$\text{PROBABILITY OF NOT WINNING} = 1 - 0.00000001 = 0.99999999$$

Hence, to find the probability that there will be a *matching birthday* in a group of 23 people, we first find the probability of its complement, i.e., the probability *everyone has different birthdays*.

First, note the probability that 2 persons have different birthdays is

$$\left(\frac{365}{365}\right)\left(\frac{364}{365}\right) \doteq 0.997$$

since the first person can have his or her birthday any day of the year, hence the first factor $365/365 = 1$, whereas the second person must have his or her birthday on one of the *remaining* 364 days (out of 365 days), giving rise to the second factor $364/365$. Hence the probability that both persons have different birthdays is the above product 0.997, which yields the complement probability $1 - 0.997 = 0.003$ that two persons have matching birthdays. We are not saying *what* day of the year they have matching birthdays, just that they have matching birthdays on either January 1, January 2, ... December 30, December 31. Also for 3 or 4 persons and more, see Tables 1 and 2.

Table 1	
Number of Persons	Probability of distinct birthdays
2	$\left(\frac{365}{365}\right)\left(\frac{364}{365}\right) \doteq 0.997$
3	$\left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right) \doteq 0.992$
4	$\left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right)\left(\frac{362}{365}\right) \doteq 0.98$
5	$\frac{365 \cdot 364 \cdot 363 \cdot 362 \cdot 361}{(365)^5} \doteq 0.97$
6	$\frac{365 \cdot 364 \cdot 363 \cdot 362 \cdot 361 \cdot 360}{(365)^6} \doteq 0.96$
10	$\frac{365 \cdot 364 \cdots 357 \cdot 356}{(365)^{10}} \doteq 0.88$
15	$\frac{365 \cdot 364 \cdots 352 \cdot 351}{(365)^{15}} \doteq 0.75$
20	$\frac{365 \cdot 364 \cdots 347 \cdot 346}{(365)^{20}} \doteq 0.59$
→ 23 ←	$\frac{365 \cdot 364 \cdots 350 \cdot 349}{(365)^{23}} \doteq 0.49$
30	$\frac{365 \cdot 364 \cdots 337 \cdot 336}{(365)^{30}} \doteq 0.29$
35	$\frac{365 \cdot 364 \cdots 332 \cdot 331}{(365)^{35}} \doteq 0.19$
50	$\frac{365 \cdot 364 \cdots 317 \cdot 316}{(365)^{50}} \doteq 0.03$
100	$\frac{365 \cdot 364 \cdots 367 \cdot 366}{(365)^{100}} \doteq 0.0000004$

We now compute the probability that at least two persons have *matching* birthdays by using the complement rule and taking one minus the numbers in Table 1, giving the probabilities in Table 2.

Table 2		
Number of People	Probability of no matching birthdays	Probability of at least one matching birthday
2	0.997	0.003
3	0.992	0.008
4	0.98	0.02
5	0.97	0.03
6	0.96	0.04
10	0.88	0.12
15	0.75	0.25
20	0.59	0.41
23	0.49	0.51← over half
30	0.29	0.71
35	0.19	0.80
50	0.03	0.97
100	0.0000004	0.9999996

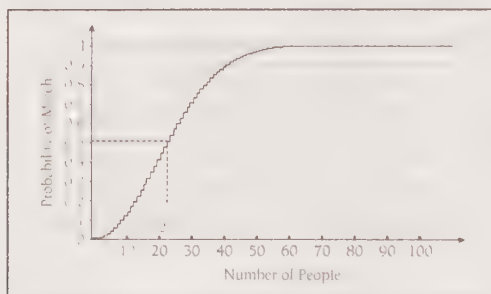
So, let's get back to our original question. We have shown that in a group of 23 people, the probability of matching birthdays is 0.51. This could be considered a paradox² since most people guess in the neighborhood³ of $23/365 \doteq 0.06$. In fact, with 50 people there is a 97% chance at least two persons have the same birthday, and for 100 people, the probability of matching birthdays is 0.9999996³. This probability

² It is not really a paradox since the problem is completely understood. It is simply counterintuitive, at least to most people.

³ The Birthday Problem is standard fare in all beginning probability classes, and the author knows from experience that most people will guess far less than 0.51 for 23 people.

is so large that if you were to walk into a room of 100 people every day for the next 3 thousand years you would expect to find matching birthdays every day *except* one!

Of course, in a group of 366 people (no leap year, please) or larger the probability is 1 (100% certain) that there are matching birthdays (right?). This is called the *Pigeonhole principle*; if you put 366 pigeons in 365 boxes, there will be one box with more than one pigeon. This principle also implies the interesting fact that there are two persons in Chicago with *exactly* the same number of hairs on their heads.⁴



50% chance two people have the same birthday.

Reason why: The reason the probability of matching birthday is greater than what our intuition tells us is due to the fact that we *overlook* all of the ways birthdays can match. When we say “matching birthdays,” we mean matching birthdays of *any* two persons, *any* day of the year, and there are more of these possibilities than we consider.

⁴ Assuming the maximum number of hairs on a person’s head is 1 million and since the population of Chicago is over 1 million, the Pigeonhole principle yields the result.

Sometimes the Birthday Paradox is posed as the Presidents' Paradox, which asks the probability that two former presidents of the U.S. had the same birthday.⁵ At the time of writing this book, there have been 44 presidents of the United States. We know for 44 people the chances are

$$1 - \frac{365 \cdot 364 \cdots 323 \cdot 322}{(365)^{44}} = 0.92$$

In other words there is a 92% chance at least two presidents had the same birthday. That doesn't mean for sure they did, after all there is an 8% chance they didn't. But, checking the historical records, we find that James Polk (11th president) and Warren G. Harding (29th president) both had birthdays on November 2, which as of 2013 were the only matching birthdays.

More than a parlor game: The Birthday Paradox can serve as more than a parlor diversion, and also serve as a warning to computer programmers who often assign *hash tables* to memory locations. The idea behind a hash table is to establish a mapping between the members in the hash table (called keys) and memory locations in a computer.

⁵ An honest-to-goodness *probabilist* (expert in probability) might object to asking for the probability of something that has already happened since it either happened or it didn't. In other words, there is no probability involved. When we talk about the probability of something happening, we generally refer to something that is *going* to happen. The same is true for a roomful of people: they already had their birthdays and so a purist would argue we shouldn't be talking about the chance of matching birthdays. But, we are not purists, so we will.

One enters a key and out comes information from a given memory location (i.e., a retrieval process). However, the Birthday Paradox in the context of hash codes illustrates the fact that the probability that two hash codes refer to the same memory location is higher than intuitively imagined, thus hashing strategies should be handled with care.

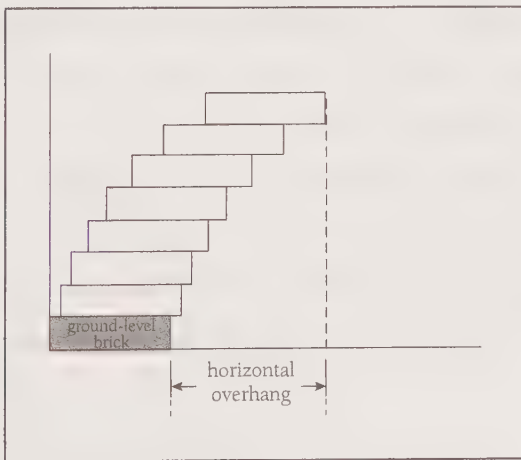
$$\Phi\Delta\Pi\Omega\Xi\Psi$$

Mathematical Masonry

Some mathematical paradoxes are logical in nature, some deal with counting, some deal with infinity (which can be very nonintuitive to say the least), others deal with geometry, but the one we present here involves both geometry and what might be called “higher-level arithmetic.” The result is so nonintuitive you probably won’t believe the outcome even after it is proven to you. And if you don’t, you’re not alone.

We start with a collection of bricks of identical size and uniform density, which for simplicity’s sake we assume has a length of 1 foot.

Our goal is to stack them on top of each other much like those in Figure 1, careful so they don’t topple, but stacking them so that the

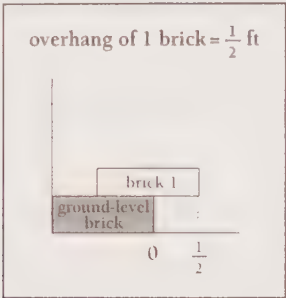


How far can you extend the bricks?

Figure 1

horizontal overhang, or distance the top brick extends out horizontally, is as large as possible.

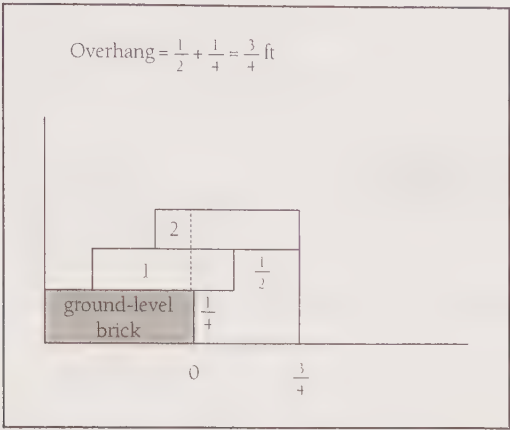
We start by placing a brick on the floor or any flat surface, then place a brick, which we call brick 1 on top of it to maximize the overhang, which is clearly $\frac{1}{2}$ foot. In Figure 2, we label 0 as the location of the right edge of the ground-level brick.



Stacking one brick
Figure 2

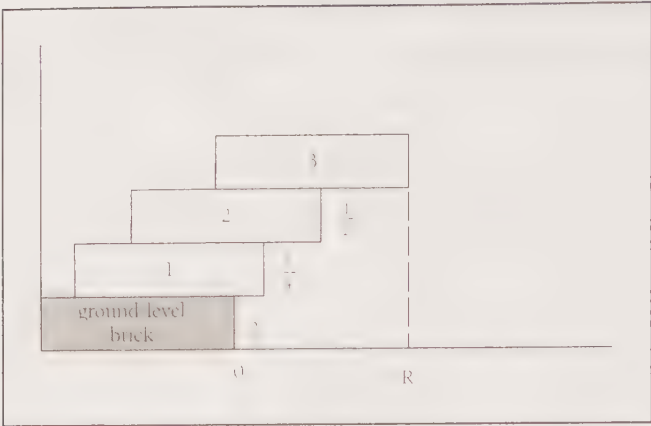
Let's now place a new brick, called brick 2, on top of brick 1 in such a way that it is as far to the right as possible without tipping. In other words, the center of gravity of bricks 1 and 2 is directly over the right edge (location 0) of the ground-level brick. In Figure 3, we see that brick 2 should have an overhang of $\frac{1}{2}$ ft over brick 1. Note the dotted vertical line going up from the right edge of the ground-level brick and observe that half the volume of the bricks lies to its left and half to its right and so bricks 1 and 2 will balance with the maximum overhang of two bricks being

MAXIMUM OVERHANG OF TWO BRICKS = $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ feet



Maximum Overhang $\frac{3}{4}$ ft with 2 Bricks
Figure 3

Now onward and upward. Now, by adding another brick is where it starts to get interesting, or complicated might be a better word. As for all future bricks, we slide in the *new* brick *directly* on top of the *ground-level brick* (not the top brick) with the previously placed bricks on top of it, as shown in Figure 4.



What is the overhang of brick 1? (second brick from the bottom)
Figure 4

We do this in such a way that all the bricks resting on the ground-level brick have their (total) center of gravity at location 0 (over the right edge of the ground-level brick). Every time you add a new brick directly over the ground-level brick and below the previously placed bricks, the previously placed bricks do not move relative to one other and can be imagined as being glued together. The \$64,000 question then arises: How far can you stack these bricks without toppling? When we stacked the first two bricks it was an easy problem, but for later bricks it isn't so easy.

To determine the total number of bricks we can stack without tipping the stack over, observe the bricks in Figure 5, which shows the first three stacked bricks, where **R** denotes the *total* overhang, keeping in mind that the length of each brick is 1 foot. From this figure, it is not hard to determine the locations of the left and right edges of the three bricks to be

RIGHT END POINT OF BRICK 1 = $R - \frac{3}{4}$

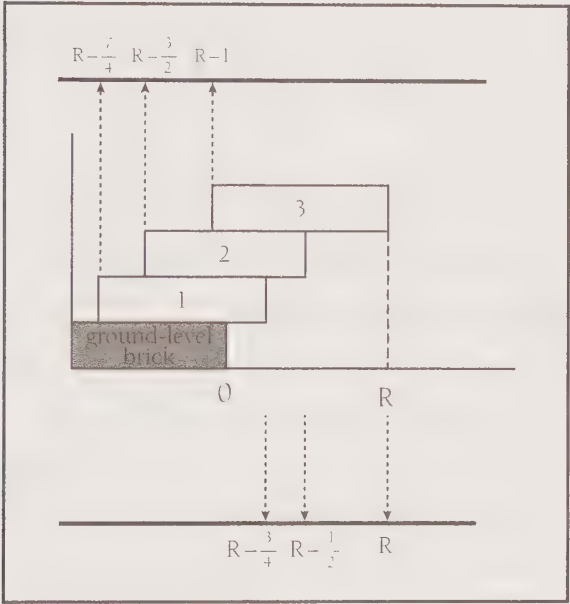
RIGHT END POINT OF BRICK 2 = $R - \frac{1}{2}$

RIGHT END POINT OF BRICK 3 = R

LEFT END POINT OF BRICK 1 = $R - \frac{7}{4}$

LEFT END POINT OF BRICK 2 = $R - \frac{3}{2}$

LEFT END POINT OF BRICK 3 = $R - 1$



Locations of the left and right ends of the three stacked bricks
Figure 5

Now, we are getting somewhere! We find the center of gravity of the three (stacked) bricks, which we know must be location 0 (right end of the ground-level brick). From Figure 5 we see that the center of gravity of the three bricks is the average of the center of gravities of the three stacked bricks, which are

$$\text{CENTER OF GRAVITY OF BRICK 1} = \frac{(R - 3/4) + (R - 7/4)}{2} = R - 5/4$$

$$\text{CENTER OF GRAVITY OF BRICK 2} = \frac{(R - 1/2) + (R - 3/2)}{2} = R - 1$$

$$\text{CENTER OF GRAVITY OF BRICK 3} = \frac{R + (R - 1)}{2} = R - 1/2$$

which yields the center of gravity of the three bricks taken as a collection to be

$$\begin{aligned} \text{CENTER OF GRAVITY OF THE 3 BRICKS} &= \frac{(R - 5/4) + (R - 1) + (R - 1/2)}{3} \\ &= R - \frac{11}{12} \end{aligned}$$

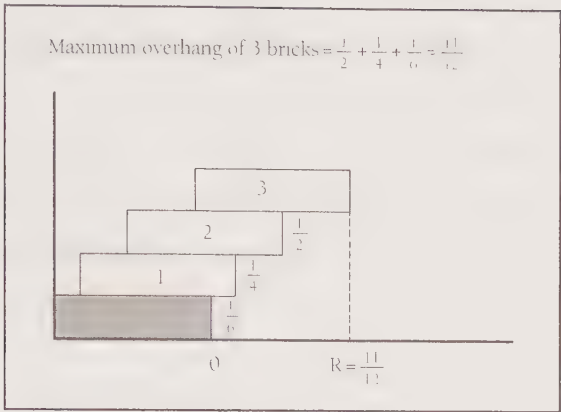
Since the center of gravity must be at location 0 for the bricks to balance with maximum overhang, we have

$$R - \frac{11}{12} = 0 \Rightarrow R = \frac{11}{12} \text{ feet}$$

which means the total overhang is now $\frac{11}{12}$ ft. (i.e., 11 inches). We can now determine how far to slip in brick 1 by computing

$$\text{brick 1 overhang} = R - \frac{1}{2} - \frac{1}{4} = \frac{11}{12} - \frac{1}{2} - \frac{1}{4} = \frac{1}{6} \text{ feet (2 inches)}$$

See Figure 6.



Three stacked bricks
Figure 6

Continuing this process, the strategy is to add bricks one at a time in a similar manner; directly on top of the ground-level brick, sliding it to the right so the center of gravity of the bricks is directly over the right edge of the ground-level brick. We have found the overhangs of the first three bricks to be $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{6}$ foot. If we were to continue by a similar process we would find future overhangs of the new bricks to be

OVERHANG OF BOTTOM BRICK = $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \frac{1}{12}, \dots$ ft

So, the answer to the question of the maximum overhang can be found as the sum of the (infinite) series¹

MAXIMUM OVERHANG = $\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{10} + \dots = \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \right)$

¹ We are *mathematical* masons and thus have an infinite supply of bricks.

But the infinite series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

well known to mathematicians is called the harmonic series and it is well known (even among beginning calculus students) to have no sum. That is to say, if we interpret its “sum” as the *limiting value* of this series as we add more and more terms, the series will *eventually* be larger than any number you choose.

The first person to answer this question was the French mathematician, Nicole d’Oresme (*circa* 1323–1382). who showed that the series grows without bound (diverges, as we say in mathematics) by combining the terms of the series in groups of 2, 4, 8, 16,... and observing the following:

$$\begin{aligned} 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ &\quad + \left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16} + \dots\right) + \dots \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) \\ &\quad + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}\right) + \dots \\ &= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \\ &= \infty \text{ (grows without bound)} \end{aligned}$$

The moral to our story is that given an unlimited number of bricks, we can extend the bricks horizontally as far as we please, all the way to China.² For example, to stack bricks with a total overhang of 100 feet would require n bricks, where n can be determined from the (calculus) equations³

$$\begin{aligned} \int_1^n \frac{dx}{x} &= 100 \Rightarrow \ln n = 100 \\ \Rightarrow n &= e^{100} \\ &= 26881171418161354484126255515800135873611119 \end{aligned}$$

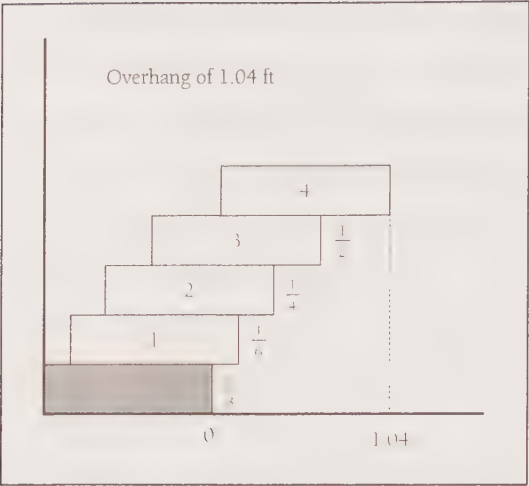
(Thank you, www.wolframalpha.com.)

Epilogue: Of course, the world of pure mathematics not always agrees exactly with the real world and this is a case in point. Some people have been known to construct a pile of dominoes this way, generally having a maximum of about 15. If you can't do that, you can try to stack four bricks with overhangs of

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} \doteq 1.04 \text{ ft}$$

which makes the left end of the top brick positioned at 0.04, which is further to the right than the right end of the ground-level brick located at 0 as drawn in Figure 7. Good luck.

² However, it would take quite a few bricks to get to China since 500 bricks, each 1 foot long, has a horizontal overhang of only 3.4 feet.
³ If you haven't studied calculus and do not understand these equations, just say a *large* number of bricks and that is just as good.



Left end of the top brick farther to the right than
the right end of the bottom brick

Figure 7

ΔΚΘΠΩΞΨ

Aaaaaaaaaaaaaagh: The Curry Paradox

If ever there was someone who deserves to be bowstrung and hung by his thumbs, it is Mr. Paul Curry¹ from New York City, an amateur magician responsible for what is called the Curry Paradox, an enigma so dastardly ingenious as to drive you insane. And to make matters worse, after you unmask it, you want to kill yourself for not thinking of it yourself! He gets you coming and going!

The best paradoxes are the easiest to state and the hardest to resolve, and the Curry Paradox is a prime example of such a paradox. Here's the problem. Figure 1a) shows a right triangle² with legs having lengths of 8 and 13. Inside this triangle are

right triangle A with legs of lengths 3 and 8

right triangle B with legs of lengths 2 and 5

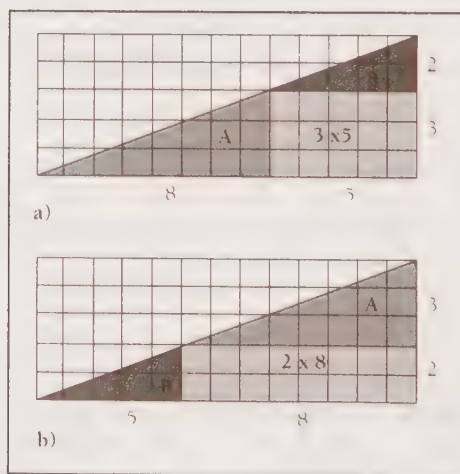
Left over after we remove these triangles is a $3" \times 5"$ rectangle having an area of 15. We have overlaid the triangle with grid lines so you can be

¹ Paul Curry (1917–1986) was a vice president of the Blue Cross Insurance Company of New York, and a well-known amateur magician. His most well-known creation was the *Out of This World* card trick, a trick he created at the age of 25, which many magicians say is the most creative and imaginative card trick invented in the past 100 years.

² Just in case you forgot what a right triangle is, it is a triangle in which one of the angles is 90 degrees. The side opposite the right angle is called the hypotenuse and the two sides adjacent to the right angle are called the legs of the triangle.

sure we aren't cheating about the size of the triangles. We don't want to be like one of those devious magicians who wave a rabbit around in the air with one hand and carry out some secret flimflam with the other.

OK, now here is where everything starts to get weird. If we interchange triangles A and B we arrive at the drawing in Figure 1b), which leaves a $2" \times 8"$ rectangle, which has an area of 16, which means something is wrong! The areas of the rectangles in Figures 1a) and 1b) must be the same, since we have simply interchanged triangles A and B, which should result in equal remaining areas. But the area of the rectangle in Figure 1a) is $3" \times 5" = 15$, whereas the area of the rectangle in Figure 1b) is $2" \times 8" = 16$. And all Mr. Curry did was wave his magic wand, interchange the two triangles, and arrive at different areas! Mr. Curry then bows and asks the audience, exactly how did he create the extra square?



Interchanging triangles in a) and b) should leave rectangles with equal areas.

Figure 1

But since pictures are often misleading, let us methodically compute the areas of all the relevant triangles and rectangles in Figure 1 and find the discrepancy between the drawings in Figures 1a) and 1b). Doing this routine task, we get

$$\text{AREA OF LARGE TRIANGLE} = \frac{1}{2}(\text{base} \times \text{height}) = \frac{1}{2}(13 \times 5) = 32.5$$

$$\text{AREA OF TRIANGLE A} = \frac{1}{2}(\text{base} \times \text{height}) = \frac{1}{2}(8 \times 3) = 12$$

$$\text{AREA OF TRIANGLE B} = \frac{1}{2}(\text{base} \times \text{height}) = \frac{1}{2}(5 \times 2) = 5$$

$$\text{AREA OF RECTANGLE IN FIGURE 1a)} = 3 \times 5 = 15$$

$$\text{AREA OF RECTANGLE IN FIGURE 1b)} = 2 \times 8 = 16$$

Hence, the total areas of the two larger triangles in Figures 1a) and 1b) are

$$\begin{aligned} \text{Figure 1a): total area} &= \text{triangle 1} + \text{triangle 2} + \text{rectangle} \\ &= 5 + 12 + 15 = 32 \end{aligned}$$

$$\begin{aligned} \text{Figure 1b): total area} &= \text{triangle 1} + \text{triangle 2} + \text{rectangle} \\ &= 5 + 12 + 16 = 33 \end{aligned}$$

which should be the same. But, and this is the cruncher, that total area of the triangle is

$$\frac{1}{2}(5 \times 13) = 32.5$$

which is different from BOTH 32 and 33. In other words, things are just getting worse and worse! The mathematics seem to be all messed up!

OK, give up? We will now draw a simple picture that should be light to the darkness. You see, the Curry Paradox isn't really a paradox at all. *It's an illusion! A fraud!* The paradoxes we saw earlier, the Paper-Folding Paradox, the Rope-Around-the-Earth Paradox, now they *are* paradoxes, or at least counterintuitive phenomena. In the case of the Curry Paradox, oops illusion, the large triangle in Figure 1 is *not* a triangle, it just looks like one.

So, what do we mean it's not a triangle? Well, note that triangle A in Figure 1 has a slope (rise/run) of $\frac{3}{8} = 0.375$, whereas the slope of triangle B is $\frac{2}{5} = 0.40$. In other words, the larger triangle with legs 8 and 13 is *not* a triangle. It's a fraud! Mr. Curry has fooled you by finding a figure that *looks* like one. It is just close enough to a triangle to trick you.

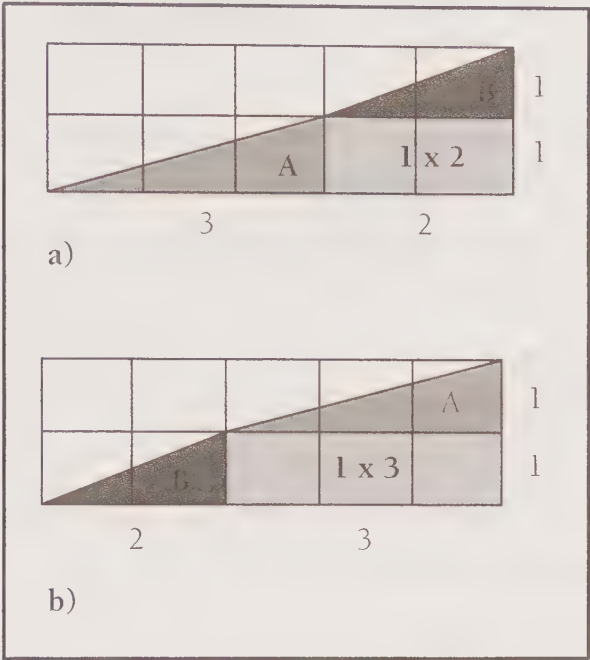
We can illustrate Mr. Curry's fallacy more clearly in by drawing a very small triangle whose legs have lengths 2 and 5, and the two smaller triangles A and B have different lengths:

triangle A has legs of lengths 1 and 3

triangle B has legs of lengths 1 and 2

For this smaller triangle, you can see that the "larger triangle" in Figure 2a) is "bowed in" whereas the larger triangle in Figure 2b) is "bowed

out”. Hence, the area of the rectangle in Figure 2b) is one more than the area of the rectangle in Figure 1a).



Big “triangle” not a triangle: a) bowed in, b) bowed out
Figure 2

Connection with the Fibonacci Sequence: There is an interesting connection between this illusion and the Fibonacci sequence

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \dots$$

This sequence obeys the simple rule that after selecting the first two numbers as 1, thereafter each number is the sum of the previous two. Mathematically, we write

$$F_1 = 1$$

$$F_2 = 1$$

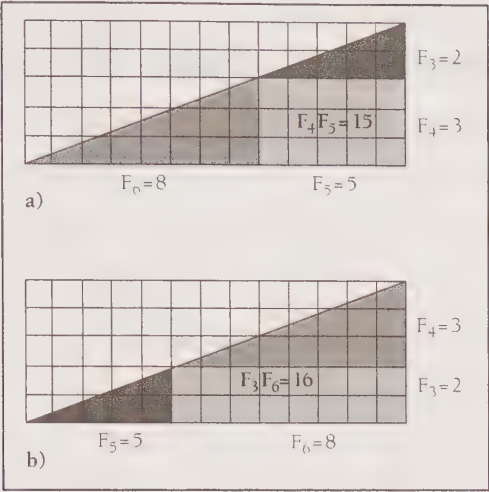
$$F_n = F_{n-1} + F_{n-2} \quad n = 3, 4, \dots$$

This seemingly simple sequence has intrigued mathematicians for centuries. We have redrawn the triangle from Figure 1 as Figure 3, denoting the fact that the lengths of the legs of triangle B are the Fibonacci numbers $F_3 = 2$, $F_5 = 5$ and the lengths of the legs of triangle A are Fibonacci numbers $F_4 = 3$, $F_6 = 8$. We now make the observation that we could take any four consecutive Fibonacci numbers and construct triangles A and B in an analogous manner. The larger the Fibonacci numbers the better Curry's illusion becomes since the slopes of triangles A and B are closer together.

We will not prove it, but one of the many interesting properties of the Fibonacci sequence is the relationship

$$F_3 F_6 = F_4 F_5 + 1$$

which says mathematically that the rectangle in Figure 3b) is larger by one than the rectangle in Figure 3b).

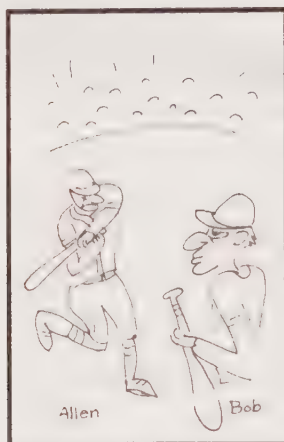


Connection between the Curry Paradox and the
Fibonacci Sequence
Figure 3

XEΛΩΞΨΘ

The Not-So-Uncommon Anomaly of Mr. Simpson

Allen and Bob are baseball players who play for a minor league baseball team.



In 1923 their season ran from April to September and every month Allen had a higher monthly batting average than Bob, which of course means that at the end of the year Allen's yearly average was higher than Bob's. Or was it? Since baseball fans are obsessed with statistics, maybe we should think about that a little longer. Can you be positively 100% sure if Player A has a higher *monthly* batting average than Player B *every* month of the season, then he will have the higher *season* average?

The answer to this question is, contrary to most people's beliefs, no. It is possible for Player B to have a higher season average than Player A even though his monthly average was always lower. This is just one of many interesting occurrences of what is called the Simpson Paradox, named after E.H. Simpson, who first described the phenomenon in a 1951 paper in the journal *Nature*.

So, how does Bob manage to get a higher yearly average than Allen if his monthly averages are always lower? Well, the numbers in Table 1 show how this can happen. Suppose in April, Allen was injured so he played only one game and went 3 for 4, hence he batted 0.750 (750 as they say in baseball). On the other hand, Bob was healthy and played the entire month, getting 50 hits in 100 at bats for a 0.500 average. Allen then recovered from his injury and played from May 1 to the end of the season, getting 30 hits in 100 at bats each month. Unfortunately, Bob came down with elbow tendinitis on May 1 and after having a great April, getting 50 hits in 100 at bats, played only one game a month for the rest of the season, each game going 1 hit in 4 at bats. Table 1 shows Bob's monthly and annual averages.

On the other hand, Allen remains healthy the entire season and his monthly and annual averages are also shown in Table 1. Note that Allen's monthly averages are always higher than Bob's monthly averages, but Bob had the larger season average (0.458 versus 0.303). How is that for a baseball statistic? We are not saying it has ever happened in real baseball, but it could.¹

	April	May	June	July	Aug.	Sept.	Average over season
Allen	$\frac{3}{4}$	$\frac{30}{100}$	$\frac{30}{100}$	$\frac{30}{100}$	$\frac{30}{100}$	$\frac{30}{100}$	$\frac{153}{504} = 0.303$
Bob	$\frac{50}{100}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{55}{120} = 0.458$

Allen has higher monthly averages but a lower yearly average.

Table 1

¹ The author is unaware of the actual occurrence of such an episode, but if one were found, it would be popular in the world of baseball trivia.

Although our baseball example is somewhat contrived, the Simpson Paradox comes up surprisingly often in other arenas, like in medical and social data. For example, suppose a clinical trial is carried out to determine the effectiveness of two new drugs, Drug 1 and Drug 2, in the treatment of a given ailment. In the study, patients are subdivided into two classes, those in the beginning stages of the ailment and those in advanced stages. A total of 1,000 patients take part in the study, 500 are given Drug 1 and 500 are given Drug 2.

Table 2 shows the results of the study: The fraction (and percentage) of patients whose condition improved as a result of treatment from the two drugs. The experiment indicates that Drug 1 is *more* beneficial in the treatment of both the onset and the advanced disease but *less* effective overall, as indicated by the ratios in bold type.

	Fraction of people with advanced disease helped	Fraction of people with onset disease helped	Totals
Drug 1	$\frac{116}{124}$ (94%)	$\frac{274}{376}$ (73%)	$\frac{390}{500}$ (78%)
Drug 2	$\frac{334}{386}$ (87%)	$\frac{79}{114}$ (69%)	$\frac{413}{500}$ (83%)

Fraction (percentage) of people helped by treatments
Table 2

Looking more carefully at the numbers in Table 2, the cause for the paradoxical result is due to the ratios

$$\frac{116}{124} > \frac{334}{386} \text{ and } \frac{274}{376} > \frac{79}{114} \text{ but } \frac{116+274}{124+376} < \frac{334+79}{386+114}$$

In general, the arithmetic basis for the Simpson Paradox is the non-result

$$\frac{a}{b} > \frac{c}{d} \text{ and } \frac{A}{B} > \frac{C}{D} \text{ does not imply } \frac{a+A}{b+B} > \frac{c+C}{d+D}$$

which is often intuitively believed as true.

Example: Berkeley Sex Case A case illustrating Simpson’s Paradox came about in 1973 when the University of California at Berkeley was sued for bias against women who were applying to graduate school. The data showed that, overall, men were accepted at a higher percentage than women, and at a greater rate than was due to simple chance. However, when the data was studied more carefully, it was discovered that no individual department was biased against women. In fact, there was a slight bias against men. The explanation for the overall bias was due to the fact that women tended to apply to academic departments that had low admission rates, whereas men tended to apply to departments with high admission rates.

Table 3 shows how this situation can happen, where we have a hypothetical university with two departments, the medical college and the college of physical education. At this university, women, are more likely to apply to medical school, which has a low acceptance rate, while men are more likely to apply to physical education, which has a high acceptance rate. Looking carefully at the data, we see Simpson’s Paradox at work.

	Acceptance in Applying to P.E.	Acceptance in Applying to Medical School	Total Admittance Rate
Men	$\frac{90}{100}$ (90%)	$\frac{3}{10}$ (30%)	$\frac{93}{110}$ (85%)
Women	$\frac{9}{10}$ (90%)	$\frac{30}{100}$ (30%)	$\frac{39}{110}$ (35%)
Admittance Rates to Different Departments			

Simpson’s Paradox in action
Table 3

The physical education department admits 90% of applicants whereas the medical school admits 30%, and both departments admit men and women at the same rate, hence no bias.

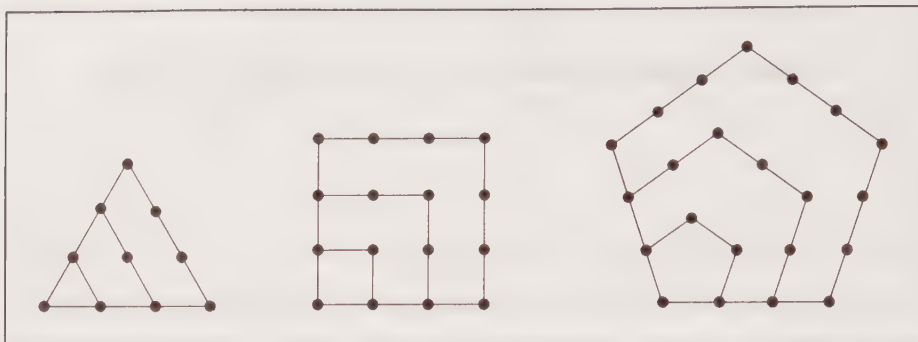
However, of the 39 women who collectively applied to these two departments, most (30 out of the 39) applied to medical school, which, being more difficult to get accepted to made the overall acceptance rates appear as biased toward men at a rate of 85% for men and 35% for women.

ΦΓΠΘΩΞΨ

Some Paradoxes Can (Literally) Kill You

Some people don't solve paradoxes, they *create* them. Such was the case of the Greek mathematician Hippiasus of Metapontum in the 5th Century B.C., who thought he had created a paradox for the ages, but unfortunately, it was taken as blasphemy, which led to his death sentence.

The Pythagoreans were a secretive religious society that flourished in Greece in the 5th Century B.C. who worshiped numbers and considered them their masters. (I wish some of my students had a little Pythagorean philosophy in them.) Founded by the Greek philosopher Pythagoras of Samos, it was the members' belief that all quantities of nature could be explained by integers or ratios of integers, like 1, 2, 3, $\frac{4}{3}$, $\frac{1}{2}$, $\frac{7}{3}$, $\frac{25}{11}$, Unlike earlier cultures in Egypt and Mesopotamia who used numbers for counting or tabulating, the Pythagoreans were interested in numbers as an opening to understanding the world around them. They studied even numbers (2, 4, 6, ...), odd numbers (1, 3, 5, ...), the idea of divisibility, prime numbers, composite numbers, and so on. They studied square numbers 1, 4, 9, 16, ... triangular numbers (1, 3, 6, 10, 15, ...), pentagonal numbers (1, 5, 12, 22, 35, ...), and patterns they observed in the real world that reflected numbers.



Patterns for finding triangular, square, and pentagonal numbers

Figure 1

It was their core belief that the fundamental qualities of humankind: rationality, masculinity, femininity, justice, morality, all had logical interpretations through numbers. Pythagoras said it himself. *“Everything is number.”*



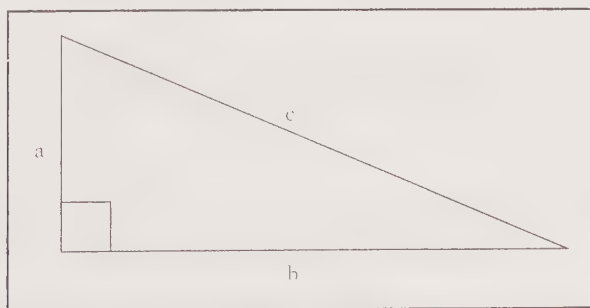
Pythagoras (560–480 B.C.)

Unlike today, where researchers publish their findings in scholarly journals, the Pythagoreans were secretive and guarded their discoveries. It is ironic then, that their main discovery, the Pythagorean theorem,¹ was the instrument from which Hippasus created his famous (or infamous at the time) paradox, one that shook the Pythagoreans' belief system to the core.

The Pythagorean theorem

$$a^2 + b^2 = c^2$$

is probably the most well-known equation in mathematics, possibly the mathematical equivalent of Einstein's equation $E = mc^2$ in physics. The Pythagorean theorem states that the square of the hypotenuse of a right triangle is equal to the sum of squares of the two legs of the triangle. See Figure 2.



The Pythagorean theorem: $a^2 + b^2 = c^2$

Figure 2

¹ Although the Pythagorean theorem is generally associated with the Pythagoreans, it was probably known at the time by Chinese and Babylonian mathematicians.

The funny thing about the Hippasus Paradox is how it seemed like a paradox to the Pythagoreans, but now, 2,500 years later, we view it as a monumental discovery, especially the method of proof, and fundamental to today's mathematics.

So, what is the Hippasus Paradox? The idea is simple: we draw a square with sides 1 unit long, then draw a diagonal connecting two opposite corners, like the square drawn in Figure 3.



Unit square with irrational diagonal

Figure 3

Now, we apply the Pythagorean Theorem, getting the equation

$$d^2 = 1^2 + 1^2 = 2$$

which means d is a number whose square is 2, which in today's notation is written as

$$d = \sqrt{2}$$

Knowing $\sqrt{2}$ must be a fraction according to Pythagorean beliefs. Hippasus writes

$$\sqrt{2} = \frac{p}{q}$$

where the goal is to find the natural numbers p and q . He begins by squaring each side of the equation, getting

$$2 = \frac{p^2}{q^2} \quad \text{or equivalently} \quad p^2 = 2q^2$$

At this point, you must understand that all natural numbers, like 8, 27, 45, 130, 5010,... which are *composite numbers* (i.e., not prime numbers) can be factored into their prime factors, like

$$8 = 2 \times 2 \times 2$$

$$105 = 3 \times 5 \times 7,$$

$$44 = 2^2 \times 11$$

$$169,400 = 2^3 \times 5^2 \times 7 \times 11$$

The prime numbers are natural numbers 2,3,5,7,... whose only factors are 1 and itself, like $3 = 3 \times 1$, $7 = 7 \times 1$,....

Making this observation, we now realize that both sides of the equation $p^2 = 2q^2$, that is p^2 and $2q^2$, are composite numbers, the reason being that with *any* square, i.e., 2^2 , 3^2 , 4^2 ,... when factored into primes, the prime factors will always come in pairs, like

$$3^2 = 3 \times 3$$

$$4^2 = 4 \times 4 = (2 \times 2)(2 \times 2)$$

$$21^2 = 21 \times 21 = (3 \times 7)(3 \times 7) = (3 \times 3)(7 \times 7)$$

Knowing this, we will now see that the equation

$$p^2 = 2q^2$$

is a *fraud* if p and q are required to be natural numbers $1, 2, 3, \dots$

To understand this argument, if we factor the left-hand side p^2 into prime factors, there must be *two copies* of each prime factor, which means the number of 2s in the prime factorization of p^2 will be either $0, 2, 4, 6, \dots$ (right?). If we now turn our attention to the right-hand side $2q^2$, we conclude that the number of 2s in the prime factorization of this is either $1, 3, 5, 7, \dots$. But in order that the equation hold, it must contain the same number of 2s on each side of the equation, which means the equation $p^2 = 2q^2$ cannot *hold* as long as p and q are natural numbers $1, 2, 3, \dots$. But this means the equations

$$2 = \frac{p^2}{q^2} \quad \text{and} \quad \sqrt{2} = \frac{p}{q}$$

also are invalid for integers q and p .

At this point, Hippasus realized² he had made a major discovery, namely that the square root of 2, i.e., $\sqrt{2}$, or the diagonal of the unit square, is *not* a fraction,³ or what we now call a rational number. Today, we call real numbers that are not rational numbers *irrational number*, and Hippasus discovered the very first one.

The story goes that Hippasus did not get the glowing reception he expected, and according to legend, his discovery was made at sea, and the only award he got for his efforts was being thrown overboard.⁴

So, where is the paradox? To the Pythagoreans, it was a real conundrum, maybe the work of some unknown evil force, but with the advantage of 2,500 years of mathematical history, there *is* no paradox. But, to the Pythagoreans, the discovery of a physical quantity (the diagonal of the unit square⁵) *was* an enigma, a paradox. How could something exist in their world outside their faith?

To fast-forward to the late 1800s, the German mathematician, Georg Cantor, proved there are an *infinite* number of irrational numbers, and in fact there are *more* irrational numbers than rational numbers. The rational numbers are said to be *countably infinite*, whereas the

² We don't know of course, but it would be nice to know.

³ Today, we call real numbers that are fractions rational numbers, and real numbers that are not rational numbers irrational numbers. Hippasus discovered the first irrational number.

⁴ The author would not bet his life on the accuracy of the Hippasus story, but the story is the one told by math historians. So, that's my story and I'm sticking to it.

⁵ The irony of the discovery that the square root of 2 is irrational is that the proof used the crowning achievement of the Pythagoreans, the Pythagorean theorem!

irrational numbers have cardinality of the *continuum*. Today, we know many more irrational numbers, like

$$\pi, \sqrt{p}, p \text{ prime}, e (\text{Euler's Constant}), \phi (\text{golden ratio}), \dots$$

In the history of mathematics, the Pythagoreans were not alone in resisting new number systems. The number “zero” was not accepted as a “number” in European mathematics until the 1300s. and without 0, there would be no negative numbers. It wasn’t until the latter part of the 1400s that negative numbers were widely accepted into the family of numbers. No matter what the discipline, new ideas are often resisted in favor of the status quo.

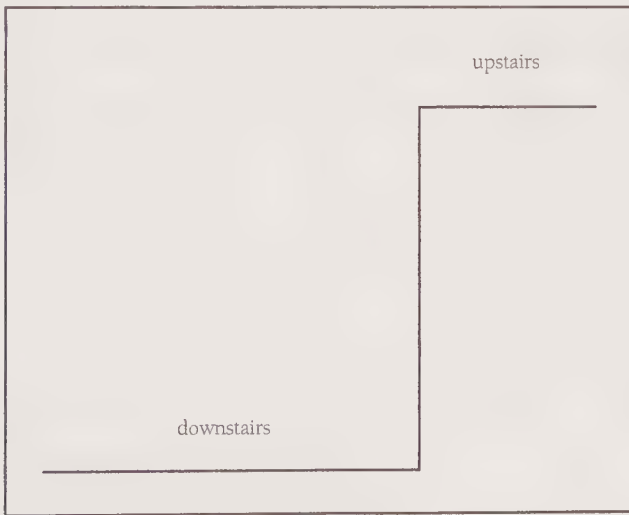
Reference: A great book that describes the lives and times of the Pythagoreans is *How Plato and Pythagoras Can Save Your Life: The Ancient Greek Prescription for Health and Happiness* by Nicholas Kardaras, Ph.D, Conari Press, 2011.

ΦΔΠΘΞΨΩ

The Curve That Shook the World

We are going to approach this paradox differently from the others in the book. We are going to let *you* find it. We will tell you a story and somewhere in the middle of the story you will have an “aha” moment and think to yourself, hmmmmmm, something strange is going on here.

The story is very simple. John is building a house and is in the process of building a staircase from the ground floor to the upstairs. Figure 1 shows an embarrassingly crude diagram of the downstairs and upstairs before any stairs have been built.

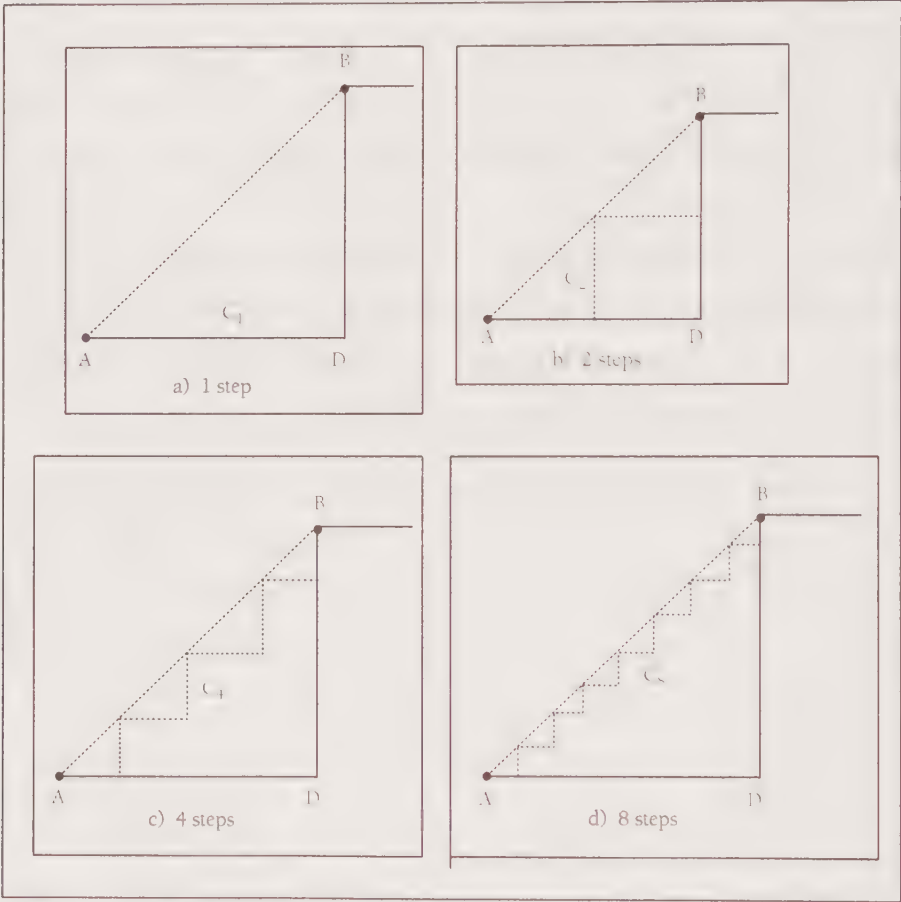


Downstairs and upstairs before the stairs are built

Figure 1

John would like to cover the steps of the staircase all the way from the bottom to the top with continuous carpeting, but being frugal he

wants to determine the number of steps that will minimize the cost of the carpeting. To do this, John draws a diagram showing 1,2,4, and 8 steps as shown in Figure 2, where C_1, C_2, C_4 , and C_8 denote jagged curves where the carpeting will be laid.



1,2,4, and 8 steps
Figure 2

Suppose the second floor is 10 feet above the ground floor and John wants the rise of the steps to be 45 degrees. Given this, John can then find the length of carpeting required for the four different staircases

in Figure 2. That is, the length of the curves C_1, C_2, C_4 , and C_8 . For the staircase in Figure 2a) which has one giant step, the length of carpeting, or length of curve C_1 , is 10 feet horizontally from Point A to Point D, and another 10 feet straight up from Point D to Point B (2×5) for a total length of

Length of carpeting for 1 step = $10 + 10 = 20$ ft

John now sees how much carpeting he saves by making 2 steps, as shown in Figure 2b). Here, we have two horizontal treads and two vertical risers, each having a length of 5 feet for a total of

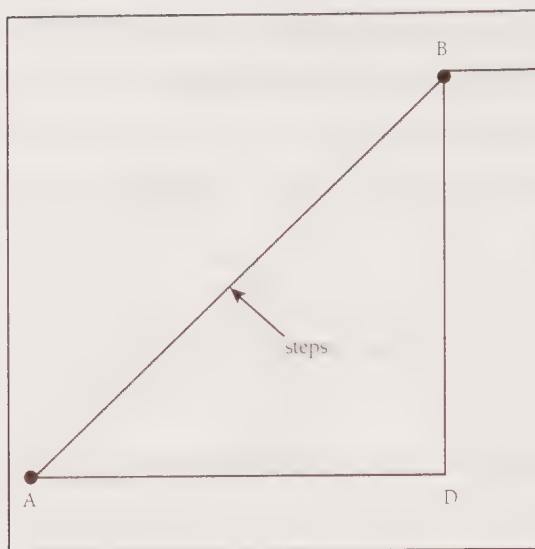
Length of carpeting for 2 steps = $5 + 5 + 5 + 5 = 20$ ft

Hmmmmmmm, the same. Adding more steps it is not difficult to see that the length of carpeting does not change. For example with 4 and 8 steps, we discover

Length of carpeting for 4 steps = $\underbrace{2.5 + 2.5 + \cdots + 2.5}_{8 \text{ terms}} = 20 \text{ ft}$
Length of carpeting for 8 steps = $\underbrace{1.25 + 1.25 + \cdots + 1.25}_{16 \text{ terms}} = 20 \text{ ft}$

John concludes the length of carpeting is always the same: no matter how many steps are built, it will always be 20 feet.

So, just for the fun of it, John builds a staircase with a million steps, which we have drawn in Figure 3.



A million stairs

Figure 3

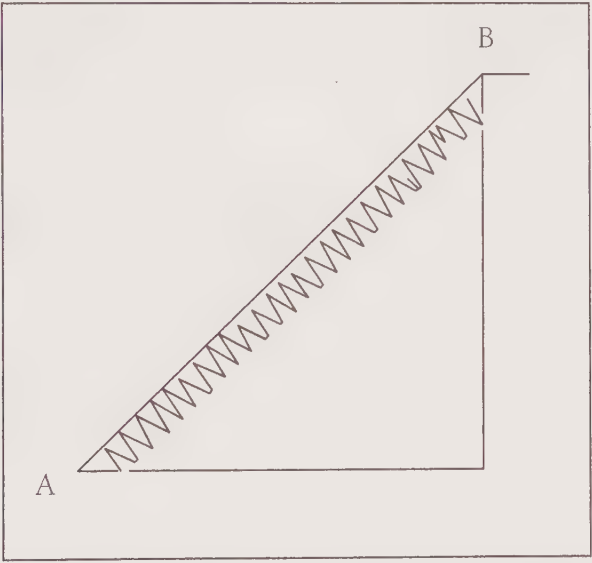
The stairs look like a jagged line very close to the straight line between Points A and B. So, are we having that “aha” moment yet? Exactly what profound idea are we trying to illustrate?

Well, if we apply the Pythagorean theorem, we see that the length of the *dotted line* between Points A and B is

$$\text{Length of dotted line between A and B} = \sqrt{10^2 + 10^2} = \sqrt{200} \doteq 14.1 \text{ ft}$$

whereas all the curves C_1, C_2, C_3, C_4 , have a length of 20 feet. In fact, even John’s pride and joy of a million steps with carpeting curve $C_{1,000,000}$ has a length of 20 feet. In other words (do I hear an “aha?”), it is possible for a curve to be close to another curve, but the lengths of the curves are not close. Don’t you think that is worth just a small “aha”?

In fact, John *could* have added more stairs like the ones drawn in Figure 4, where the treads are not horizontal and the risers are not vertical. Stairs of this type may not pass any housing code, but they are interesting mathematically. If you were to carry out the necessary computations, you would find out it is possible to construct a sequence of jagged curves $C_1, C_2, \dots, C_n, \dots$ getting closer and closer to the straight line connecting Points A and B, which has a given finite length, but the lengths of the curves $C_1, C_2, \dots, C_n, \dots$ approach infinity. Now, that has to be an eye-opener!



Curve is close to a straight line but arbitrarily long
Figure 4

If you are *still* not yet impressed with stair steps, consider the *Koch Snowflake*, named after the Swedish mathematician Nils Helge von Koch (1870–1924), which can be drawn by carrying out the following steps.

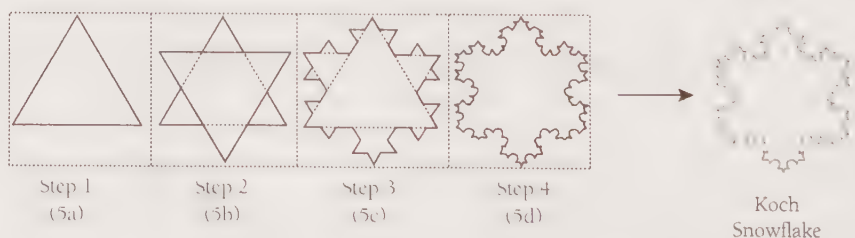
Construction of the Koch Snowflake

Step 1. Begin with an equilateral triangle as shown in Figure 5a where each side of the triangle has length 1.

Step 2. On the middle third of each side of the triangle, draw a new equilateral triangle with sides of length $\frac{1}{3}$, erasing the base of each new triangle. See Figure 5b.

Step 3. Now, on the middle third of each of the 12 sides, form another equilateral triangle with sides of length $\frac{1}{9}$, and again erase the base of each of these new triangles. See Figure 5c.

Step 4. Repeat the process with the 48-sided figure you formed in Step 3. A likeness of a snowflake will emerge. If you were to continue this process over and over you would approach the Koch Snowflake. Of course you never really arrive at the Koch Snowflake since you would have to carry out these steps an infinite number of times, but you can still get a pretty good idea what it will look like if you could.



Construction of the Koch Snowflake

Figure 5

So, why do we mention the Koch Snowflake? Well, if you look carefully at construction of each collection of triangles, you will discover that each new collection adds more than 1 to the total length of the previous curve, and since the original triangle has length 3, the total length of the limiting Koch Snowflake must be greater than

Length of the Koch Snowflake is greater than $3 + 1 + 1 + 1 + \dots$

which, of course, is infinite. In other words, it must be infinite. You see, the length of the Koch curve you see in Figure 5, which could be placed on a postage stamp, is longer than a trip to China and back. In fact, by 1,000 times.

$\Delta\chi\Gamma\Phi\Omega\Xi\Psi\text{BZ}$

Mary's Paint Can Conundrum

We all learned about area and volume back in the third grade. Area is what is on the outside of something and volume is on the inside. Area is a two-dimensional animal and volume is three-dimensional: they measure the size of something in their respective dimensions. There is an interesting paradox that involves areas and volumes, which can be described in terms of any object, be it a cube, cylinder, or sphere, but it is most easily described for a cube.

Mary has taken a job with the EZ-On Paint Company where her duty is to pour paint coming out of a machine into paint cans. (See the kind of job you can expect if you don't take math in college.) However, the company is not your run-of-the-mill paint company. *They are innovative!* First of all, they sell *cubical* cans of paint rather than the usual cylindrical cans, and secondly, their paint comes in an infinite number of different-sized cubes whose sides have lengths:

Giant size = 1 ft side

Next smaller size = $\frac{1}{\sqrt{2}} \doteq 0.70$ ft sides

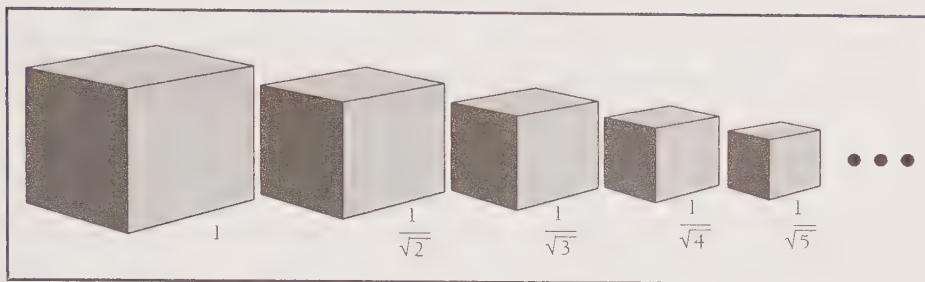
Next smaller size = $\frac{1}{\sqrt{3}} \doteq 0.58$ ft sides

Next smaller size = $\frac{1}{\sqrt{4}} = 0.50$ ft sides

... ..

... and so on

Figure 1 shows a few of these cubical cans.



Product line of the EZ-On Paint Company

Figure 1

Well, if you think the story is getting weirder and weirder, just wait, it gets even weirder.

After a few days on the job, Mary is transferred to a new department where her new job is to paint the cans. But, being the smart girl she is, she makes an interesting observation. She notices she uses more *paint* in painting the cans than she did in her earlier job when she *filled* the cans. To get to the bottom of this anomaly, she decided to compare the total volume and total surface areas of *all* the infinite cans.

Since the surface area of a cube with sides of length L is $6L^2$ (6 sides right?), and using the fact that the cans have sides of length

$$1 \text{ ft}, \frac{1}{\sqrt{2}} \doteq 0.70 \text{ ft}, \frac{1}{\sqrt{3}} \doteq 0.58 \text{ ft}, \frac{1}{\sqrt{4}} = 0.50 \text{ ft}, \frac{1}{\sqrt{5}} \doteq 0.45 \text{ ft}, \dots$$

the surface of *all* the cans is

$$\begin{aligned}\text{Total surface area of all cans} &= 6 \left(1 + \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{4}} \right)^2 + \dots \right) \\ &= 6 \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \right)\end{aligned}$$

But we have seen this series earlier in the connection of laying bricks, and discovered that a French mathematician, Nicole d'Oresme (*circa* 1323–1382), showed that the series grows without bounds, by combining the terms of the series in groups of 2, 4, 8, 16, ... and observing

$$\begin{aligned}1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4} \right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right) \\ &\quad + \left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16} \right) + \dots \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4} \right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right) \\ &\quad + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} \right) + \dots \\ &= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \\ &= \infty \text{ (grows without bounds)}\end{aligned}$$

Hence, the total surface area of the cans is *infinite*, which you might say is not all *that* surprising since, after all, there *are* an infinite number of cans. Mary did think she was using a LOT of paint.

Now, here's where things start to get weirder. Mary then decided to determine the total amount of paint required to fill all the infinite number of cans. Since the volume of a single can with sides having length L is L^3 , the total volume of all infinite cans is

$$\begin{aligned}\text{Total volume of the cubical cubes} &= 1 + \left(\frac{1}{\sqrt{2}}\right)^3 + \left(\frac{1}{\sqrt{3}}\right)^3 + \left(\frac{1}{\sqrt{4}}\right)^3 + \dots \\ &= 1 + \frac{1}{2^{3/2}} + \frac{1}{3^{3/2}} + \frac{1}{4^{3/2}} + \dots\end{aligned}$$

So, we have another infinite series, which is slightly different from the harmonic series since here the numbers in the denominator are raised to the $\frac{3}{2}$ power. The effect of doing this makes the terms in the series smaller than the corresponding term in the harmonic series, the net result being that although we are adding an infinite number of terms, the limiting sum approaches a finite number, which is approximately 2.612.

So, what does Mary conclude? Well, she was so flabbergasted that she sent off a memo to the company president, stating it is impossible to paint all the cans, no matter how much paint was used, but surprisingly, possible to fill the entire infinite collection of cans with a small amount of paint, namely about 2.6 cubic feet!

Epilogue: So how can an object have an infinite surface area, but its insides have finite volume? The answer is we can't compare sizes of

things in different dimensions; at least when we do, we sometimes get into trouble. Area measures things in two dimensions and volume measures things in three dimensions.

ΧΔΦΛΠΘΒΞ

The China Paradox

We are not sure this little narrative classifies as a paradox, but we have discovered some people classify it as such. especially mothers who tell their kids to stop digging up the backyard, or else they will end up in China. We hope you find it entertaining and educational. Who knows? From what you learn here, you might even win a bar bet sometime.

Here's the problem. Suppose you dig a hole through the center of the earth, say from where you are now sitting. The question then arises, where would you arrive at the other end of the earth? In other words, your antipodal point. To find the latitude of this point, you simply interchange north and south if you live in the northern hemisphere, and vice versa for those of you in the southern hemisphere. In other words, if you are at 30° north latitude, your antipodal point will have latitude of 30° south. For longitude, if you are at west longitude, then you simply replace your L degrees west by $180-L$ east (vice versa if your longitude is east). For example, Bangor, Maine sits at 45° north latitude, and 68° west longitude, hence the location on its antipodal point resides at 45° south latitude and $180 - 68 = 112^\circ$ east longitude.

So, why the paradox? It is nothing more than simple arithmetic. Well, it is conventional wisdom in some circles in the U.S. that digging a hole

through the center of the earth you will end up in China. But of course this is nonsense since both the U.S. and China are in the northern hemisphere.

So, as an aid to mothers who continually warn their kids about digging holes in the backyard, or else they'll end up in China, we have drawn a map showing the real location of where they'll end up. It is not China. It is far wetter, it's in the middle of the Indian Ocean.



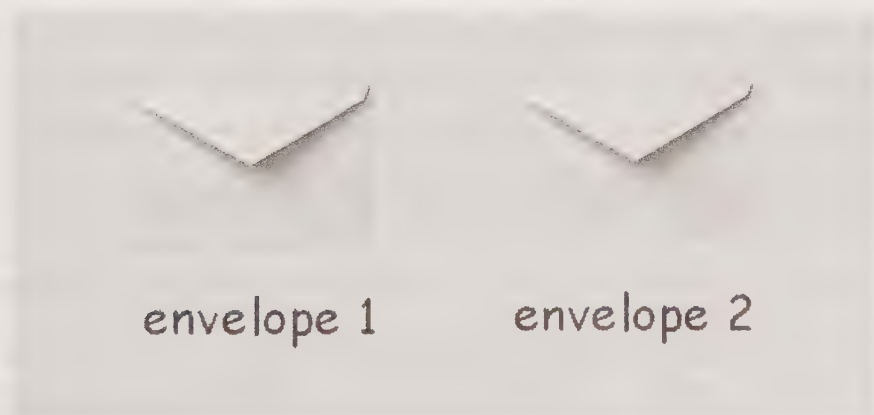
Reflection of U.S. through the center of the earth

Figure 1

$\Delta X \Gamma \Phi \Omega \Xi \Psi B Z$

The Two-Envelope Paradox

Very few paradoxes are so easy to state but as difficult to resolve as the two-envelope paradox, or sometimes called the exchange problem. In this paradox, your rich uncle places two envelopes before you and tells you that one envelope contains *twice as much* money as the other, but being the curmudgeon that he is, doesn't tell you which one. He then says that you should choose either envelope and keep its contents.



So, wherein lies the paradox? Well, so far there isn't one, but suppose you pick Envelope 1, and suppose it contains \$500, and then your uncle offers to let you exchange it for Envelope 2. The \$64,000 question is, should you switch envelopes? What would be the *expected take* if you switch, and what exactly do we mean by *expected take*?

Expected Value

The expected value of a random process is an easy but important concept in probability. You simply sum all possible gains (or losses) multiplied by the probability that they occur. For example, suppose a casino charges \$40 to spin a wheel, where the house pays \$1,000 when a patron wins and \$10 (booby prize) when the patron loses. Suppose the probability of winning is 0.02 (hence, the probability of losing is 0.98). For this random process, a player's *expected winnings* (before the \$40 charge is taken out) per spin is

$$\text{Expected winnings} = \$1,000(0.02) + \$10(0.98) = \$29.80.$$

Subtracting the \$40 charge to spin the wheel, the player's *expected winnings* will now be $-\$10.20$. That is, the player should expect to lose (on average) \$10.20 on each spin of the wheel.

So, back to your rich uncle. Your uncle has told you that one envelope contains twice as much as the other, but we don't know which envelope is which, and since Envelope 1 contains \$500, Envelope 2 must contain either \$250 or \$1,000, each with a probability of $1/2$. Hence, if you switch to Envelope 2, your expected take will be

$$\begin{aligned} \text{Expected take when switching envelopes} &= \$250\left(\frac{1}{2}\right) + \$1,000\left(\frac{1}{2}\right) \\ &= \$625 \end{aligned}$$

which is a gain of \$125. In general, if you had found N dollars in Envelope 1, and if you switch to Envelope 2, your expected take will be

$$\text{Expected take in general when switching} = \left(\frac{N}{2}\right)(0.5) + (2N)(0.5) = \frac{5}{4}N$$

which is a gain of $N/4$ from Envelope 1.

So, where is the paradox? Before you think you've solved the problem, ponder for a moment. Your decision to switch does not depend on the amount of money in Envelope 1, hence you could have made your decision before even looking inside! What's more, the two envelopes are completely symmetrical so if you had selected Envelope 2, your decision would be to switch to Envelope 1, and therein lies the paradox. You have found yourself in a quandary, where your best strategy is to waffle back and forth an infinite number of times between the two envelopes.

So, how is this paradox resolved? Well, some people say that they have solved the paradox, while others say it has not been resolved. In other words, a true paradox!

We leave it up to a clever reader to resolve the paradox.

ΔΧΓΦΩΞΨΒΖ

St. Petersburg Paradox

There are paradoxes whose conclusions are understandable from a purely mathematical point of view, but their interpretation in the real world is downright idiotic. One such paradox is the St. Petersburg Paradox, a 300-year-old enigma proposed in 1713 by Nicolaus Bernoulli, a Swiss mathematician living in St. Petersburg at the time, and solved by his cousin, Daniel Bernoulli in 1738.

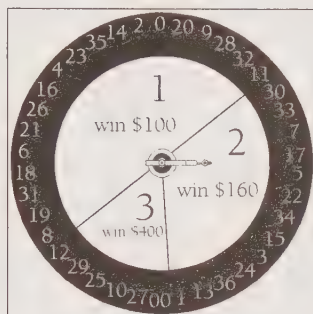
The paradox involves playing a game of chance, where the player tosses a coin until the coin turns up HEADS, after which the game concludes. If the coin turns up HEADS on the first toss, the player wins \$2 and abruptly stops. If the coin lands TAILS on the first toss and HEADS on the second toss, the player wins \$4; if the coin lands TAILS on the first 2 tosses and HEADS on the third toss, the player wins \$8; and so on. In general, if the coin lands TAILS on the first $n-1$ tosses and HEADS on the n th toss, the player wins 2^n dollars.

The paradox arises when we try to determine how much you would be willing to play this game. Would you pay \$25, \$50, \$100? What about \$1,000? Any takers?

To get a better handle on this problem, we make a brief introduction of the expected value of a game of chance.

Value of a Game of Chance

The *value* of a game of chance, or the amount of money one would be willing to pay to play the game, is found by summing the possible gains times the probability of obtaining those gains. For example, for purchasing this seminal treatise on paradoxes, the author would like to give you the opportunity to win back the exorbitant cost of the book by spinning the roulette wheel shown below.



The wheel has three possible outcomes: 1, 2, and 3 with respective payoffs, \$100, \$160, and \$400. To spin the wheel, the author charges a modest fee of \$200. So, is it a fair price? Hint: The probability the wheel will land on 1 is $\frac{1}{2}$, landing on 2 is $\frac{3}{8}$, and landing on 3 is $\frac{1}{8}$. Hence, the value of the game (which can be thought of as your average payoff if you spin the wheel an infinite number of times) is

$$\begin{aligned}
 \text{Value of the game} &= \$100\left(\frac{1}{2}\right) + \$160\left(\frac{3}{8}\right) + \$400\left(\frac{1}{8}\right) \\
 &= \$50 + \$60 + \$50 \\
 &= \$160
 \end{aligned}$$

In other words, a fair price to both the author and the reader would be \$160. A reader who pays \$200 to play the game will lose on *average* \$40 a spin.

To determine the expected value of the coin-flipping game, we list the probability of getting the different outcomes and the rewards gained in getting those outcomes. The possible outcomes of the game, having probabilities of $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots$ and respective payoffs of \$2, \$4, \$8, \$16, \$32,.... are summarized in Table 1.

Number of tosses (n)	Your prize	Probability of n tosses	Your gain for n tosses
1	\$2	1/2	$\$2 \times (1/2) = \1
2	\$4	1/4	$\$4 \times (1/4) = \1
3	\$8	1/8	$\$8 \times (1/8) = \1
4	\$16	1/16	$\$16 \times (1/16) = \1
5	\$32	1/32	$\$32 \times (1/32) = \1
6	\$64	1/64	$\$64 \times (1/64) = \1
7	\$128	1/128	$\$128 \times (1/128) = \1
8	\$256	1/256	$\$256 \times (1/256) = \1
9	\$512	1/512	$\$512 \times (1/512) = \1
....	...	...	...

Possible outcomes from playing the St. Petersburg paradox game
Table 1

Hence, the value of our coin-tossing game is

$$\begin{aligned} \text{Value of the game} &= \$2\left(\frac{1}{2}\right) + \$4\left(\frac{1}{4}\right) + \$8\left(\frac{1}{8}\right) + \$16\left(\frac{1}{16}\right) + \dots \\ &= \$1 + \$1 + \$1 + \dots \\ &= \infty \text{ (infinity)} \end{aligned}$$

In other words, a fair asking price to play the St. Petersburg game is infinite. More than a hundred dollars, more than a million, more than a trillion! But who in their right minds would pay that much? Who in

their right minds would pay even \$100? Generally, people are willing to play \$20–\$30 and that’s it. As they say, it’s a paradox. It is not a mathematical paradox, it’s a paradox of economics.

The classically accepted solution to this paradox, from Daniel Bernoulli, goes outside classical probability theory, where he places the problem in the context of economics and sociology, which some people might argue is cheating since it is a pure mathematical problem. But one might argue paradoxes prompt one to enlarge one’s thinking “outside the box” to play with new ideas and theories.

ΔΧΓΦΩΞΨΒΖ

Through Cantor's Looking Glass

It is not surprising that many paradoxes in mathematics involve the infinite, although since Cantor's seminal research in the late 1800s, many of the old conundrums are now part of modern set theory and have long left the realm of the paradoxical. However, paradoxes still abound in the study of infinity, and for those not versed in Cantor's basic theory of sets, working with infinite sets seems strange, to say the least. For that reason, we present a few of the enigmas and paradoxes of the past.

Throughout much of the history of mathematics, the subject of infinity has been taboo and belonged more in the realm of religion. The Greek philosopher Aristotle (*circa* 360–280 B.C.), one of the first mathematicians to think about the subject, felt there were two varieties of infinity, the *potential* and the *actual*. He said the natural numbers 1, 2, 3, ... are *potentially* infinite since they go on forever and have no greatest number. He felt they were not *actually* infinite in size since they were not a completed entity.

In the 1600s, the Italian scientist, mathematician, and astronomer Galileo made an interesting observation about the natural numbers 1,2,3,... when he subdivided them into two types: the perfect squares: 1, 4, 9, 16, 25, ... or numbers that are squares of other numbers. Since the perfect squares are a *subset* or subcollection of all natural

numbers 1,2,3,... that would seem like there are fewer of them, as illustrated in Figure 1.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1			+					9							16	

More natural numbers than perfect squares
Figure 1

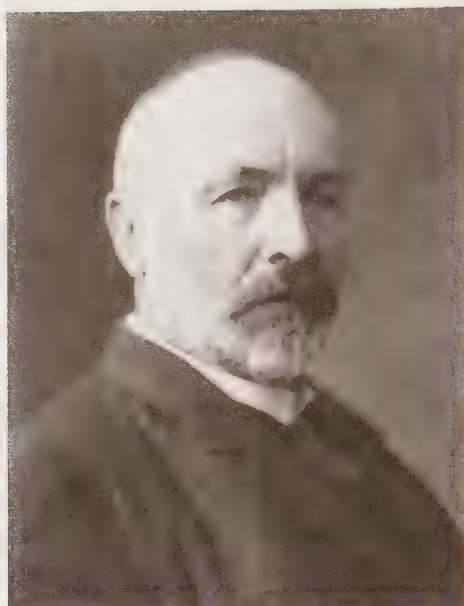
However, as Galileo observed, if you *line up* the perfect squares as Figure 2 shows, you can match any perfect square with a natural number, and any natural number with a perfect square, suggesting the two sets have the same number of members.

1	2	3	4	5		n	...
↑	↑	↑	↑	↑		↑	⋮
↓	↓	↓	↓	↓		↓	⋮
1	4	9	16	25	⋮	n^2	...

Just as many perfect squares as natural numbers
Figure 2

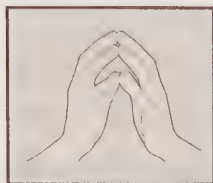
Galileo's conclusion was that when it comes to infinite sets, things are not the same as for finite sets.

The Godfather of Infinity: Georg Cantor: The subject of infinity as we know it today had its origins in the work of the German mathematician. Georg Cantor (1845–1918), whose groundbreaking insights rest upon a very simple idea (as do many seminal insights). Suppose you are unable to count and would like to know if your hands have the same number of fingers.



Georg Cantor

Not being able to count, you do something more basic. You simply place the thumb of one hand against the thumb of your other hand, then place your index finger of one hand against the index finger of your other hand, and do the same for the remaining fingers. When you are finished, your fingers are all matched up in a *one-to-one correspondence*: every finger on each hand has exactly one kindred soul on the other. *You may not know how many fingers you have on each hand, but you know they are the same.*



Now, what if you had an infinite number of fingers? In this case, it doesn't matter if you can't count, no one else can either. I mean, no one can count that high! Cantor's inspiration was that, even if we can't *count* infinite sets, we can tell if they have the *same* number of members by applying the "finger rule" used when determining that our hands had the same number of fingers.

Cantor called the *size* of a set the *cardinality* of the set, and if a set can be placed in a one-to-one correspondence with the natural numbers 1,2,3,... the "size" of the set is called *countably infinite*, and these types of sets are said to have *cardinality* $\aleph_0$ (aleph- null).¹

Cantor then asked if all infinite sets have the same cardinality; that is, are there larger infinities? The next infinite set Cantor tested was the rational numbers (ratios of natural numbers), and if you were a guessing person you might be tempted to say there are "more" rational numbers than integers since the natural numbers are a subset of the rational numbers. If you did, however, you would be wrong since Cantor found an ingenious matchup (like the finger matching, but with more fingers) between the natural numbers 1,2,3,... and all the positive rational numbers illustrated in Figure 3.

¹ $\aleph$ is the first letter in the Hebrew alphabet. The subscript of 0 on $\aleph_0$ denotes it is the smallest infinity.

Larger Infinities? Cantor then looked around for sets that might have “more” members, the most obvious set to try being the real numbers. The answer was astounding. Cantor proved that there are “more” real numbers than natural numbers: $1, 2, 3, \dots$, that it is impossible to match up the real numbers with the natural numbers, and that since the natural numbers are a subset of the real numbers, the cardinality of the real numbers must be larger than that of the natural numbers. So how did Cantor do this?

Proofs that show something cannot happen are often proved by what is called *proof by contradiction*, where one assumes the contrary, that it can happen. In this case, we assume the rational numbers and the real numbers between 0 and 1 *can* be matched up in a one-to-one manner, and then arrive at some type of contradiction, from which our conclusion is there is no correspondence between the rational numbers and these real numbers.

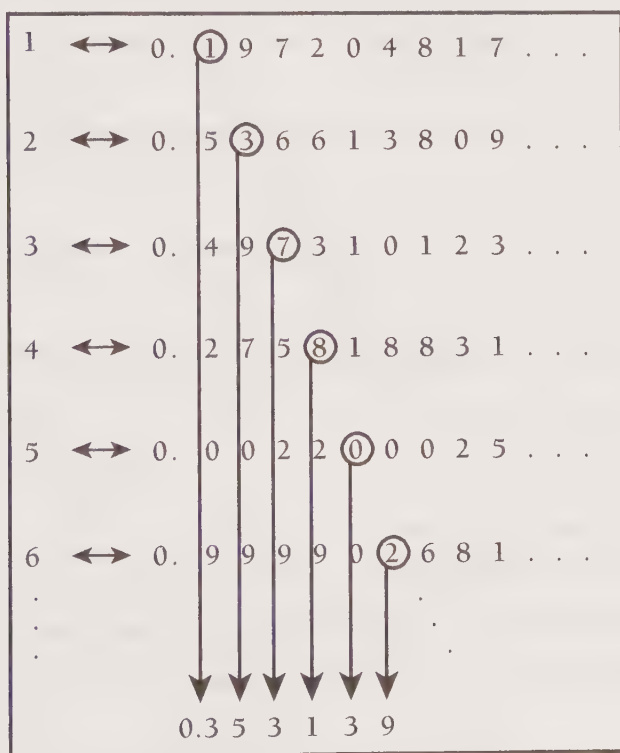
Cantor used an ingenious proof called the *diagonalization process*, where he hypothesized that it was possible to match up the natural numbers with all positive real numbers. A typical matching is shown in Figure 4 where the natural numbers $1, 2, 3, \dots$ are in the left column, and typical real numbers, expressed in decimal form, on the right.

1	↔	0. 1 9 7 2 0 4 8 1 7 . . .
2	↔	0. 5 3 6 6 1 3 8 0 9 . . .
3	↔	0. 4 9 7 3 1 0 1 2 3 . . .
4	↔	0. 2 7 5 8 1 8 8 3 1 . . .
5	↔	0. 0 0 2 2 0 0 0 2 5 . . .
6	↔	0. 9 9 9 9 0 2 6 8 1 . . .
.		.
.		.
.		.

Cantor’s hypothesized one-to-one correspondence
between the whole numbers and the real numbers
Figure 4

The assumption Cantor expresses in Figure 4 is that as one runs down the whole numbers 1,2,3, ... on the left, every real positive number will eventually show up in the right column. But Cantor showed that no matter how one chooses the real numbers in Figure 4, there will always be another real number not on this list, hence there are more real numbers between 0 and 1 than natural numbers. To illustrate how Cantor does this, look at the columns of real numbers in the right column of Figure 4. To find a real number $0.d_1d_2d_3\dots, d_n\dots$ not on this list, we pick the first digit d_1 to be *anything* other than 1 (like 0,2,3,4,5,6,7,8, or 9). By doing this we are sure our new real number $0.d_1d_2d_3\dots, d_n\dots$ will not be the same as the first real number in the list, i.e., 0.197204817.... We then pick the second digit d_2 to be anything other than 3 since we want it to be different from the second real number on the list, i.e., 0.536613809.... We continue this diagonal process,

selecting numbers $d_3, d_4, \dots, d_n, \dots$ so the n -th digit d_n is different from the n -th real number given in Figure 4, thus arriving at a real number $d_3, d_4, \dots, d_n, \dots$ not listed in the real number column of Figure 4. Hence, we contradict the claim that there is a correspondence between $1, 2, 3, \dots$ and the real numbers between 0 and 1, hence we cannot find a correspondence between the two sets. Although for simplicity in Figure 4 we only displayed real numbers in the interval $(0, 1)$, we could have just as well taken the decimal form of any real number, and thus the same conclusion; more real numbers than rational numbers.

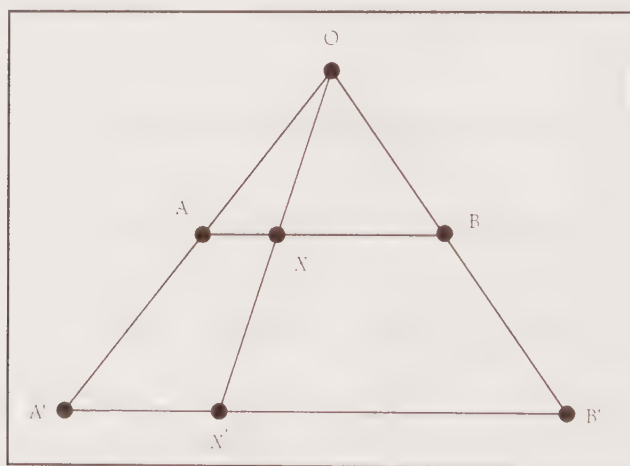


Cantor's diagonalization process in action

Figure 5

Cantor then proved the natural numbers cannot be matched with all real numbers, thus the real numbers have a larger “size” or cardinality than the natural numbers. Cantor now has two infinities, the countable infinity $\aleph_0$ of the natural numbers (and rationals), and a new cardinality of the real numbers, which he called the *cardinality of the continuum* denoted by c .

Cantor then asked if there were other sets that had the same cardinality as the real numbers, and he found many. For example, consider the two line segments **AB** and **A'B'** of different lengths drawn in Figure 6.



Segment **AB** has the same number of points as segment **A'B'**.

Figure 6

Clearly, segment **A'B'** is longer than segment **AB**, but does that mean it contains more points? Normally, you would say yes, but the one-to-one

correspondence $x \leftrightarrow x'$ between points x on the line segment **AB** and points x' on the line segment **A'B'** says otherwise. Hence, in the world of Cantor, the line segments **AB** and **A'B'** have the same cardinality, which, by the way, is the same cardinality as the entire real line which is infinite in length.

One of the most surprising things Cantor showed was that the real line has the same cardinality as the plane; they each contain the same number of points.³

By this time, questions about infinity were starting to get more numerous than the answers! For instance, Cantor proved that $\aleph_0$ was the smallest infinity and that $\mathfrak{c}$ was a larger infinity, which gave rise to the obvious questions: first, are any infinities between $\aleph_0$ and $\mathfrak{c}$, and second, are there any infinities larger than $\mathfrak{c}$? The second question was answered by Cantor and his solution is sometimes called Cantor's Paradox. The paradox being that there is no largest infinity, that you can always find bigger and bigger infinities, so the collection of "infinite sizes" is itself infinite.

So, how does Cantor find bigger and bigger infinities? Although the proof is not trivial, the idea is basic. Let's take a very simple set $\{a, b, c\}$ with 3 elements, and let's write down the collection of its subsets, i.e.,

³ In 1878 after Cantor proved that the plane contained the same number of points as the real line, he sent a letter to his good friend, the mathematician Richard Dedekind, stating "I see it but I don't believe it."

sets whose elements are also in $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$. Clearly, there are $2^3 = 8$ of them⁴ and they are:

$$\phi, \{\mathbf{a}\}, \{\mathbf{b}\}, \{\mathbf{c}\}, \{\mathbf{a}, \mathbf{b}\}, \{\mathbf{a}, \mathbf{c}\}, \{\mathbf{b}, \mathbf{c}\}, \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$$

The conclusion is that if a set has 3 elements, its collection of subsets has $2^3 = 8$ elements, i.e., the key word being *more*. In fact, it was well known during Cantor's time that there are always more members in the collection of subsets of a finite set than the set itself. The question that Cantor asked was, is the same thing true for *infinite* sets? Well, to make a long story short, the answer is yes. Cantor proved⁵ it was possible to find "bigger and bigger" infinite sets by simply forming the family of subsets of infinite sets. Granted, these sets get horribly complicated but who cares, they exist. So in effect Cantor found an infinite number of infinities

$$\aleph_0, \aleph_1, \aleph_2, \aleph_3, \aleph_4, \dots$$

called *transfinite numbers*.

OK, for the other question that drove Cantor mad, is there an infinity *between* $\aleph_0$ and $\mathbf{c}$, or is the next infinite number larger than $\aleph_0$ equal to $\mathbf{c}$, the cardinal or in fact is $\mathbf{c} = \aleph_1$ the next larger infinity? Cantor

⁴ The symbol ϕ refers to the "empty" set, a set that has no elements. The empty set is always considered a subset of any set for convenience.

⁵ Although Cantor is generally credited as the first person to identify this property of cardinal sets, some mathematicians give the distinction to the English logician, Bertrand Russell, who proved a similar theorem in 1899.

believed c to be the next infinity, but was unable to prove it, one way or the other. The hypothesis that the cardinality of the continuum c is the next larger infinite number up from $\aleph_0$ is called the *continuum hypothesis*, and in a sense can be thought of as a paradox even today since the problem has still not been resolved. In 1930 the Austrian logician Kurt Gödel proved that the continuum hypothesis can't be *disproved* from the axioms of set theory, and in the 1960s the American mathematician Paul Cohen proved it can't be *proven*, the net result being that the continuum hypothesis is *undecidable*, which means you can assume it's true (which is called *Cantorian set theory*), or you can assume it's not true (which is called *non-Cantorian set theory*). In either case, you have a valid set of axioms, albeit different. Very strange indeed!

ΧΛΦΠΘΞ

Aristotle's Wheel Paradox

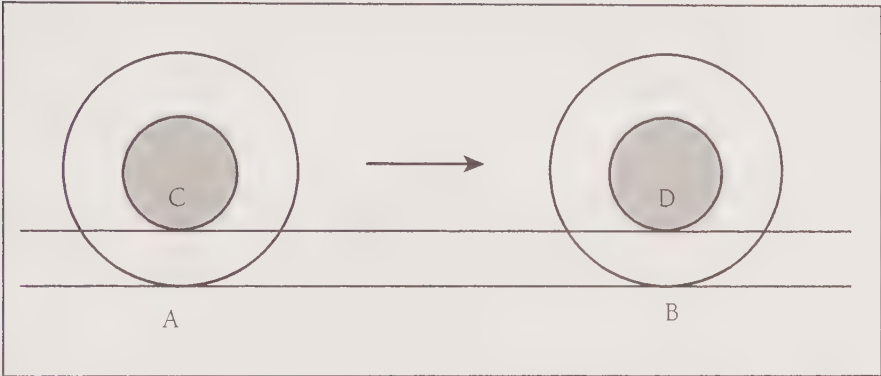
People like Aristotle (384–322 B.C.) just don't come along often enough. Although today people generally think of Aristotle as a mathematician due to his systemization of deductive logic, he also made major contributions to the physical world, which made a lasting influence on Western philosophical thinking.



Aristotle (384–322 BC)

The current paradox, known as Aristotle's Wheel Paradox, dates back to Aristotle's seminal treatise *Mechanica* and involves not only mathematics, but physical ideas as well. In this strange little enigma, we have a wheel with a smaller wheel attached solidly to it as shown in

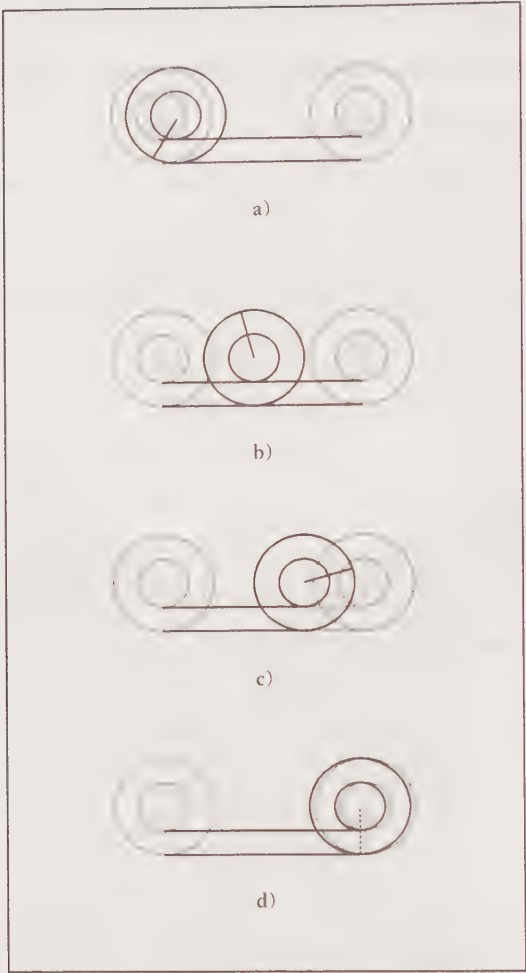
Figure 1. The wheel is then rolled one complete revolution (without slipping) from Point A to Point B, where we trace the paths of the *bottoms* of both the larger and smaller wheel, the larger wheel yielding the straight line from Point A to B, the smaller the straight line from Point C to D. But, the two lines have the same length which means the two wheels have the same circumference, which they obviously do not. A paradox.



Aristotle's amazing wheel
Figure 1

Figure 2 gives a live animation of the two wheels, where for the purpose of demonstration, we have wrapped tiny strips of red carpet on the outside of each wheel. If you look carefully while the wheels turn you will conclude that the carpets are rolled out on equal lengths. Hence, again we seem to have proven that the larger and smaller wheels have the same circumference. So what is wrong?

So, are you buying into Aristotle's argument? The interesting thing about this paradox is that it has both a mathematical and physical explanation. We will let you ponder your thoughts before unraveling the paradox.

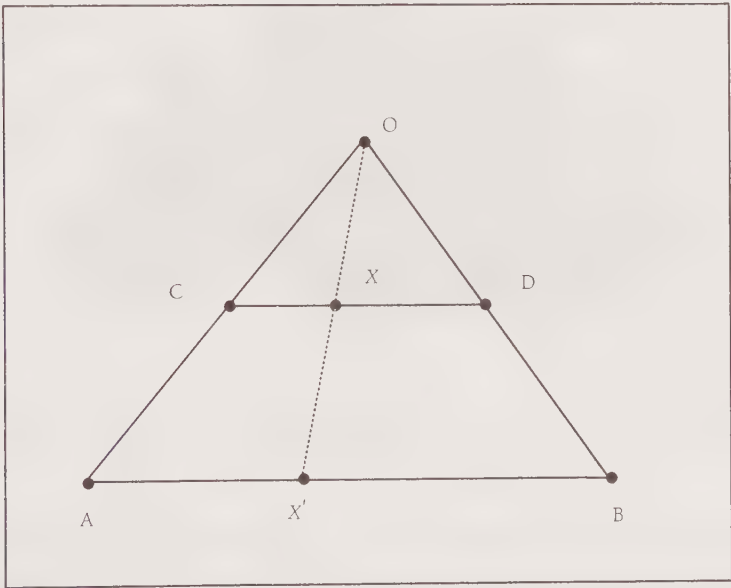


Animation of Aristotle's Wheel
Figure 2

As we said, the fallacy of the paradox is twofold, although it contains mathematical elements, it is more of a physical paradox. From a *physical* point of view, what makes you think that while the larger wheel rolls smoothly from Point A to Point B, the smaller wheel will roll smoothly from Point C to Point D *without slipping*? It won't! If a cog system were put on the smaller wheel to keep it from slipping, nothing would move.

It is physically impossible for both wheels to move without slippage. If this seems counterintuitive to you, imagine the larger wheel being the size of the earth and the smaller wheel has a radius of a foot. With these dimensions, our intuition starts to behave as it should.

Now, what about the mathematical explanation? Well, mathematically, if you buy into Aristotle's Wheel Paradox, you are accepting the fallacy that any two lines which can be placed in one-to-one correspondence, like the lines AB and CD in Figure 3, have the same length, which of course they don't.



One-to-one correspondence between lines of different length
Figure 3

ΧΔΓΘΠΩΞΨ

Did I Ever Tell You About My Uncle Charlie?

Do you mind if I snoop around in your family tree? Don't worry. I'm not going to reveal any skeletons you've got stashed away in there. However, I do want to make the point that if you start aggrandizing about a celebrated pedigree, I will come forward with a number of *other* kin you had at the time, just to put things in proper perspective.



In other words, I would like to determine exactly how many ancestors you had at different times in the past, and to make the arithmetic simple, I will assume on average there are 25 years to a generation, which means every 25 years the number of your ancestors doubles. So 25 years before you were born, your mother and father were born, so 50 years before you were born your 4 grandparents were born, 75 years before you were born your 8 great-grandparents were born, and 100 years before you were born your $2^4 = 16$ great-great-grandparents

were born. What this means is the number of your ancestors alive 100 years ago, if you include your great-great-grandparents, who were just born, and their parents and grandparents (we assume their consecutive generations were alive) would be

$$\begin{aligned}\text{Number of your ancestors alive 100 years before} \\ \text{your birth} &= 2^4 + 2^5 + 2^6 \\ &= 16 + 32 + 64 \\ &= 112\end{aligned}$$

Going back 8 generations, or 200 years before your birth, the number of your “grandparents”¹ increases to $2^8 = 256$, and including their parents and grandparents, the number of ancestors alive at the time increases to

$$\begin{aligned}\text{Number of your ancestors alive 200 years before} \\ \text{your birth} &= 2^8 + 2^9 + 2^{10} \\ &= 256 + 512 + 1024 \\ &= 1792\end{aligned}$$

So don't start pontificating about that ancestor who came over on the Mayflower in 1620, or else I just might bring up the character of some of your other ancestors alive at the time. To find out this number, if you were born in 1970, then there are 14 generations between 1620 and 1970, which means the number of your “grandparents,” their

¹ We use “grandparents” in quotes to refer to grandparents, great-grandparents, great-great-grandparents, and so on.

parents, and their grandparents alive at the time would be, 2^{14} , 2^{15} , 2^{16} respectively, which means the total number of ancestors alive at the time was (roughly, of course)

Number of your ancestors alive in 1620 is = $2^{14} + 2^{15} + 2^{16}$

= 16,384 + 32,768 + 65,536

= 114,688

This leads us to examine the number of ancestors at various times of history. For example, how many of your ancestors were alive

Table 1 Counting Your Great-Grandparents		
Generation	Number of Ancestors	History Time Line
1	2	1925 quantum mechanics
2	4	1900 Einstein, theory of relativity
3	8	1875 4-cycle engine invented
4	16	1848 Lord Kelvin, absolute zero
5	32	1825 Ohm's law of electricity
6	64	1800 first presidential election
7	128	1775 Constitution framed
8	256	1750 Benjamin Franklin, lightning
9	512	1725 Peter the Great
10	1024	1700 probability, Bernoulli
11	2048	1675 calculus discovered
12	4096	1650 barometer invented
13	8192	1625 Pilgrims on Plymouth Rock
14	16,384	1600 Kepler's laws of motion
15	36,768	1575 Drake circles the earth
16	65,536	1550 earth revolves around sun
17	131,072	1525 Ottomans conquer Hungary
18	262,144	1500 Columbus, New World

when Columbus landed in the New World in 1492? Table 1 shows (roughly of course) the number of your parents, grandparents, great-grandparents, ... living in past times, starting back at 1925.

For example, when Columbus arrived in the New World in 1492, the number of your 18-generation “grandparents” is 262,144, and if you include their parents and grandparents, the number of your direct ancestors alive in 1492 is

$$\begin{aligned}\text{Ancestors alive at the time of Columbus} &= 2^{18} + 2^{19} + 2^{20} \\ &= 262,144 + 524,288 \\ &\quad + 1,048,576 \\ &= 1,835,008\end{aligned}$$

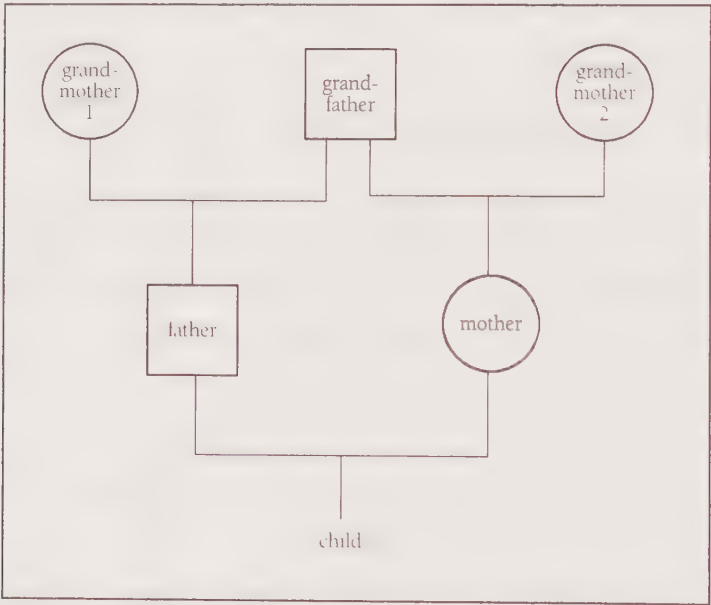
Now, if you really want to find your family tree, let’s go back 2,000 years ago to the year 0 of the Common Era, which is 80 generations ago, and let’s say 78 generations before you. Hence, the number of your “grandparents” of that vintage would be

$$2^{78} = 302,231,450,343,454,534,501,024$$

Whooooooooo! Before you start thinking you had three hundred and two thousand quadrillion “grandparents,” which is a couple of zillion times larger than the population of the earth at the time, you might contemplate the possibility that one or two of your great-great-great-great-grandparents on your mother’s side of the family tree was also a great-great-great-great-grandparent on your father’s side. In other

words, after you go back a few generations, the number of ancestors doesn't simply double with each passing generation. (So, after a few generations, don't take the numbers in Table 1 all that seriously.) Generally, a person has eight great-grandparents, but if someone marries his or her first cousin, then the number of great-grandparents drops to six because two of them are duplicates, genealogically speaking.

Figure 1 shows a situation where a child's parents are stepbrother and stepsister, and hence the child has three grandparents, 2 grandmothers and one grandfather in this case.



Child with three grandparents
Figure 1

Geneticists call the circumstance where the number of ancestors doesn't double at each generation, *pedigree collapse*, and there is more pedigree

collapse in the world than people realize. In fact, geneticists estimate that *no one* on earth is more than the 15th cousin from anyone else.

However, the moral of family trees is the undeniable fact that we are more related than some would choose to believe.

Counting ancestors brings up an interesting phenomenon regarding the evolution of humankind. Anthropologists believe humans first came to the North American continent across the Bering Strait around 12,000 years ago. Generally, we consider this length of time so great it is almost impossible to comprehend on an intuitive level. But when you consider four generations per century, this means humans have been in the New World only $\frac{12,000}{25} = 480$ generations. From that point of view, it doesn't seem that long ago. In fact, if we assume humans evolved from chimpanzees four million years ago, this means we humans are only

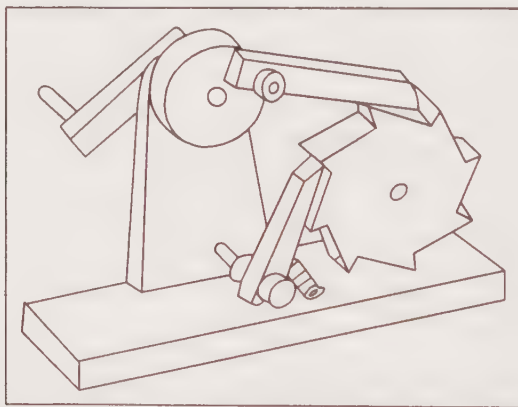
$$\frac{4,000,000}{25} = 160,000$$

generations old.

Reference: An utterly fascinating book about family trees and pedigree collapse is *The Mountain of Names: A History of the Human Family* by Alex Shoumatoff, Simon & Shuster, New York (1985).

Parrondo's Paradox: A Paradox for Losers

Some paradoxes go back 2,500 years. A new kid on the block that is causing considerable stir, not only to mathematicians but to physicists, economists, biologists, social scientists, and others is the Parrondo Paradox. The paradox goes back only to 1996 when Spanish physicist, Juan Parrondo, made an interesting observation about the workings of the common ratchet. That's right, those familiar sawtooth devices, which allow movement in one direction but prevent movement in another, and are used in everything from common carpentry to self-winding wristwatches. What he did was translate the mechanical behavior of the ratchet to the mathematical theory of games.



Ratchet

The paradox is so counterintuitive most people would probably bet their lives it couldn't happen. It involves two games of chance, each of which results in a player having steady losses over time, but when

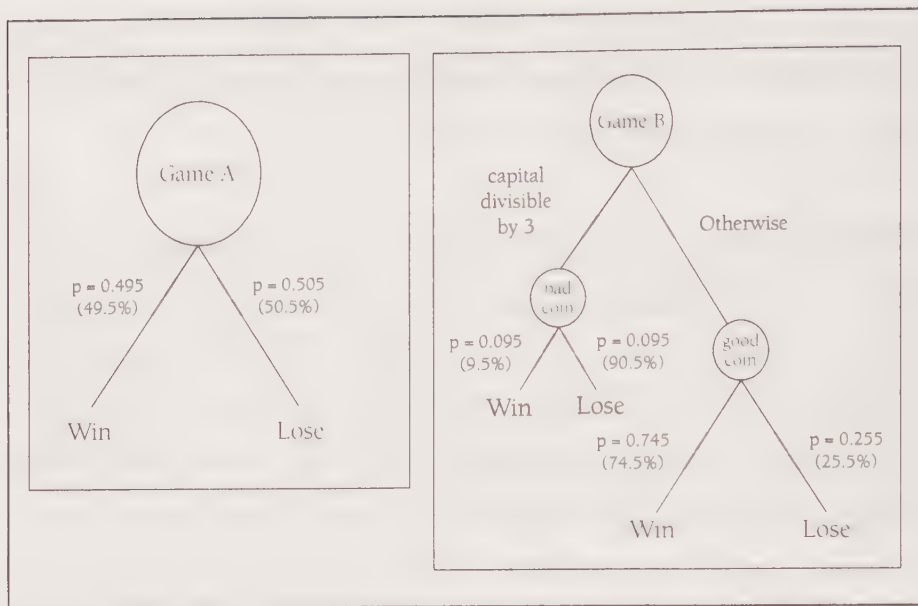
the games are alternated in some fashion, either in a deterministic or random order, surprisingly, the player has a winning expectation. A very strange phenomenon indeed!

Game A. Game A is very simple. It consists of the player flipping a biased coin that turns up HEADS 49.5% of the time and TAILS 50.5% of the time on average. If the coin turns up HEADS, \$1 is added to the player's capital, if the coin turns up TAILS, \$1 is subtracted from the player's capital. Negative capital means the player is in the hole. Clearly, any player will lose Game A in the long run.

Game B. This game is more involved and requires the use of two coins, a "good" coin and a "bad" coin. The "good coin" is biased in *favor* of the player, turning up HEADS 74.5% of the time on average, and TAILS 25.5% of the time. (As in Game A rolling HEADS means the player wins \$1 and TAILS means the player loses \$1.) The second coin is the "bad coin" and is biased *against* the player, turning up HEADS 9.5% of the time and TAILS 90.5% of the time. The determination of which coin to flip depends on the current capital. The player flips the bad coin if the present capital is a multiple of \$3; otherwise the player flips the good coin. That is

Flip the "bad" coin if the current capital is ... - \$9, - \$6, - \$3, \$0, \$3, \$6, \$9, ... Otherwise flip the "good" coin.

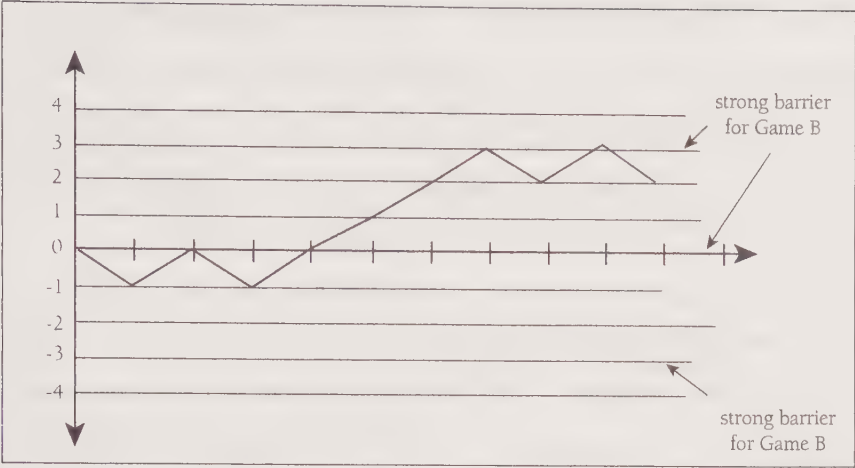
Figure 1 gives diagrams of the two games.



Rules for Game A and Game B

Figure 1

To get a feel for Game B, let's simulate a few tosses of the coins. Starting with \$0 capital, the player flips the bad coin, which most likely results in a TAIL. Hence, the player's capital most likely decreases to -\$1. (Of course the player *could* win which means the capital increases to + \$1.) The player then flips the good coin since the capital is now \$1 or -\$1. When the capital is a multiple of 3, the player must toss the bad coin resulting in a probable loss; hence the capital decreases by \$1. Suppose the capital is \$9, then when flipping the bad coin the capital most likely goes to \$8, after which the player flips the good coin, resulting most likely with new capital back up to \$9, after which it most likely goes back to \$8. The net result is that there tend to be oscillations of capital from \$9, \$8, \$9, \$8, ..., and similar oscillations around other multiples of \$3.



Simulation of Game B; oscillations around multiple of 3 barriers
Figure 2

To analyze Game B, we must determine the fraction of times a coin, regardless of whether it is a good or bad coin, turns up HEADS. If the game were simulated on a computer for 1,000,000 coin tosses (either the good or bad coin), the fraction of times the coin turns up HEADS is roughly 49.4% of the time. What that says is that on average you will lose \$1 more than you win \$1, hence Game B is also a losing game.

THE GOOD NEWS!

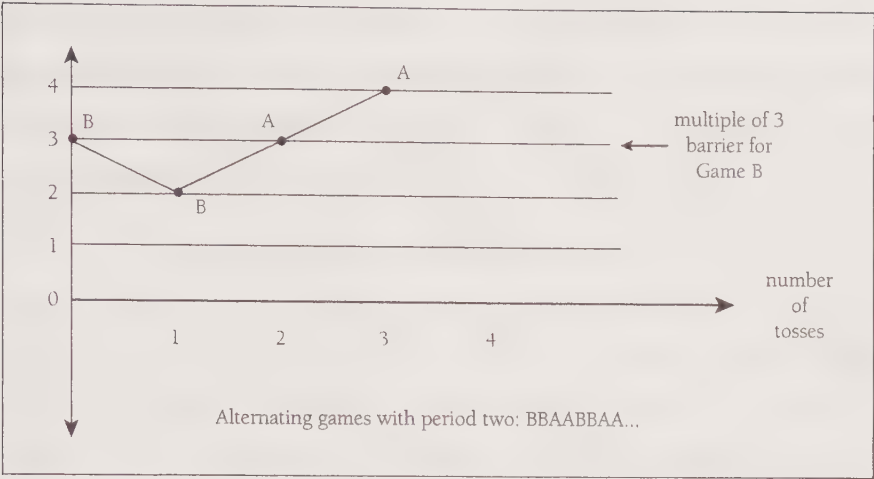
Now, for the good part. We have seen that both Game A and Game B are losing propositions, but we will now see that by alternating games in some manner, the player will win over time. To see why this happens, first recall that the reason Game B is a losing game due to the fact that it is hard to get past the multiple of \$3 barriers, when the

capital is ... -\$6, -\$3, \$0, \$3, \$6, Hence, the player wins on average only 9.5% of the time. But by alternating play between games, let's see how this barrier is overcome. Suppose we alternate play by playing Game B two times, then Game A two times, and so on, which we denote by BBAABBAA ... Look what happens in Figure 3.

Flip n (Playing Game B): We start by playing Game B with a capital of \$3 (normally we start at \$0, but we start at \$3 for illustration). The probability of losing is 0.905 and so our capital most likely goes back down to \$2. (This is the dreaded "multiple of three" barrier for Game B.)

Flip $n+1$ (Playing Game B): We now play Game B again (remember we are playing the order BBAABBAA...) and the odds are now in our favor since the capital is not a multiple of \$3. Let's assume we win and our player goes back up to \$3.

Flip $n+2$ (Playing Game A): We are now back to the dreaded multiple of 3 barrier, but this time we are playing Game A, which isn't bothered by the barrier (only Game B is bothered by the barrier). Here, the odds are slightly against us but basically 50-50. Hence, we have a good chance of breaking through the \$3 barrier.



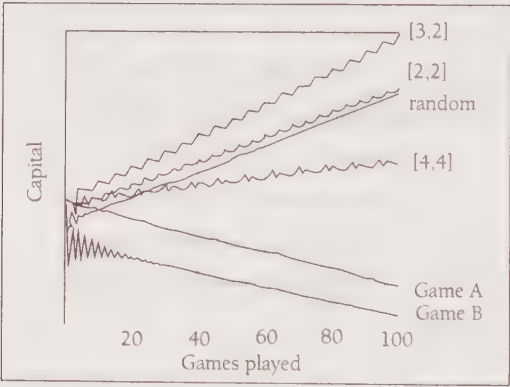
Alternating games wins the day
Figure 3

This is why alternating the games results in a winning strategy. When Game B is in dire straits because of one of the multiple of \$3 barriers, Game A comes to the rescue. Game A isn't a winning game, but it has the ability to get through the multiple of \$3 barriers more than does Game B.

For a sports analogy, consider two tennis players, Mary and John. Mary makes 49.5% of her shots (hence, she misses 50.5%), which means she loses in the long run. (Think of Mary as Game A.) John is like Game B, he is a pretty good player, making 75.5% of normal shots, but has a glaring weakness in that he only makes 9.5% of his shots when he has to rush the net, and with that weakness he too loses in the long run. But together, by alternating play of the ball, they are winners: on occasion when a player has to rush the net, exactly those times when John normally misses the ball, Mary is there to make the play.

The same phenomenon may be at work in medicine when two drugs, each not very effective in fighting a disease, become effective when pre-scribed alternately. So-called “cocktails” of drugs that are sometimes prescribed in fighting diseases might to some extent be behaving like the games in the Parrando Paradox.

Figure 4 shows the results of a computer simulation by Parrondo, in which he switched between games, playing Game A a total of a times and Game B a total of b times. The notation $[a,b]$ at the right of the graph denotes this. For example $[3,2]$ refers to playing the games in order AAABBAABB, ... Random order means that a fair coin flip determines which game to play.

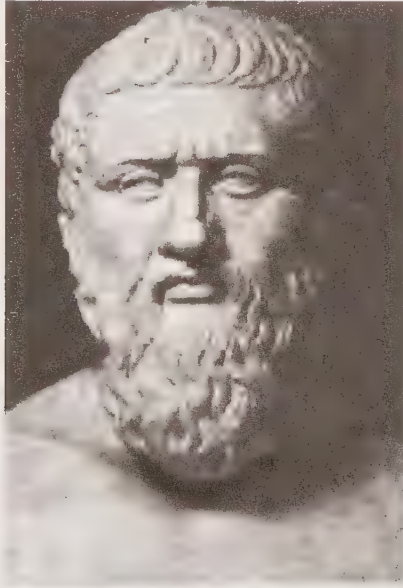


Simulation of different alternations of games
Figure 4

ΧΔΛΣΘΑΑΩΞ

Zeno's Paradox

Some of the oldest paradoxes in mathematics go back to the Greek philosopher Zeno of Elea (circa 450 B.C.) who is credited with paradoxes involving space and time, the most famous being Achilles and the Tortoise Paradox (Achilles being the Greek hero of Homer's *The Illiad*). In this paradox, Zeno "proves" that *anything that moves can never catch up with anything that moves slower.*



Zeno

The paradox takes the form of a dialogue between Achilles and a very smart Tortoise, where the Tortoise challenges the great Achilles to a footrace, claiming he would win if Achilles would give him a small head start. Achilles, of course, laughs at the thought of the idea since he

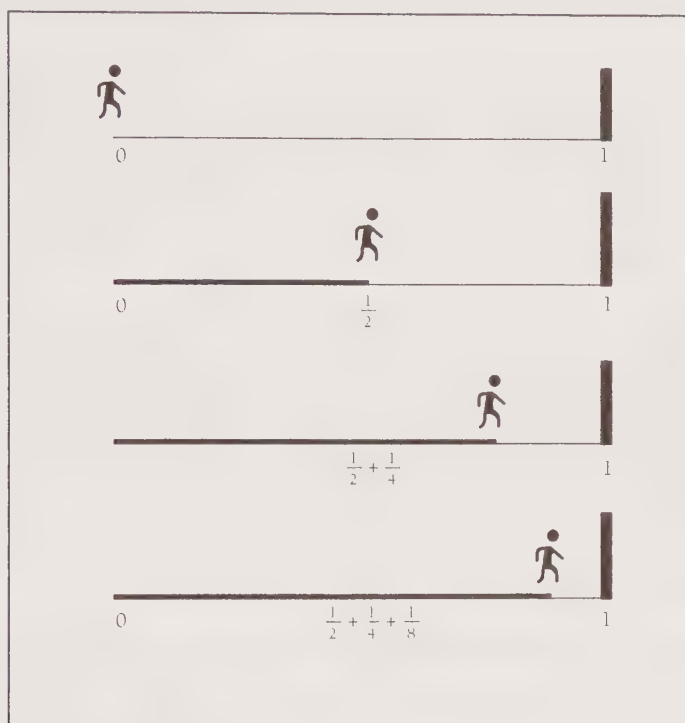
was a great Greek warrior, and the Tortoise's track accomplishments were not exactly legendary. Achilles was even more astounded when the Tortoise told him he needed only a 10-meter lead. The Tortoise then shocked Achilles further when he told him there wasn't any reason to race since he could *prove* that he (the Tortoise) would win before they even started! It just gets weirder and weirder.

Achilles was amused at the braggadocio of the Tortoise so he told him to make his case. The Tortoise said if he were given a 10-meter head start, then by the time Achilles ran the first 10 meters, the Tortoise would have traveled a small distance, the net effect being that Achilles would still be behind the Tortoise.

"And how long would it take you to make up the extra distance?" the Tortoise asked Achilles. Achilles responded boastfully that it would take almost no time at all. But the Tortoise said in the time Achilles made up that little distance, he (the Tortoise) would have gone a little farther, so Achilles would still be behind, and must make up another distance.

"Uh, yes," Achilles looked puzzled. The Tortoise then said that while Achilles made up the new distance, he (the Tortoise) would again have gone a little farther. Hence, each time Achilles made up the extra distance, the Tortoise would have moved a little farther ahead. The net effect being that Achilles would never catch the Tortoise, whereupon Achilles agreed and conceded the race to the Tortoise.

This paradox is clearly a “precalculus” enigma. Any student of calculus would (hopefully) laugh his or her head off at it. To resolve it, we rephrase it in a different way. Suppose you cross a room to the opposite wall, 1 unit away as illustrated in Figure 1.



Walking to a wall 1 unit away in halves.

Figure 1

You first cover half the distance to the wall: then you cover half the remaining distance after that, and so on forever; the consequence, if you believe Zeno's thinking, is that you never reach the wall, since you will be traversing an infinite number of distances. In fact, if you buy into that line of reasoning, you would have to agree that *all* motion is impossible!

However, adding up the infinite number of half-distances, we know that the sum is 1, the distance to the wall:

$$\text{Distance to the wall} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots = 1$$

So, now we see the flaw in Zeno's Paradox: simply because the distance Achilles must run to catch the Tortoise can be subdivided into an infinite number of smaller distances, that doesn't mean the sum of those distances is infinite. In fact, it's a finite number!

Suppose for example that Achilles runs 10 times faster than the Tortoise, and Achilles gives the Tortoise a 10-meter lead. In this case, arguing as before, the total distance Achilles runs before catching the Tortoise is¹

$$\begin{aligned} \text{Distance to catch the Tortoise} &= 10 + 1 + \frac{1}{10} + \frac{1}{100} + \cdots + \frac{1}{10^n} + \cdots \\ &= 10 + \left(1 + \frac{1}{10} + \frac{1}{100} + \cdots + \frac{1}{10^n} + \cdots \right) \\ &= 10 + \frac{10}{9} \\ &= 11\frac{1}{9} \text{ (meters)} \end{aligned}$$

ΦΓΔΠΘΩΣΨ

¹ We use the identity $1 + x + x^2 + x^3 + \cdots = 1 / (1 - x)$ when $-1 < x < 1$ to get this result.

Will the *Real* Numbers Please Stand Up?

There have been paradoxes in the history of mathematics that have been resolved so completely that today these paradoxes of the past are completely forgotten. The history of numbers is rife with paradoxes where mathematicians were dragged, kicking and screaming, into new number systems.

Zero: There once was a time when there was no zero. Of course, people *knew* when they had nothing, they just didn't think of giving a number to their nothing. After all, numbers were supposed to count things, like 1 wife, 2 goats, 3 sons, and so on. If you didn't have a wife, you simply said no wife.

It is generally thought by math historians that zero originated independently in three different cultures: the Babylonians in the Tigris-Euphrates Valley, the Maya of Central America, and in India. The question of exactly who first used zero as a number arises from the fact that zero was used in different ways by different peoples. Sometimes zero was simply a "placeholder" as in 102 to distinguish 102 goats from 12 goats. This was how the Babylonians first used zero between 400 and 300 B.C. To Indian mathematicians, however, it was used as a number in computations, and for this reason most math historians say zero had its origins in India.

In early cultures, numbers were thought of as concrete things rather than the abstract entities which they are today. It is a giant leap from thinking of no pigs to 0 “things” to the abstract concept of 0. Zero was brought to the West around 1200 A.D. by the Italian mathematician Fibonacci (aka Leonardo of Pisa), who brought it along with the rest of the Arabic numerals from his travels in north Africa.

Mathematicians in Europe resisted the use of zero up to the 1600s. Even the Italian mathematician Gerolamo Cardano, who solved the cubic and quartic equations, did so without the use of zero. It wasn't until the Hindu-Arabic number system we use today was accepted, that zero became an important number in arithmetic. (There is a difference between 15 and 1,005, right?)

Negative Numbers: The concept of a “negative” number was not universally accepted until the 14th Century. At the time, people didn't accept as a number anything less than nothing. A 14th-century French mathematician, F.C. Busset, said a failing of the French education system was the acceptance of negative quantities. Subtracting 3 from 2 was a paradox.

$$2 - 3 = \text{paradox}$$

Often to overcome “negative” solutions to problems, they would restate the problem so the result came out positive. Mathematicians would

only give positive solutions to equations like $x^2 = 1$. saying the solution was +1, but that was the only solution, not -1.

One mathematician argued “How can a larger number to a smaller number be the same as a smaller number to a larger number,” referring to the new found identity

$$\frac{1}{-1} = \frac{-1}{1}$$

The hostility to negative numbers was brought out in a 1759 essay by English mathematician Francis Maseres, who commented on “-”, the newly created negative sign

“They are useful only in so far as I am able to judge, to darken the very whole doctrines of the equations and to make dark of the things which are in their nature excessively obvious and simple. It would have been desirable that negative roots were never allowed in algebra and that they were discarded.”

Only over time did negative numbers take their place in the pantheon of the number system of mathematics. Today, we don’t think of numbers without including negative numbers.

Complex Numbers: After negative numbers were accepted, mathematicians tried to find numbers that when squared yielded negative numbers. Not finding any, they gave up. In the 1500s, speculation about square roots of negative numbers was brought up in the research of

Italian mathematicians Niccolo Tartaglia and Geralamo Cardano, who found the roots of 3rd- and 4th-order polynomial equations.

In 1545, Cardano published a book *Ars Magna*, where he solved the equation

$$x^2 - 10x + 40 = 0$$

getting $x = 5 \pm \sqrt{-15}$. He said working with these kinds of numbers was “mental torture.”

Later, in 1637, the great French mathematician Rene Descartes, who also objected to complex numbers, came up with the standard way of writing complex numbers: $a + ib$. It was Descartes’ opinion that if complex numbers were involved in a problem, the problem had no solution. Even the great Isaac Newton, the coinventor of calculus, agreed with Descartes.

It was a paradox to mathematicians in those days of how i multiplied by i could be -1 , and how $-i$ multiplied by i could be $+1$. Over time, however, complex numbers were accepted and the focus placed on understanding them. In 1806, Jean-Robert Argand showed how complex numbers could be represented as points in the plane, and in 1831, the eminent Carl Friedrich Gauss made Argand’s idea popular. By this time, there was no question that complex numbers constituted numbers that could be added, subtracted, multiplied, and divided, just like the real numbers.

Today, complex numbers and complex analysis are the cornerstones of much of mathematics, and their applications to science, engineering, and mathematics are too numerous to mention. It is safe to say that the modern world we live in today would be vastly different without the invention of $i = \sqrt{-1}$.

One wonders what ideas we are resisting today will be the cornerstones of mathematics tomorrow.

$$\Delta\Phi\Pi\Theta\Xi\Omega\Psi$$

The Four Bears Paradox

Mama and Papa Bear decide to raise a family and have decided upon four baby bears. Papa Bear wants boy bears and Mama Bear wants girl bears. Papa Bear says, "We will both be happy since most likely we will have two bears of each sex." But Mama Bear doesn't agree, saying it was more likely there will be three bears of one sex and one of the other. Who do you think was right, Papa Bear or Mama Bear?

WHO IS CORRECT?



People generally say it's more likely there will be two bears of each sex than three bears of one sex and one of the other. But Mama Bear had taken a course in probability at B.U. (Bear University) and gave the

following argument. She said the probability of something happening can be found by adding the desired number of times that something actually occurs, divided by the total number of things happening. For example, if you flip a coin, the probability of turning up a HEAD is $\frac{1}{2}$ because there is one favorable outcome (HEAD), out of two total outcomes {**HEAD, TAIL**}.

“So, let’s add up the total number of ways we can have four bears,” Mama Bear said. In Table 1 she enumerated the 16 possible ways to have 4 baby bears, all the way from 4 girl bears (GGGG) to 4 boy bears (BBBB). The combination GGBG in Brood 3 means (reading from the right) the first baby is a girl, the second a boy, and the third and fourth are girls.

Brood	Sexes	Brood	Sexes
1	GGGG	9	BGGG
2	GGGB	10	BGGB
3	GGBG	11	BGBG
4	GGBB	12	BGBB
5	GBGG	13	BBGG
6	GBGB	14	BBGB
7	GBBG	15	BBBG
8	GBBB	16	BBBB

16 possible broods for 4 baby bears
Table 1

Note that in Broods 4, 6, 7, 10, 11, and 13 contain two boy bears and 2 girl bears, whereas in Broods 2,3,5,8,9,12,14 and 15 contain either 3 boy bears and 1 girl bear or 3 girl bears, and one boy bear. Hence, the probability of having 2 boy bears and 2 girl bears is $\frac{6}{16} = 0.375$ (37.5%

chance), whereas the probability of getting 3 of one sex and 1 of the other sex is $\frac{8}{16}$ (0.5% chance).

If you guessed wrong, you most likely didn't take into account the fact that there are *two* ways of having 3 offspring of one sex and 1 of the other: 3 boys and 1 girl, and 3 girls and 1 boy.

ΦΠΨΩΞΣΘ

Uncle George's Will

Uncle George, a very rich uncle, who made his millions by writing math books, has left his priceless collection of 59 high-carat diamonds, each valued at \$1 million dollars, to his three nephews and two nieces, dispersed in the following manner:

Buffy gets $\frac{1}{2}$ the diamonds

Egbert gets $\frac{1}{4}$ the diamonds

Wilfred gets $\frac{1}{6}$ the diamonds

Audrey gets $\frac{1}{20}$ the diamonds

Garwin gets $\frac{1}{60}$ the diamonds

The will stipulates that the diamonds should not be cut into smaller pieces, and that each beneficiary must receive the *exact* fraction of diamonds specified in the will, otherwise they get nothing.



After some thought, Buffy realizes that Uncle George has played one of his nasty tricks since it is impossible to portion out 59 diamonds in fractional amounts of $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{20}$, and $\frac{1}{60}$ since

$$\text{Buffy gets } 59\left(\frac{1}{2}\right) = 29\frac{1}{2} \text{ diamonds}$$

$$\text{Egbert gets } 59\left(\frac{1}{4}\right) = 14\frac{3}{4} \text{ diamonds}$$

$$\text{Wilfred gets } 59\left(\frac{1}{6}\right) = 9\frac{5}{6} \text{ diamonds}$$

$$\text{Audrey gets } 59\left(\frac{1}{20}\right) = 2\frac{19}{20} \text{ diamonds}$$

$$\text{Garwin gets } 59\left(\frac{1}{60}\right) = \frac{59}{60} \text{ diamonds}$$

Finally, Garwin suggests they see a math professor at the local university, who had a reputation as being a whiz with fractions.

“Oh, this is quite simple,” Professor Birdhatcher told them after a few minutes. “Here, let me loan you *my* own diamond for a few minutes, and I’ll solve this problem to everyone’s satisfaction.” The professor then took out a shiny object from his pocket and placed it alongside the other 59 diamonds spread out on a velvet pad.

“With my diamond, there are now 60 diamonds on the table,” he said. “Now, I want each of you to take the *exact* fraction of diamonds as specified in the will.” So, with that:

Buffy takes 30 diamonds from the 60, which is her allocated $\frac{1}{2}$

Egbert takes 15 diamonds from the 60, which is his allocated $\frac{1}{4}$

Wilfred takes 10 diamonds from the 60, which is his allocated $\frac{1}{6}$

Audrey takes 3 diamonds from the 60, which is her allocated $\frac{1}{20}$

Garwin takes 1 diamond from the 60, which is his allocated $\frac{1}{60}$

After this was done, they have collected a total of

$$\text{diamonds collected} = 30 + 15 + 10 + 3 + 1 = 59$$

leaving one diamond remaining on the table.

“And that’s the diamond I loaned you,” Professor Birdhatcher said, putting it in his pocket. “It’s been a pleasure doing business with you,” he said as he walked out the door, leaving everyone extremely pleased.

So, how did the professor pull off this math-a-magic trick?

RESOLVE THE PARADOX

This paradox has been around for over a thousand years in various forms¹. It is really a fraud since the fractions in Uncle George’s will *do*

¹ There is an old Arabian paradox that involves horses instead of diamonds, but the idea is the same.

not add up to 1. Uncle George was either very poor at arithmetic, or he was a sly old devil. The fractions add up to²

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{20} + \frac{1}{60} = \frac{59}{60}$$

In other words, the fractional amount of diamonds specified in the will is completely bogus, and so the professor resorts to a nonmathematical solution by realizing that although 59 is not divisible by the desired numbers 2, 4, 6, 20, or 60, the number 60 is, hence he simply observes that if he momentarily added his own diamond to Uncle George's 59, getting 60 diamonds, it would be possible to divvy out the diamonds in the specified fractional amounts of $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{20}$, and $\frac{1}{60}$. Yielding

Buffy's fraction $\frac{1}{2}$ gives her $60\left(\frac{1}{2}\right) = 30$ diamonds

Egbert's fraction $\frac{1}{4}$ gives him $60\left(\frac{1}{4}\right) = 15$ diamonds

Wilfred's fraction $\frac{1}{6}$ gives him $60\left(\frac{1}{6}\right) = 10$ diamonds

Audrey's fraction $\frac{1}{20}$ gives her $60\left(\frac{1}{20}\right) = 3$ diamonds

Garwin's fraction $\frac{1}{60}$ gives him $60\left(\frac{1}{60}\right) = 1$ diamond

And what's more, the professor realizes that the allocated diamonds to the five heirs will add up to 59, not 60, since

² If there were 2 heirs and the will specified that one person got $\frac{1}{2}$ of diamonds and the other $\frac{1}{4}$ the diamonds, you would have objected and said what about the other $\frac{1}{4}$, but in this example one never thinks to ask if the fractions add up to 1.

$$60\left(\frac{1}{2}\right) + 60\left(\frac{1}{4}\right) + 60\left(\frac{1}{6}\right) + 60\left(\frac{1}{20}\right) + 60\left(\frac{1}{60}\right) = 60\left(\frac{59}{60}\right) = 59$$

There was no reason to think they should add up since the original fractional amount of diamonds willed to the heirs was mathematically invalid. So, after the 59 diamonds are dispersed to the heirs, everyone is happy, and the professor takes his diamond and goes home.

In other words, the solution is not mathematical. It worked because the 59 and fractions $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{6}$, $\frac{1}{20}$, and $\frac{1}{60}$, just happened to line up for a nice solution.

Epilogue: You can make up a similar paradox by looking for equations of the form

$$\frac{n}{n+1} = \frac{1}{k} + \frac{1}{l} + \frac{1}{m}$$

where k , l , m , and n are integers. The number of terms on the right does not have to be three, and in fact in Uncle George's paradox, we had five terms on the right:

$$\frac{59}{60} = \frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{20} + \frac{1}{60}$$

Here, 59 is the number of diamonds, and to make a good paradox you seek a number that is preferably prime, like 59, in which the next larger number of 60 is divisible by many small whole numbers, like 2, 4, 6, 20, and 60. Good luck.

Milos and the Tainted Wine

Milos is between a rock and a hard place. As the king's official sommelier, his duty is to keep the wine flowing at the royal galas. The king has just announced a celebration commemorating his years on the throne and has ordered Milos to break out 1,000 bottles of the king's finest. However, the day before the big event, a castle informant tells Milos that someone has spiked one of the bottles with poison, the exact bottle being a mystery.

Milos knows that if one of the guests croaks over a bad bottle of wine, Milos' life and career as a sommelier will probably come to an end. However, the king has 1,000 prisoners in jails and dungeons around the kingdom, and Milos, being a friend of the royal jailer, has the authority to request that any prisoner take a sip from any of the 1,000 bottles.

However, with only 24 hours before the celebration, there is no time to round up all the prisoners, and so Milos must devise a plan to find the tainted wine with the fewest number of prisoners.

To do this, Milos contacts the king's mathematician for assistance. Milos tells the mathematician it takes six hours for a person to die after drinking the tainted wine.

Milos was surprised when the mathematician provided him with a plan. We pose the problem to you, the reader. What is the *fewest*

number of prisoners required to determine the tainted wine? Can you get by with fewer than 1,000 prisoners? If so, how many? 500? 250?. ... smaller?

MAKE YOUR GUESS, OR BETTER, YOUR ANSWER

To answer this question, we use an age-old technique in mathematics; solve the *same* problem with smaller numbers. Assume we have eight bottles of wine with one tainted. We will see that three prisoners are sufficient to find the bad bottle, but that two are not. To see this, we number the eight bottles 0, 1, 2, ... 7 rather than the more customary 1, 2, 3, ... 8. This is one of those problems where a picture is worth a thousand words. See Table 1.

Bottle	Prisoner		
	1	2	3
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Distribution of 7 bottles of wine to 3 prisoners
Table 1

Table 1 shows the results of giving wine to three prisoners, each prisoner getting a sip from 4 out of the 8 bottles. A “1” under a prisoner in

a given row means the prisoner has drunk from the bottle in the given row, while a “0” means the prisoner did *not* drink from the bottle in the given row.

For example

Bottle 0: 000 = none of the prisoners taste bottle 0

Bottle 1: 001 = prisoner 3 tastes bottle 1

Bottle 2: 010 = prisoner 2 tastes bottle 2

Bottle 3: 011 = prisoners 2 and 3 taste bottle 3

Bottle 4: 100 = prisoner 1 tastes bottle 4

Bottle 5: 101 = prisoners 1 and 3 taste bottle 5

Bottle 6: 110 = prisoners 1 and 2 taste bottle 6

Bottle 7: 111 = all three prisoners taste bottle 7

In other words, if no prisoner dies, bottle 0 is the tainted bottle (Right?) and if prisoners 1 and 3 die, then bottle 5 is tainted (see it?) and if prisoners 2 and 3 die, bottle 6 is tainted, and so on. Observing the prisoners that die will tell us one which of the 8 bottles is tainted.

From Table 1 we can design an experiment so each prisoner only needs to drink from one bottle. Milos could prepare a wine cocktail for each prisoner; the first cocktail would include wine from bottles 4–7 which would be given to prisoner 1, a cocktail of wines from bottles 2, 3, 6, and 7 given to prisoner 2, and a cocktail from bottles 1, 3, 5, and 7 given to prisoner 3. This strategy has both good and bad news. On one hand, each prisoner gets to imbibe from a blending of 50% of the king's finest stock, and the bad news is the chance each prisoner imbibes from a poisonous bottle is also 50%.

We also see in Table 1 that two prisoners are not enough for 8 bottles of wine since there are only *four* combinations; 00, 01, 10, and 11 of a two-digit binary number. In other words, two prisoners would be sufficient to find a bad bottle within four bottles, but not more than four.

If the king had $2^4 = 16$ bottles of wine, then four prisoners would be necessary; if the king had $2^5 = 32$ bottles of wine, five prisoners would be required; if the king had $2^{10} = 1,024$ bottles, ten prisoners are required. In Milos' problem, the number of bottles is not an exact power of 2, but between $2^9 = 512$ and $2^{10} = 1024$, hence 10 prisoners are sufficient, but 9 is not enough.

In Table 2, we have listed each ten-digit *binary* number from 0000000000 to 1111101000 (the binary number for the decimal number 1000). As it was in Table 1, a "1" means a prisoner had sipped from the bottle described in a given row. Milos will carry out the experiment by bringing the 10 prisoners into a tasting room. He then places the first bottle aside (bottle 0) so no prisoner tastes it, the second bottle (bottle 1) is given to prisoner 10 (as specified by 0000000001); the third bottle given to prisoner 9 (as specified by 0000000010); the fourth bottle to prisoners 9 and 10 (as specified by 0000000011) and so on. Since it takes six hours for a prisoner to die, after tasting the tainted wine, it is necessary for Milos to distribute the 1,000 bottles to the prisoners as quickly as possible to find the bad bottle within 24 hours.

Bottle	Prisoner 1,2,3,4,5,6,7,8,9,10	
0	0000000000	
1	0000000001	
2	0000000010	
3	0000000011	
...	...	
548	1000100100	← bottle 548 given prisoners 1,5,8
...	...	
999	1111100111	

Distribution of 1,000 bottles of wine to 10 prisoners
Table 2

Milos administers minuscule portions of wine from each of the 1,000 bottles to the 10 prisoners, each prisoner getting tastes from 500 bottles, in a manner described by Table 2. For example, bottle 0 is set aside and given to no one, bottle 1 is given to prisoner 10, and so on, until bottle 999 is given to prisoners 1,2,3,4,5,8,9, and 10 (as specified by 1111100111). Milos then just sits back, relaxes, and waits for a prisoner or prisoners to drop dead. If prisoners 2, 3, and 9 drew the short straw, then since 0110000010 has 1s in the 2nd, 3rd, and 9th positions and since the binary number 0110000010 is equal to the decimal¹

$$2^8 + 2^7 + 2^1 = 386$$

the poisonous bottle is 386.

¹ The conversion of binary numbers to decimal numbers is very simple. As an example, the binary number 1101 is the same as the decimal number $1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = 1 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 13$. In other words, the 1.1.0.1 in the binary number are the *coefficients* of 2^3 , 2^2 , 2^1 , 2^0 . Using this logic, the binary number 0110000010 is the same as the decimal $2^8 + 2^7 + 2^1 = 386$.

In general, the number of prisoners required to detect the bad bottle grows at the rate of logarithm to base 2 of the number of bottles. For example, to test 1,000 bottles would require n prisoners where n satisfies the equation $2^{10} = 1024$, hence $n = \log_2 1024 = 10$ prisoners are required. If there had been 500 bottles, the number of prisoners required would be the value of n , that satisfies $2^n = 512$, hence $n = \log_2 512 = 9$ prisoners are required. We took 512 bottles since it is the next power of 2 larger than 500. It seems surprising that the determination of one bad bottle out of 60,000 could be accomplished with only $n = \log_2 65,536 = 16$. Of course, each prisoner probably doesn't care about his fate since each person would have taken sips from 30,000 bottles.²

ΧΔΓΦΩΞΒΨΠ

² Of course, if a person dies *instantly* after drinking the poison, then the bad wine has been found and the experiment can be stopped. We haven't considered that aspect of the problem.

Space-Filling Curve

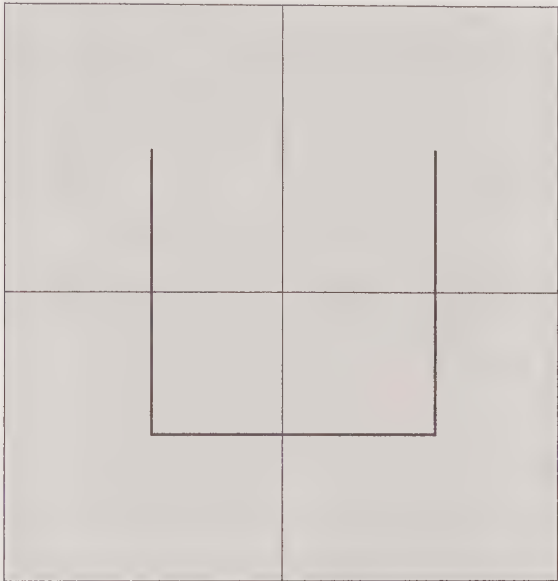
There have been few paradoxes in mathematics which have shaken the mathematical community to the core as the one devised by Italian mathematician, Giuseppe Peano in 1890. Not everyone would call it a paradox, but it is certainly counterintuitive. To give you an idea of the stir it caused, here is what a couple of notable mathematicians said at one time of Peano's "monster."

Naum Ya. Vilenkin, 1965: "Everything has come unstrung! It's difficult to put into words the effect that Peano's result had on the mathematical world. It seemed that everything is in ruins, and that all the basic mathematical concepts have lost their meaning."

Hans Hahn, 1956: "It cannot possibly be grasped by intuition: it can only be understood by logical analysis."

You would think that simply drawing a curve in the plane would not have the effect it did. But this was not an ordinary curve. You see, curves do not have any width, only length. But, nevertheless Peano found a way to draw a curve in the plane that filled the entire plane, without a single point left over! A plane-filling curve, if you wish. After Peano made this amazing discovery, other mathematicians jumped in and drew their own plane-filling curves. One was due to the eminent German mathematician David Hilbert (1862–1943).

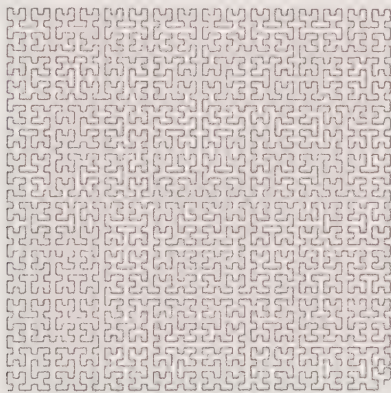
Instead of covering the entire plane with a curve, we will just fill a square. If you can cover a square with a curve, it is an easy matter to cover the entire plane since the plane is just made up of squares. We start with a basic “U-shape” drawn in Figure 1.



First step in the plane-filling curve
Figure 1

We now double the number of grid lines so instead of the square being subdivided into 4 smaller squares as in Figure 1, the square is subdivided into 16 smaller squares, and draw our basic “U’s,” the top two sitting on their sides, and the bottom two “U’s” turned up the same way as in Figure 1. After you draw those 4 “U’s” you connect them by lines in Figure 2. Note that a couple of the connecting lines are vertical and a couple are horizontal.

Of course Peano doesn't stop with just three iterations, he hasn't even gotten a good start. He continues this process an infinite number of times, and although we can't draw his final curve, we can get a good idea of the final product by looking at drawing in Figure 4 showing the results after 16 iterations. What Peano proved was that if this process is continued indefinitely, then no matter what point in the original square (or plane) is chosen, Peano's curve will eventually cover the point, thus qualifying it as a *space-filling curve*.



Peano Curve after 16 iterations

Figure 4

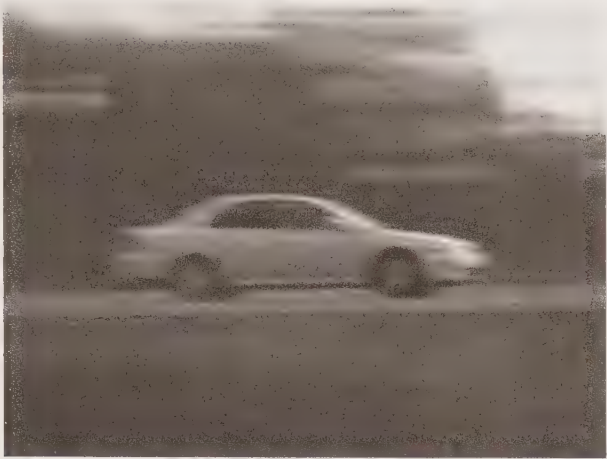
ΧΔΓΦΩΞΒΨΠ

Wheels: More Than Just Round and Round

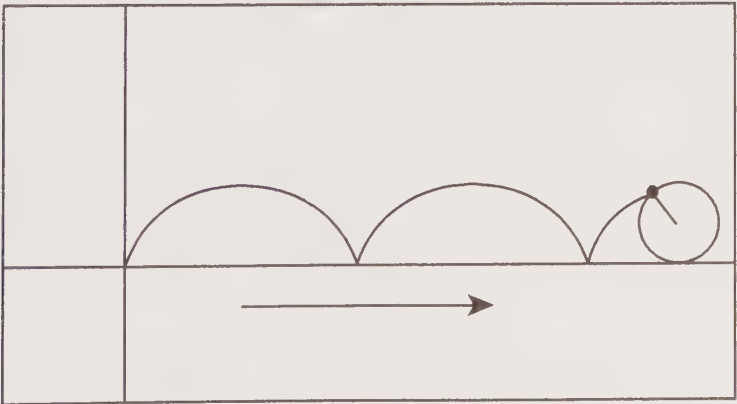
No doubt we go about our daily routine unaware of all sorts of interesting phenomena going on around us. There was a time not so long ago when we were completely unaware of most electromagnetic radiation, such as radio and television waves, which bombard us day and night. Who knows what physical presence is whirling right under our noses, about which we are completely oblivious.

One such phenomenon, agreed not on the scale of importance as electromagnetic radiation, but nevertheless something we can all relate to, has to do with the wheel, one of the oldest human inventions. To place this phenomenon in the context of our daily lives, are you aware that when you are driving your car, part of the car isn't moving in the same direction as you are? In fact, part of the car isn't moving at all! And what's more, part of the car is moving twice as fast as you! It's true!

Let's say the tires on your car have a circumference of 6 feet, which is a good estimate for most passenger cars, and suppose you take a leisurely drive for an hour at a constant speed of 60 mph. Under these conditions, over the course of that hour, every point on your tires, where the rubber meets the road, will go from 0 mph to 120 mph and back to 0 mph a total of 52,800 times. And what's more, this little jerk-and-stop routine performed by your wheels is enacted 16.6 times per second.



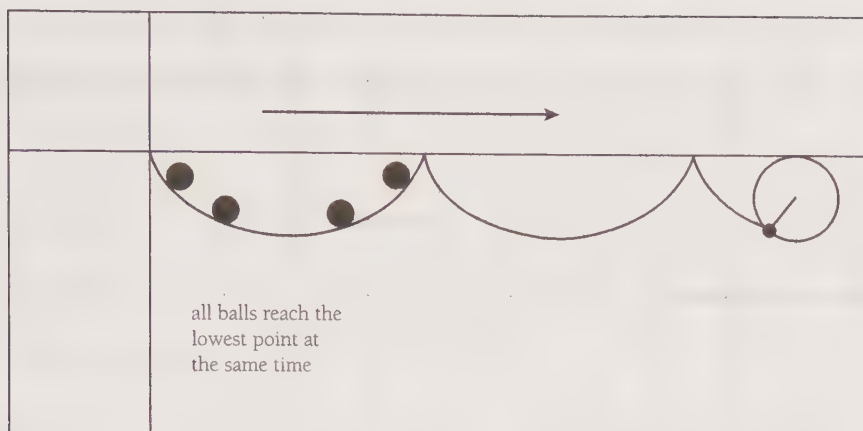
Archaeologists claim the wheel was invented in Mesopotamia over 5,000 years ago, and its mathematical idealization, the circle, is arguably the most studied geometric shape in all of mathematics. One fascinating property of a circle, not known to most people outside of mathematics, has to do with the path traced out by a point on a circle as the circle rolls on a flat surface. See Figure 1. The curve is called a *cycloid* and, according to math historians, was first studied by *Nicholas of Cusa* (1401–1464), a German cardinal in the Catholic Church.



Cycloid traced out by a point on a rolling circle
Figure 1

The cycloid was later studied by Galileo who gave the curve its name. During the 1600s, the cycloid was studied by many of the great mathematicians of the day, including G.P. de Roberval, Christopher Wren, Christian Huygens, James Bernoulli, Johann Bernoulli, Pierre de Fermat, Blaise Pascal, and many others. Some of the major discoveries at the time:

- a) In 1634, G.P. de Roberval showed that the area under a cycloid is three times the area of the circle that creates the cycloid.
- b) In 1658, Christopher Wren showed that the length of one arc of a cycloid is four times the diameter of the circle that creates the cycloid.
- c) In 1658, Christian Huygens discovered experimentally that when balls placed anywhere inside an *inverted* cycloid (See Figure 2), and released at the same time, they all reach the lowest point of the cycloid at the same time, showing experimentally that the cycloid is the solution to the *isochrone* problem, which is the problem of finding the curve with the property that objects placed on the curve at the same time all reach the bottom at the same time. Huygens went on to invent the pendulum clock, constructed in a way that the bob of the pendulum followed the path of a cycloid, thus making the period of each swing independent of the amplitude of the bob, a must have property for an accurate pendulum clock.

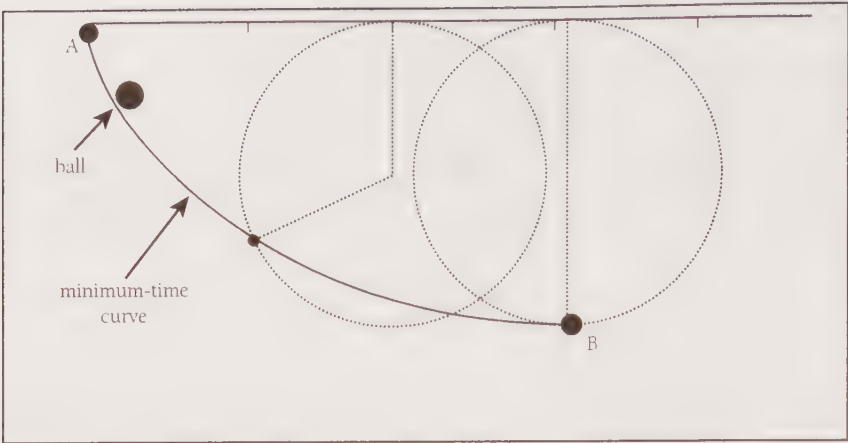


Wheel rolling on a ceiling gives the inverted cycloid

Figure 2

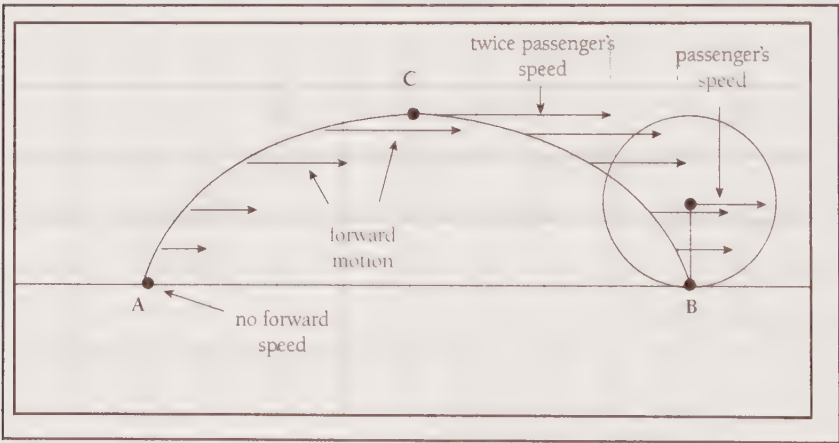
d) In 1690, James Bernoulli proved *mathematically* what Huygens showed experimentally, that the inverted cycloid is the solution of the isochrone problem.

e) In 1696 Johann Bernoulli, the brother of James Bernoulli, showed the inverted cycloid is also the solution of the *brachistochrone* problem, which is the problem of finding the curve that minimizes the time taken for a ball to roll down the curve. The cycloid would be the perfect childhood slide, giving faster descents than a straight line or a circular path. See Figure 3.



Inverted cycloid is the minimum-time curve
Figure 3

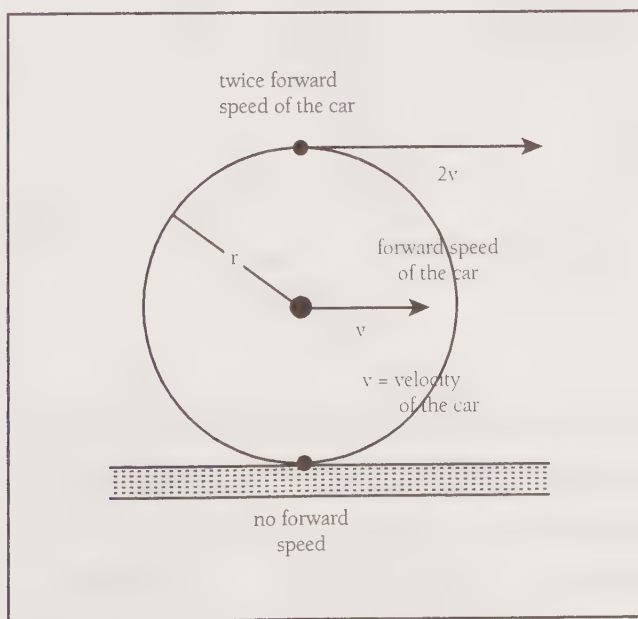
Back to the Wheels on Your Car: Returning to your car, Figure 4 shows the cycloid traced out as the tire makes one complete revolution.



Forward speeds of a wheel
Figure 4

Note that at Points A and B where a given point on the wheel touches the ground, the cycloid goes *straight up*, with no forward movement to the right. In other words, when a part of the wheel touches the road, it doesn't have any forward motion. It does have motion, of course, but the motion is straight down, then straight up. In fact, the tangential motion of points on the outer edges of the wheels is the same as the forward speed of the car.

Now, when a point on the wheel is at its maximum height, its motion is all forward (i.e., no vertical motion) and at that point its forward speed is twice the speed of the car. (It has both the translation motion of the car, and also a rotation motion.) Only the center point of the wheel always travels at the same forward speed as the car, as shown in Figure 5.

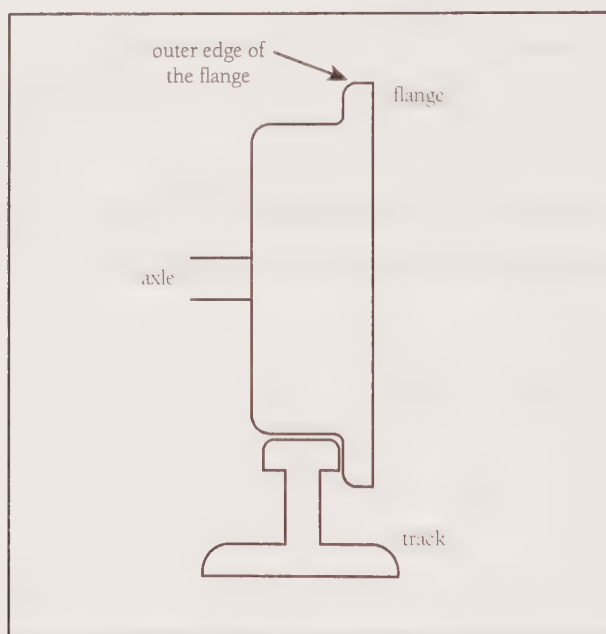


Forward motion of different parts of the wheel

Figure 5

So, do you have a new respect for the wheels of your car? The next time you drive at 60 mph, imagine that each point (outside edge) of your tires has a forward speed that goes from 0 mph to 120 mph and back to 0 mph, and at 16 times each second.

Well, if you are impressed with the wheels of your car, wait till you study the wheels of a train. See Figure 6.



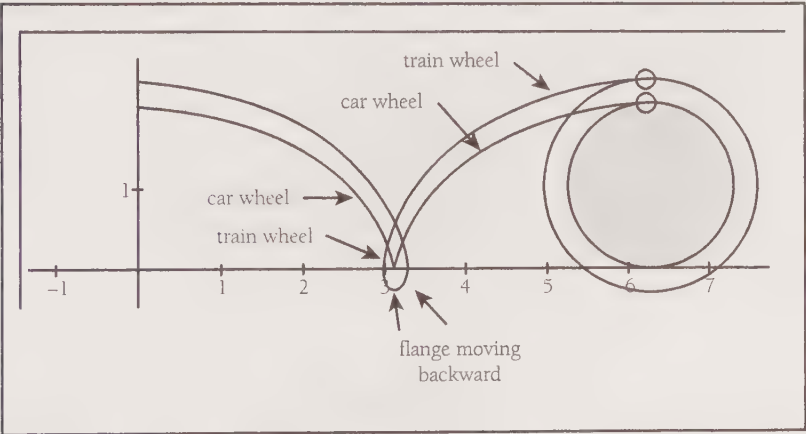
Train wheel

Figure 6

As we have seen, the cycloid is the curve traced out by a point on the *edge* of the wheel as it rolls on a flat surface. If we now trace a point lying on the outside edge of the flange (i.e., that part of the train wheel that sticks down over the track to keep the train on the track), then we arrive at a new path, called the *prolate cycloid*. Figure 7 shows the

“regular” cycloid created by a car wheel and the prolate cycloid formed by a train wheel.

If you follow along the prolate cycloid, you see it moves to the left when the curve is below the train tracks. In other words, points on the outer edge of the flange momentarily travel in the opposite direction to that of the train, and they do this every rotation of the wheels. The wheels of a train moving at 120 mph will go through the four-cycle movement of “going forward, stopping, going backward, stopping” a total of roughly 50 times a second. The exact number would depend on the size of the wheels.



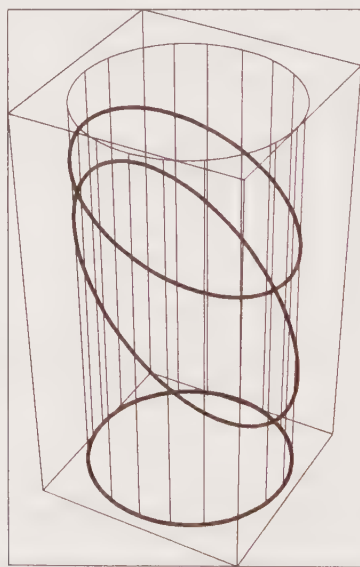
Prolate cycloid traced out by a point on the flange wheel.

Figure 7

References: There are several excellent websites that animate the rolling of a wheel on a flat surface. Simply enter words like cycloid, rolling wheel, rolling circle, wheel paradox, and you will find many pages.

Space Travel in a Wineglass

We have known since the time of the German astronomer and mathematician Johannes Kepler (1571–1630) that the planets revolve about the sun in elliptical paths, which are kind of elongated circles. The ellipse is one of the most interesting curves in all mathematics, having so many fascinating properties it would require a large book with very small print to describe them all. An ellipse can be drawn or interpreted in a variety of ways, a common geometric one being the intersection of a plane and a cylinder, as drawn in Figure 1.



Ellipse as an Intersection of a Plane and a Cylinder

Figure 1

Now, regarding the orbit of the earth, I just thought of a good conversation starter you might use at your next party. As you are about to

imbibe on a glass of sparkling Napa valley Syrah, look thoughtfully into your glass and announce to everyone that you are seeing the orbit of the earth. You see, if you look in the right place, you can see the orbit of the earth right in the palm of your hand. In fact, if you look more carefully, you can also see the orbits of the comets and asteroids as well, like Comet Kohoutek and the Hidalgo asteroid.

To set the scene, assume that you are drinking your wine from a cylindrical transparent wineglass and that you are looking directly down into the glass, and most important, that the glass is tilted slightly.



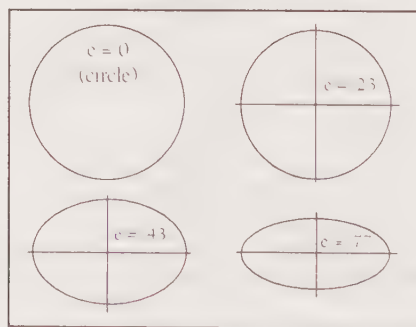
You see a circle when the glass is not titled.



Oblong appearance of the wine when the glass is titled

Since an ellipse is the intersection of a cylinder (the sides of the glass) and a plane (the surface of the wine), then as you tilt the glass various degrees, the surface of the wine takes the shape of an ellipse inside the glass. Hence, if you tilt the glass various degrees, you can see the orbits of planets, even comets for that matter.

Before continuing our party activities, we digress for a moment to discuss ellipses. Roughly, an ellipse is an oblong circle, where the amount of oblong-ness, or how much it deviates from a circle, is called the *eccentricity*¹ of the ellipse. A circle has eccentricity 0, and the more oblong the ellipse, the closer the eccentricity gets to 1. The eccentricity of an ellipse cannot actually be 1, or else it is not called an ellipse, it would be called a parabola. Figure 2 shows ellipses with different eccentricities.

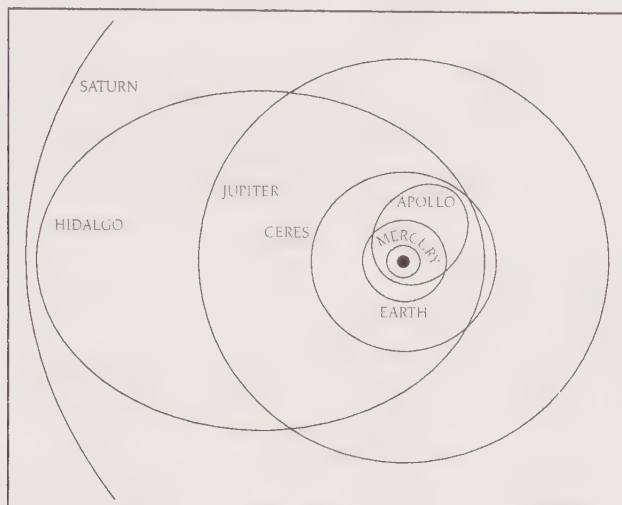


Ellipses with different eccentricities

Figure 2

¹ There are formulae for the eccentricity of an ellipse, but we need not worry about that here.

The orbit of the earth, although an ellipse, appears like a circle to the naked eye. having an ellipse of only 0.0167. On the other hand, the Hidalgo asteroid has eccentricity 0.66. which is the largest eccentricity of all the earth's asteroids. See Figure 3.



Orbit of Hidalgo and various planets
Figure 3

The Comet Kohoutek, which follows an elliptical path around the sun once every 75,000 years, is extremely oblong and has an eccentricity of 0.97, which almost looks like a straight line when drawn.

So, getting back to the party, the important question arises, how many degrees should you tilt your glass of wine to see an ellipse with a given eccentricity? We will not do the proof², but the angle θ which you should tilt the glass to achieve an eccentricity e is given by the simple trigonometric formula

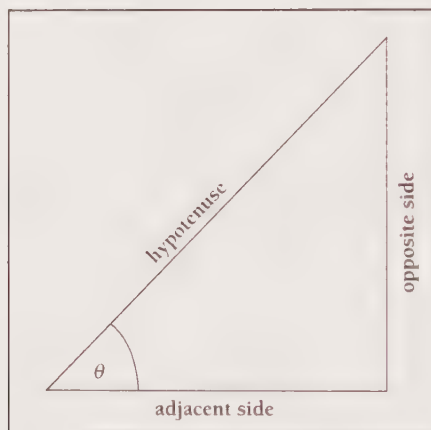
² The verification of this relationship is often asked of students in classes in analytic geometry and calculus who are familiar with the equation of an ellipse.

$$e = \sin \theta$$

where the sine function $\sin \theta$, as all high-school students of trigonometry know, is the ratio of the opposite side of an angle θ of a right triangle to the hypotenuse. That is

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

illustrated in Figure 4.



Right triangle
Figure 4

Now, it is possible to solve the equation $e = \sin \theta$ for the θ in terms of e , thus giving a formula that tells us how many degrees θ we should tilt the glass to see an orbit with eccentricity e . Solving for θ gives a well-known equation to students of trigonometry

$$\theta = \sin^{-1} e$$

where $\sin^{-1} e$ is the *inverse sine function* of e , whose values can be found on most calculators.³

So, there you have it, a simple formula that provides the number of degrees one must tilt a glass to view an ellipse of given eccentricity. Table 1 shows the tilt required to view various orbits of planets, comets, and asteroids in our solar system.

Planet	Eccentricity e	Angle of tilt θ (degrees)
Mercury	0.206	11.89°
Venus	0.0068	0.39°
Earth	0.0167	0.96°
Mars	0.0934	5.36°
Jupiter	0.0485	2.78°
Saturn	0.0556	3.19°
Uranus	0.0472	2.71°
Neptune	0.0086	0.49°
Pluto	0.250	14.48°
Asteroids and Comets	Eccentricity e	Angle of tilt θ
Ceres	0.078	4°
Hidalgo	0.66	41°
Halley	0.97	76°
Kohoutek	0.999	89°

Required tilt to view various orbits
Table 1

³ In fact, if you go to the website www.wolframalpha.com and enter `arcsin(0.0164)` in the little box, you will get the angle you must tilt the glass to get an ellipse having eccentricity 0.0167.

So, you too can be the life of the party, leading everyone on a planet-orbit expedition. You begin your journey by telling everyone to tilt their glasses 1 degree (0.96 degree to be exact), and by doing so they will see a miniature version of earth’s orbit in their wineglass. Some might say they only see a circle, which is understandable since the eccentricity of the earth’s orbit is only 0.0167, which makes it nearly a circle. You then continue your orbit expedition by having them tilt their glasses 14 degrees, after which you announce they are viewing the orbit of Pluto, the most distant planet,⁴ which has a larger eccentricity of 0.25. In this case, the orbit is clearly oblong.

One of the most interesting objects that orbits the earth is Halley’s comet, which orbits the earth every 76 years and whose orbit has an eccentricity of 0.97, which in turn means that to view the orbit in your wineglass you must tilt it

$$\theta = \sin^{-1} (0.97) = 75.93^{\circ}$$

In other words, almost 76 degrees. This orbit is so oblong that it will show up well in your glass provided, of course, that you don’t spill half your drink. It is at this point you propose a toast to Mr. Halley and his comet and proceed to drink the remainder of the wine.

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⁴ I guess you would say ex-planet, since Pluto has been demoted to “dwarf planet” status.

The Mysterious Transcendental Numbers

We saw in paradoxes involving infinity¹ how the real numbers can be subdivided into two distinct types, the *rational* numbers (ratios of integers, i.e., $\frac{3}{2}$, $\frac{16}{216}$, $\frac{1}{2}$, ...) and the *irrational* numbers (numbers that cannot be written as the ratio of integers, i.e., $\pi/2$, e , $\sqrt{2}$, ...). Another way of thinking about a rational number p/q is as the solution of a first-order polynomial equation

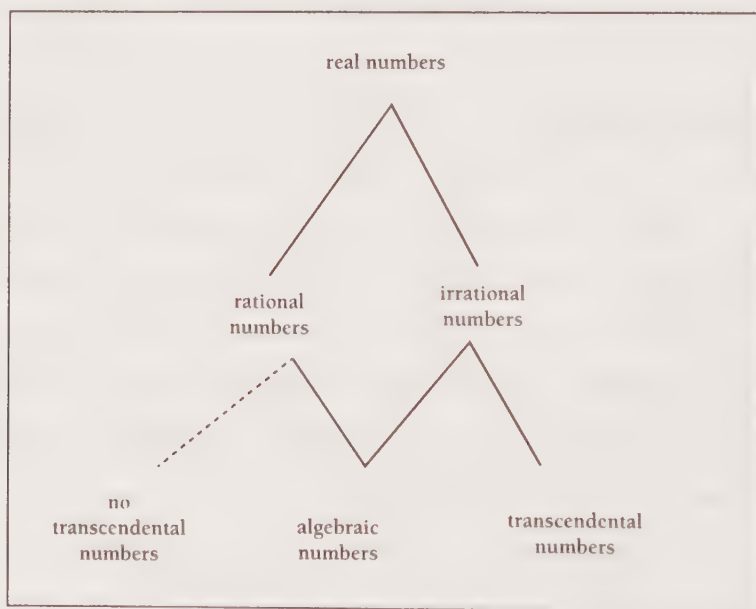
$$qx - p = 0$$

with integer coefficients p and q where $q \neq 0$. So, it is natural to enlarge the rational numbers, getting a larger class of numbers, which are solutions of both 1st-order and 2nd-order polynomial equations with integer coefficients. We could then consider 3rd-order rational numbers, which are roots of 1st-order, 2nd-order, and 3rd-order polynomial equations with integer coefficients. Continuing in this manner, we can even consider the “ultimate rational” numbers, which are real roots of some polynomial equations such as

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

¹ We are referring to Through Cantor's Looking Glass, p.80.

for *any* degree 2, 3,. Numbers like this would include the rational numbers and some irrational numbers as well. Who knows, maybe they would include all real numbers. We could call these numbers “generalized” rational numbers, although the common name for such numbers is *real algebraic numbers*, and for the fun of it, let’s suppose there *are* some real numbers left over that are *not* algebraic numbers.² We will call these numbers *transcendental numbers*. Clearly, every transcendental number is irrational. You can think of the transcendental numbers as very special irrational numbers. Figure 1 illustrates where the algebraic and transcendental numbers lie within the real number system.



Classification of real numbers

Figure 1

² We hate to keep saying the polynomial equation has integer coefficients over and over. So, when we say polynomial equation we mean it has integer coefficients.

So, are there any of these mysterious transcendental numbers? Certainly algebraic numbers seem to be more common. They include the integers since any integer n is the solution of the (trivial) polynomial equation

$$x - n = 0$$

as well as the rational numbers (ratio of two integers) since any rational number p/q is a solution of the (also trivial) polynomial equation with integer coefficients

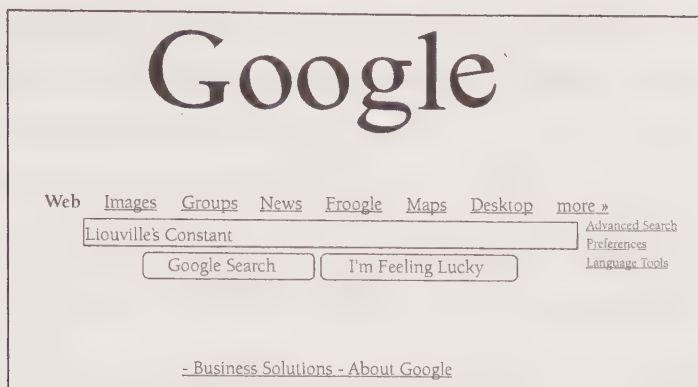
$$qx - p = 0 \Rightarrow x = p / q$$

What's more, the algebraic numbers include some irrational numbers, such as the square and cube roots $\sqrt{2}$ and $\sqrt[3]{2}$, since they are respective solutions of the polynomial equations³

$$x^2 - 2 = 0 \quad \text{and} \quad x^3 - 2 = 0$$

A few more algebraic numbers are given in Table 1.

³ Of course, many polynomial equations can have both real and complex roots. In this chapter, we are discussing only *real* transcendental and algebraic numbers. If we enlarged our discussion to complex numbers, we would say that $\pm i$ are complex algebraic numbers since they are solutions of $x^2 + 1 = 0$.



Lots of material on the Internet

So, where is the paradox in showing the existence of a *single* transcendental number? The paradox is that in 1874 Cantor proved there are an infinite number of transcendental numbers, and in fact *more* transcendental numbers than algebraic numbers, although if you asked most people (even mathematicians) to list just five, most would have a hard time. The first person to show that an “existing” number is transcendental and not a “madeup” one like Liouville’s Constant, was the French mathematician Charles Hermite (1822–1901), who proved in 1873 that Euler’s Constant $e = 2.718\dots$ is transcendental. Nine years later in 1882, German mathematician Ferdinand von Lindemann (1852–1939) showed that the holiest of all numbers π is also transcendental, although it is unknown to this day whether their sum $\pi + e$ is transcendental. Other numbers, however, have been proven (with great effort) to be transcendental, including

$$e^\pi \text{ and } 2^{\sqrt{2}}$$

although it is unknown at this time whether π^e is transcendental.

Lindemann’s proof that π is transcendental was important in the history of mathematics since it provided the first proof that many geometric problems of antiquity, which had baffled mathematicians for 2,000 years, were in fact insoluble, the most famous being the *squaring of the circle*, the problem of constructing a square with the same area as a given circle using only an unmarked straightedge and compass.

What Cantor proved in 1874 was that the algebraic numbers are denumerable (with the same cardinality as the integers), the basic idea in his proof being that the set of polynomials with integer coefficients is countable, and thus the finite number of solutions of these equations is also countable. Cantor then argued that since the real numbers are *uncountable* and the algebraic numbers *countable*, then the transcendental numbers must be *uncountable*. This means in a very real sense there are *more* transcendental numbers than algebraic numbers, even though very few transcendental numbers (or classes of them) are known. In our thinking, this classifies as a paradox.

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Gabriel's Horn Paradox

Some people just die too soon. One such person was Evangelista Torricelli, an Italian mathematician who died at the age of 39 in the year 1647. Had he lived longer, he just *might* have been the person who discovered calculus, and not Sir Isaac Newton and Gottfried Leibniz.

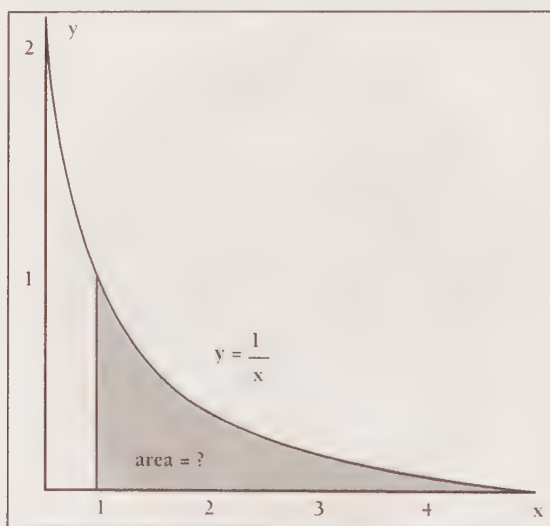


Evangelista Torricelli (1608–1647)

Although Torricelli's contributions to science and mathematics are too numerous to mention (OK, for one, he invented the barometer), it is his conundrum of comparing areas and volumes, called Torricelli's Paradox, that we describe here. You might think of it as the "continuous" version of the paint can paradox we discussed beginning on page 66. The paradox was discovered in 1641 and is even more amazing when you consider that Torricelli did not have the tool of integral calculus at his

disposal, since Newton and Leibniz had not yet uncovered it. Torricelli used a technique called the *Cavalieri method of indivisibles*, a predecessor of the integral, to demonstrate the paradox. However, the author does have integral calculus at his disposal, so the paradox is presented in the manner shown to students of calculus for the past 350+ years.

The paradox involves a surprising result related to the area under the curve $y = 1/x$, $1 < x < \infty$ shown in Figure 1.



What is the area under the curve?

Figure 1

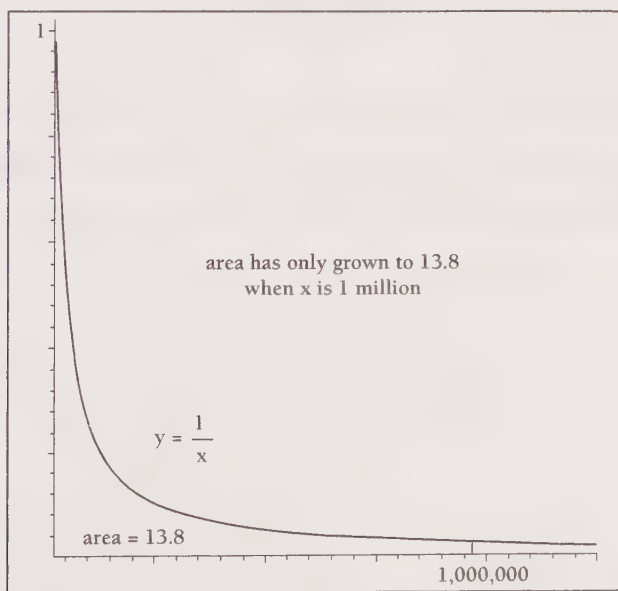
Any student of calculus knows (or should know) that the area under the curve and above the x -axis can be found by evaluating the improper integral

$$A = \int_1^{\infty} \left(\frac{1}{x} \right) dx \equiv \lim_{R \rightarrow \infty} \int_1^R \left(\frac{1}{x} \right) dx$$

Evaluating this integral gives

$$\begin{aligned}
 A &= \lim_{R \rightarrow \infty} \int_1^R \left(\frac{1}{x} \right) dx \\
 &= \lim_{R \rightarrow \infty} (\ln x) \Big|_1^R \\
 &= \lim_{R \rightarrow \infty} (\ln R - \ln 1) \\
 &= \lim_{R \rightarrow \infty} (\ln R) \\
 &= \infty
 \end{aligned}$$

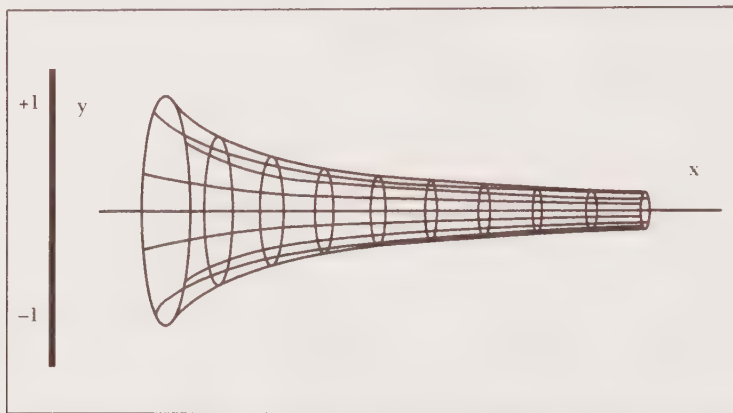
In other words, the area under the curve is infinite. What that means is that the area increases *without bound* as we move the right edge of the shaded area more and more to the right, although it reaches infinity very slowly. In fact, when we move all the way to $x = 1,000,000$, the area under the curve is only $\ln(1,000,000) = 13.8$.



Growth to infinity is *very* slow, but it *still* goes to infinity.

Figure 2

So, where is the paradox? After discovering the area under the curve to be infinite, Torricelli decided to find the volume of the region obtained by rotating this region around the x -axis as shown in Figure 3. This solid is called *Gabriel's Horn*.



Gabriel's Horn

Figure 3

Roughly, as students of calculus know, this volume can be found by adding up the areas of all the circles drawn in the horn, and since the area of a circle with radius r is $A = \pi r^2$ and since the radius of the circle at location x is $1/x$, the area of a circle at x is

$$A(x) = \pi r^2 = \pi \left(\frac{1}{x} \right)^2 = \frac{\pi}{x^2}$$

Hence, the volume of Gabriel's Horn is found by "adding" up these infinite number of areas by means of the integral

$$\text{Volume} = \int_1^{\infty} \left(\frac{\pi}{x^2} \right) dx = -\frac{\pi}{x} \Big|_1^{\infty} = -\pi \left(\frac{1}{\infty} - \frac{1}{1} \right) = \pi$$

In other words, the volume of Gabriel's Horn is π cubic units! To put this in perspective, we have rotated an *infinite area* around a line and obtained a *finite volume*.

That's like saying you fill Gabriel's Horn with $\pi \doteq 3.14$ cubic units of paint, but there's not enough paint in the world to paint its outside. If the units were in inches, that would mean you could fill Gabriel's Horn with about a pint of paint. Believe it or not!

So, why has mathematics gone awry? Well, it really hasn't. It is simply not valid to compare numbers in different dimensions. Area is a two-dimensional property and volume is a three-dimensional property.

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Russell's Paradox: Why We Need Rules

Have you ever watched a game between rival teams, where rules were relaxed and each team took the philosophy that anything goes? Unless there is mutual agreement between teams on rules of engagement, things can get out of hand in a hurry. In some areas of mathematics, notably set theory, the rules of the game were not set in stone (i.e., an axiomatic framework¹) until around 1900. Before that time, understanding of the subject was based on intuition. In an axiom system, everyone follows the same rules (axioms), hence everyone has the same starting point, and there will be agreement about what is right and what is wrong.

The concept of a set lies at the foundation of mathematics, and at first glance its basic principles offer little resistance to serious study. After all, on an intuitive level, a set is simply a collection of things, called members of the set. Typical sets can vary all the way from the members of your third-grade class to the grains of sand on a beach. Finite sets, those that contain a finite number of members, have never offered serious logical problems to mathematicians, although a student of finite probability might offer a different assessment.

¹ It is interesting that axioms for plane geometry were agreed upon by Greek mathematicians over 2,000 years ago, but set theory was only placed in an axiomatic framework 100 years ago.

Infinite sets, however, such as the real and complex numbers, collections of functions, and so on, important in the study of pure mathematics, are a different matter.

In the later part of the 1800s, cracks started to appear in the foundation of set theory, especially when it came to infinite sets. Everyone agreed that the definition of a set is simply a collection of things, which cannot be an easier concept. But over time, the problems in set theory became subtler. People started asking whether *anything* should be allowed membership in a set or should membership be restricted, something like membership in a fancy country club where only “quality” members need apply. But before you emphatically insist that anything should belong, you might want to expand your universe of possibilities.

To make sure we are on the same page, here are typical A , B , and C

$$A = \{1, 2, 3\}$$

$$B = \{a, b, c, \dots, x, y, z\}$$

$$C = \{x: 0 < x < 1\}$$

denoted by capital letters, and whose members are placed inside a brace $\{\}$. Their meaning is clear: the Set A contains the member 1, 2, 3; the Set B contains the letters of the alphabet; and the Set C contains the real numbers between 0 and 1. Rather tame sets, you must agree. So, can't you think of anything more exotic to include in your set? Anything?

As a teacher of college students, I had generally about this time some nontraditional-type student, normally sitting in the back row playing

with his laptop, come up with something out of the blue. He suggests we could allow the set *itself* to belong to a set. How can that be, you ask? Well, you just include it, he argues, just because you've never seen a set like that before, it doesn't mean it can't happen. The student then displays his candidate Set D on the board:

$$D = \{1, 2, 3, D\}$$

which contains the three members 1,2,3 and D itself.

The student then noted another set that contains itself is the largest set anyone can imagine: the *set of all sets*, a “*mother set*” or “*universal set*” if you like, which we write.

$$U = \{\text{everything}\}$$

This set certainly contains itself, after all, it contains *everything*, and by everything we mean *everything*. Think of something: well, that thing is in the Set U . Think of something else: that thing also belongs to U . Everything you can imagine and not imagine belongs to the universal set U .

About this time, a young woman in the class objects to the boy's universal *set of all sets* and makes a subtle argument in which she proves such a “universal set” cannot exist. She points out that the universal set can be subdivided into two parts: first, those sets that *do* contain themselves, which she calls *BAD SETS*; and second, those sets that do *not* contain themselves, which she calls *GOOD SETS*, or written mathematically as

$$U = \text{GOOD SETS} \cup \text{BAD SETS}$$

where

GOOD SETS = all sets that do not contain themselves

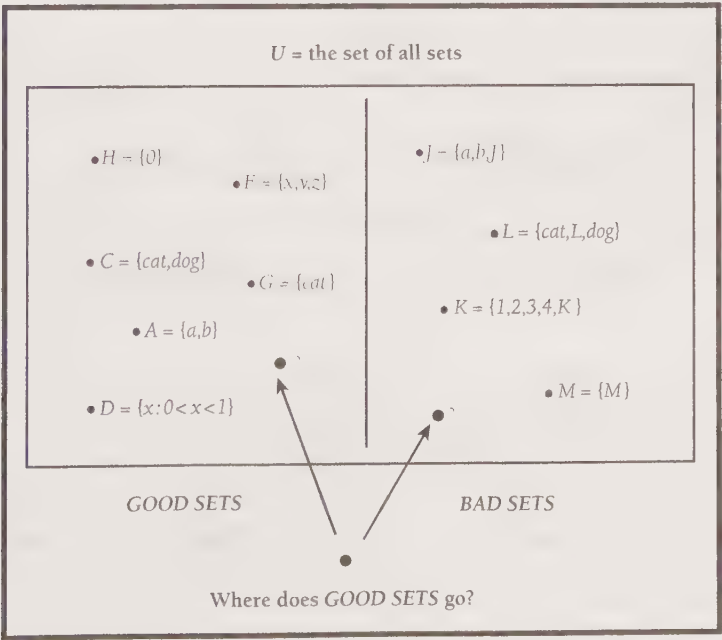
BAD SETS = all sets that contain themselves

The sets

$A = \{1,2,3\}, \qquad B = \{a,b,c,\dots,x,y,z\}, \qquad C = \{x : 0 < x < 1\}$

are typical members of the set *GOOD SETS*, and $D = \{1,2,c,D\}$ is a typical member of the set *BAD SETS*.

She then says that since the universal set U , the *set of all sets*, contains *everything*, it must contain *GOOD SETS*, but then the question arises does *GOOD SETS* belong to *GOOD SETS* itself, or does it belong to *BAD SETS*? By this time, the collective brain of the class was starting to heat up so she draws on the board the diagram of the sets. See Figure 1.



Universal set U divided into *GOOD SETS* and *BAD SETS*
Figure 1

She then asks her question once again. If *every* set lies in the universal set U , it must contain *GOOD SETS*, but where does it lie, in *GOOD SETS* or in *BAD SETS*?

She then answers her own question. Suppose first, *GOOD SETS* is a member of *GOOD SETS*. But *GOOD SETS* contains *only* sets that do *not* contain themselves and so *GOOD SETS* is not a member of *GOOD SETS*. Conversely, suppose *GOOD SETS* is *not* a member of *GOOD SETS*, which means it is a member of *BAD SETS*. But *BAD SETS* contains those sets that do contain themselves, which means *GOOD SETS* contains itself, but this is a contradiction since by definition *GOOD SETS* does not contain itself. Hence, we have proven that *GOOD SETS* belongs to *GOOD SETS* and does not belong to *GOOD SETS*. Whew, what an argument!

Or stated another way, the girl proved:

GOOD SETS is a member of GOOD SETS
if and only if
GOOD SETS is not a member of GOOD SETS

which, in the language of set theory, says that the universal set U , the *set of all sets*, makes no sense.² In other words, a set that contains itself is not considered a set in the axiomatic study of sets, since if it is considered a set, one runs into the logical difficulties discussed here.

² In the language of set theory, if $\mathbf{A}$ is the set of all sets that do not contain themselves, then $A \in A \Leftrightarrow A \notin A$.

The argument that sets cannot contain themselves is due to the English logician Bertrand Russell who, in 1902, pointed out the logical difficulties when it comes to infinite sets. His research was part of the motivation that prompted mathematicians to place set theory on an axiomatic footing. Although Russell's proof that there can be no universal set is couched in the language of sets, the same principle can be restated in natural language as the Barber of Seville Paradox, which states:

Barber of Seville Paradox: The barber (man) of Seville shaves every man in Seville who does *not* shave himself (he doesn't shave those who *do* shave themselves). The question then arises, does the barber shave *himself*?

Paradox: If he does shave himself, then he doesn't shave himself since he shaves only those who don't shave themselves. On the other hand, if he does not shave himself, he does shave himself since he shaves those who don't shave themselves. Hence, we conclude the barber shaves himself if and only if he does not shave himself. A paradox.

Resolution of the Paradox. The resolution of this paradox takes its clue from Russell's Paradox in set theory. Here, we simply point out that there cannot *be* a barber of Seville with the given attribute, since his stated characteristics engender a paradox.

The most commonly accepted axioms of set theory are the Zermelo-Fraenkel-Choice³ axioms, (ZFC) axioms, developed by German logicians Ernst Zermelo (1871–1956) and Abraham Fraenkel (1891–1965). Zermelo⁴ conceived of a set of axioms, like those of plane geometry, which specifies the types of sets that can be allowed.

Reference: Google the following words and phrases:
Russell’s Paradox, Zermelo–Fraenkel axioms, Barber of Seville Paradox, the Liar Paradox.

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³ There are other axioms of set theory than the ZFC axioms, such as the von Neumann-Bernays-Gödel axioms (which are logically equivalent to ZFC) and the Morse-Kelley axioms, which are “stronger” than ZFC.
⁴ Zermelo published his original axioms in 1908 that were modified in 1922 by Abraham Fraenkel and Thoralf Skolem, and thus today are called the Zermelo-Fraenkel (ZF) axioms.

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